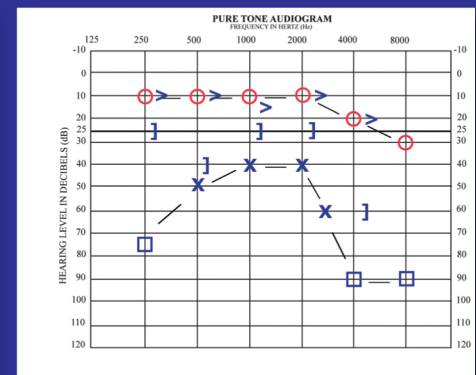
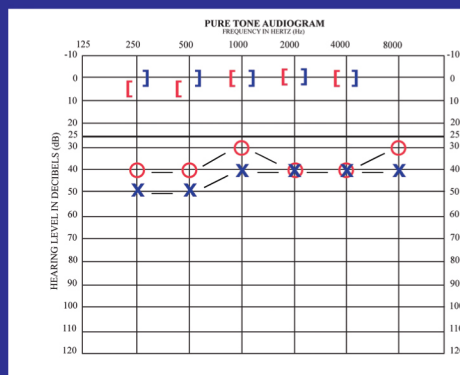
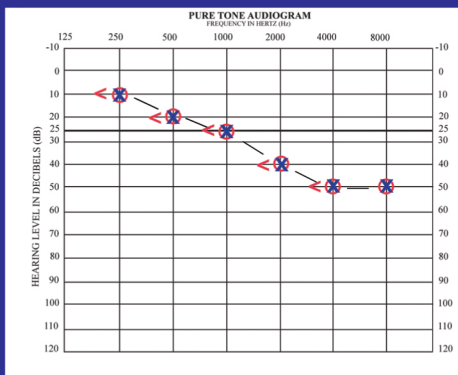


RAPID AUDIOGRAM INTERPRETATION

A CLINICIAN'S MANUAL

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RAPID AUDIOGRAM INTERPRETATION: A CLINICIAN'S MANUAL

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PREFACE

Dear Learners:

We are so happy you have found this book! It is the culmination of many years of searching, compiling, and revising to bring it to completion.

By way of introduction, the idea for this book was devised during our otolaryngology residency training at the University of California, Davis in Sacramento, California. Early in our training, we were overwhelmed by the complexity of the audiogram. Providing rapid, thorough interpretations for our patients proved quite difficult. The difficulty, however, was not with understanding the testing or the data, but rather the lack of a structured way to learn about audiogram interpretation.

Our search for a reliable, dedicated resource proved futile and frustrating. Most of the information we found was either too detailed and geared to doctorate-level audiologists or oversimplified for use by the general public. We found no interpretation guide produced for the clinicians-in-training

or other nonaudiologists who work in overlapping communication sciences who may reference audiogram information regularly.

With this in mind, we decided to bring on audiologist, Samantha McKinney, to help provide and create our own text to guide our learning, and, in the process, we developed a methodical, stepwise approach that can be applied to every audiogram, every time. Over the years, we have provided this as an informal guide to many students and residents with overwhelmingly positive results. It has provided a means for beginners to master interpretation of the audiogram. It has also served as a general review reference for more experienced clinicians.

Herein, we outline our stepwise approach for interpreting audiograms, no matter how complicated, by introducing a framework that is easy to master and will ensure a rapid, yet thorough interpretation every time. It has served us (and we hope our patients!) well throughout our training and beyond, and we hope it does the same for you.

—LS, BC, & SM

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We express our devoted gratitude to our supportive families through this long process. Our dreams have become reality through their encouragement and patience. We also wish to thank all of our teachers and mentors who helped craft our knowledge of this topic. To Professors Dr. Hilary Brodie, Dr. Karen Doyle, Dr. Ben Balough, and Dr. Rodney Diaz, for providing us with thousands of patients to learn the art and science of otology. To Dr. Robert Ivory and Dr. Rebecca Kane, for planting and cultivating the early seeds that inspired our foray into clinical audi-

ology. To all of the UC Davis expert audiologists, for offering advice and support in creating the extensive audiometric data provided herein. To our colleagues at Plural Publishing, especially Nicole Hodges and Valerie Johns, for the interminable patience and valuable insights throughout this arduous process. To Dr. Raj Dedhia, who assisted in crafting and enhancing the illustrations that adorn the pages of our text. Finally, to each of our patients, who one by one showed us how to really interpret audiometric data into meaningful clinical outcomes.

INTRODUCTION TO OUR METHOD

We envision this book to be a manual, a quick-reference guide, and a workbook, an all-in-one resource. We have tried to limit the areas of bulky text, as it is not meant to be a textbook. Rather, we have created this guide using an expanded outline model, with an appropriate depth of information provided in each section to help the reader grasp all necessary skills required to interpret the audiogram.

Our main purpose is to create a clinician-centered text, not an in-depth study into audiology. As such, its applicability is broad. Early communication disorder students (audiology, deaf & hard of hearing studies, and speech-language pathology) may also find this information particularly helpful to begin the process of learning this difficult topic. As this is meant to be an introductory handbook only, if you so choose to learn more detailed audiometric science, we urge you to continue your search once you have mastered the steps outlined in this manual.

This book's most valuable contribution is the 8-step audiogram interpretation method introduced here and used throughout the text. You will find this step-by-step guide for interpreting audiograms within this chapter. We expect that the reader will frequently utilize this page as a quick reference during initial forays in interpretation, as well as a pocket go-to reference as the reader provides patient care. As the sections unfold, the reader will be guided through a stepwise summary of the technique and reasoning behind each step in our method. In the process, these chapters cover each aspect of audiometric testing, outlining and defining common verbiage, and providing a link to the glossary of terms at the end of this manual. Words included in the glossary will be designated by bold type whenever they are initially encountered within this text.

Within each section of text, you will also find boxes that contain "PEARLS" to better understand

and apply each audiometric principle to everyday clinical practice. These tips are designed to provide additional context to the material being discussed in each section. We encourage you to return and review these important tips frequently as you begin your clinical care.

To help each of you best benefit from this manual, we have also included 65 practice audiograms to help you practice our method. We hope that, with practice, you will learn to quickly recognize common hearing loss patterns and master our method. The provided audiogram examples progressively become more complex to help guide you into higher levels of understanding. Each audiogram is accompanied with official interpretations, matching the stepwise interpretation process we recommend.

We urge you, the reader, to use this resource as you need it. Whether you are a new student to the field or a seasoned clinician brushing up on audiogram interpretation, we aim to provide you with a quick, foolproof, methodical approach that will consistently yield accurate audiogram interpretation. Best of luck!

The Method

The following is an overview of our recommended stepwise method for rapid audiogram interpretation (Figure 0-1). The steps are meant to be followed in sequence to allow for all portions of the audiogram to be reviewed thoroughly. This will ensure an accurate interpretation is completed each and every time. In order to bring context to each step, this introductory chapter will introduce the steps in sequence with a brief description. For a more thorough discussion, the chapters that follow will outline each step and its corresponding audiometric science in more detail.

The Method

- 1) Assess for hearing loss**
- 2) Determine the type of hearing loss**
- 3) Check if audiogram fits a classic pattern**
- 4) Compare speech thresholds to pure-tone audiometry**
- 5) Assess word recognition scores**
- 6) Determine tympanometry type**
- 7) Check ear canal volume**
- 8) Review acoustic reflexes**

FIGURE 0–1. Rapid audiogram interpretation method.

Prior to Starting Interpretation

One could propose that the patient's hearing history, including any recent or relevant ear-related symptoms, is the most important part of the audiologic assessment. An audiologist will have obtained this information prior to performing the test. A report of this patient-provided information will usually be included alongside the audiometric data. As a clinician interpreter, you will need to review this information carefully prior to examining the audiogram, as it may provide context for the data and may guide expected test results. If it is not available, the patient should be questioned regarding the severity or duration of their impairment. This background patient information will allow the interpreter to determine if the obtained results match what was expected based on the patient's history. If the results do not match what was expected, then one must evaluate the accuracy of the test and/or

the reliability of the patient response. More about this concept will be reviewed throughout the text. Once you begin practicing your interpretation skills in Section 4, you will notice that each example has a description of the patient's chief audiologic symptom or complaint. Be sure to review this prior to beginning your interpretation.

Step 1—Assess for Hearing Loss

An audiogram's most basic objective is to provide a measure of an individual's hearing ability. This is accomplished via a battery of tests that collectively comprise a standard audiogram. As one approaches each audiogram's interpretation, the most important question to answer must be: Is hearing loss present? Once this question is answered, the rest of the interpretation may begin. If an individual's hearing is normal, then it is possible that the remainder of the test will be normal. If an individual's hearing is abnormal, then the real challenge begins to try to measure his or her hearing loss and determine what may be causing it. Having a foundational ability to evaluate all areas of audiometric data is essential to ensure no irregularities are overlooked.

Step 2—Determine the Type of Hearing Loss

Hearing loss is considered present whenever a patient cannot hear sounds softer than 25 dB in adults and 20 dB for children. Testing is performed at many different frequencies, or pitches, ranging from 250 Hz to 8000 Hz. Three types of hearing loss exist: conductive, sensorineural, or mixed loss. Understanding which type of hearing loss is present can help guide the remaining interpretation to ensure consistency of responses and help confirm a suspected diagnosis.

Step 3—Check if Audiogram Fits a Classic Pattern

A clinician can learn to recognize patterns in the graphical data presented on a pure-tone audiogram, as certain disease states lead to similar hearing loss patterns. Once recognized, the speed and accuracy of interpretation can be increased tremendously. The natural timing to do this examination is after reviewing the pure-tone graph for type of

hearing loss. Keep in mind, while classic patterns exist with pure-tone data, a diagnosis is not ruled out as a possibility if a classic pattern is not present.

Step 4—Compare Speech Thresholds to Pure-Tone Audiometry

The next phase of the evaluation is to review the speech testing. Speech thresholds represent how well a patient can repeat spondee words at least 50% of the time. Spondee words are bisyllabic words equally emphasizing both syllables. Speech testing serves as a confirmation of the thresholds reported in pure-tone testing. Dramatic differences should raise concern for possible malingering, though many other possible explanations could exist. Speech thresholds can be obtained in two different ways, via speech recognition threshold (SRT) or speech detection threshold (SDT) testing.

Step 5—Assess Word Recognition Testing

Word recognition testing is the second part of speech audiometry. It can also be called speech discrimination testing. Prerecorded monosyllabic words are provided to the patient and the patient is asked to repeat them. A percentage score is assigned based on the number of correct responses. This testing quantifies the ability of the patients to discriminate speech with their current hearing impairment. Word recognition scoring can vary depending on the etiology of the hearing loss (conductive, cochlear, and retrocochlear). This variability can help confirm the suspected type of hearing loss.

Step 6—Determine Tympanometry Type

Tympanometry data can provide vital information regarding the capacity of the tympanic membrane and middle ear space to admit sound and transfer it into the cochlea. Objective compliance and volume data are obtained and can be compared to normative tympanometry values to suggest the presence of a conductive hearing impairment. In combination with the testing performed in the audio booth, one can deduce if the tympanic membrane and middle

Every audiogram is performed by a licensed audiologist who uses clinical discretion to measure only the areas of testing needed for each patient. A standard audiogram will include all of the areas mentioned in our method, but not every patient will need to have all of this testing performed. If an area is missing from a provided audiogram, it is likely not essential for that patient's care. Our method assumes that all testing was performed. If one portion of testing is missing, the interpreter should proceed to the next step until the analysis is complete.

ear structures have any responsibility for the measured hearing loss.

Step 7—Check Ear Canal Volume

A quick check of the static immittance, or the ear canal physical volume test (ECV), will confirm that the tympanic membrane is intact. If a perforation is suspected, the measured volume of the ear canal will be outside of the normative values for middle ear volume depending on their age (child versus adult). A normal ECV measurement has high clinical value, as it signifies the tympanic membrane is intact.

Step 8—Review Acoustic Reflexes

Acoustic reflex thresholds can serve as a final confirmation of the hearing loss type that was recognized during threshold testing. During reflex testing, changes in tympanic membrane compliance in response to loud sound stimuli are measured as either present, elevated, or absent in one or both ears. The summation of these results can be used to discriminate if pathologic lesions involving any of the structures in the acoustic reflex pathway are present and can reveal where the pathology may be located.

DEDICATION

“THESE are the times that try men’s souls . . . Tyranny, like hell, is not easily conquered; yet we have this consolation with us, that the harder the conflict, the more glorious the triumph. What we obtain too cheap, we esteem too lightly: —‘Tis dearness only that gives everything its value.”

*Thomas Paine—Crisis No. 1
December 23, 1776*

*To Cambria, Mal, Mer, Tal and Til,
Thank you for putting up with all of the late nights to make this dream a reality. You inspire me every day to be better than I am. “O be wise, what can I say more?”*

—LS

*To Mom, Dad, and Erin,
Thank you for guiding me to pursue my dreams and pushing me to stay ambitious.*

—SM

To My Family, Thank You.

—BC

Section 1

PURE-TONE AUDIOMETRY

Step 1

Assess for Hearing Loss

Step 2

Determine the Type of Hearing Loss

Step 3

Check if Audiogram Fits a Classic Pattern

Introduction

Pure-tone audiometry is the method audiologists use to determine a patient's hearing thresholds at multiple sound frequencies.¹ The testing is performed in very controlled circumstances. A standardized soundproof audiometric testing booth is needed in order to ensure accuracy of the obtained results without outside interference with the testing. Each audiologist will utilize an **audiometer** to produce sounds, such as pure tones or speech stimuli, at each of the different tested frequencies (generally 250, 500, 1000, 2000, 4000, and 8000 Hz) and at different intensities.^{2,3} Occasionally, other frequencies are tested as well if clinically indicated. These pure tones are received by the patient via speakers, headphones, in-the-ear insert earphones, or bone vibrators. A standardized environment during every audiogram ensures the testing can be reproduced at

different times, in diverse locations, and by varying providers with reliable precision.³

Air Conduction Thresholds

Air conduction thresholds are the first measurements obtained during pure-tone audiometry. With the use of headphones or insert earphones, the audiologist will play multiple sounds at each frequency and determine the lowest **sound pressure level (SPL)** at which $\geq 50\%$ of the tones are perceived.⁴ In patients who are too young or are unable to cooperate with wearing headphones during testing, sounds are played via a speaker inside the booth. This is known as sound-field testing.⁵ More information regarding **sound field testing** is discussed in Section 2: Speech Audiometry as well as in Appendix C: Other Audiometric Tests. Once

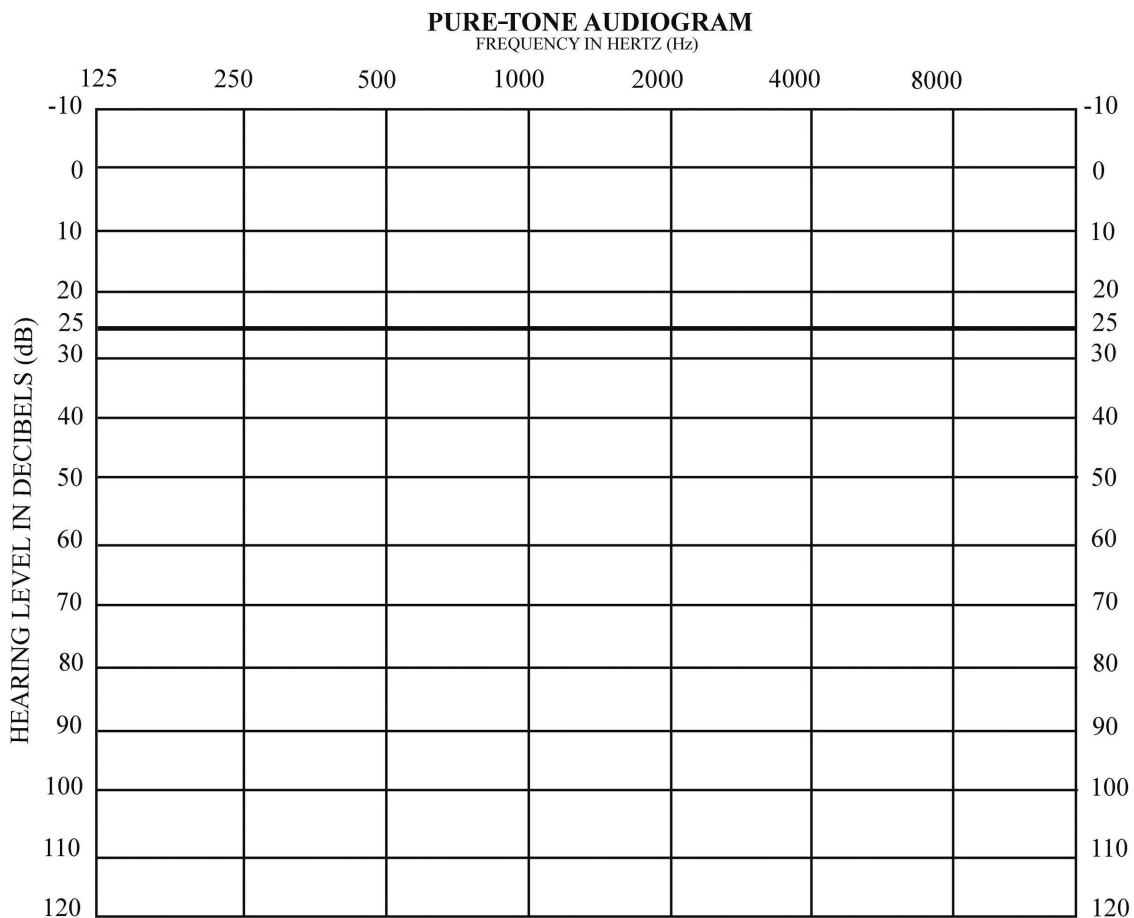


FIGURE 1-1. Standard audiogram grid.

A patient may hear sound pressures lower than the audiometric zero. How can this happen? The patient's hearing thresholds are below the standardized patients used to create the audiometric zero line.

measured, air conduction thresholds are recorded on an audiogram grid (Figure 1-1).

The standard notation used for sound pressure level is the **decibel** (dB). The decibel is calculated on a logarithmic scale using the actual sound pressure (**pascals**) relative to the **audiometric zero**. The audiometric zero is calculated at each frequency independently. It is derived from averaging the lowest sound pressure level heard at that frequency in thousands of nonpathologic ears.⁶ By using decibels instead of pascals, the sound pressure level measured at each frequency is standardized relative to the other frequencies. This allows for easier comparison of threshold data between frequencies. Thus, the hearing ranges are reported relative to the audiometric zero (dBs), not by true sound pressure levels (pascals). Graphically, the horizontal line at "0" dB on the y-axis is denoted as the standardized audiometric zero.³

The audiologist will place markings on the standard audiogram grid (Figure 1-1) to represent the sound pressure level of the air conduction thresholds at each frequency. The standard audiogram grid has sound pressure level along the y-axis, with increasing intensity from top to bottom, measured from -10 dB to 120 dB. Whenever a threshold marking is placed on the grid, it is understood that the patient can hear all pure tones louder (or lower down on the grid) than the marking. Pure tones that are softer (or higher up on the grid) than the marking will not be heard.

Sound **frequency** is found along the x-axis, and the standard audiometer allows for testing from 125 to 8000 **hertz** (Hz). Frequency is ordered with lower pitched tones on the left and higher pitched tones on the right. Each frequency is tested separately until the air threshold is obtained before moving to the next frequency. Most audiologists will start the testing at 1000 Hz as this is the frequency most easily perceived and then continue until all obtainable frequencies are tested.³

Often, an audiologist will begin testing by asking the patient to repeat bisyllabic words to obtain general range of where to begin the pure-tone testing. This initial testing with speech comprises the **speech detection threshold** (SDT) or **speech reception threshold** (SRT). Speech testing is discussed further in Section 2: Speech Audiometry.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED			△	
SOUND FIELD		S		
AIDED		A		
OTHER TESTS MASKED	*			

FIGURE 1-2. Legend of symbols used on a standard audiogram.

The patient's air conduction thresholds are mapped using coded symbols with a blue-colored X representing the left ear and a red-colored O representing the right ear (see Figure 1-2). Remembering "Red is Right" can be an easy way to identify the lines and symbols when colored markings are present.⁴ For the purposes of this book, these symbols will not be color coded in the provided examples for convenience in publishing. It should be noted, however, that often clinicians will receive black and white copies of the original audiogram. Having the ability to distinguish the ears without color designations is paramount to being an effective audiogram interpreter.

Pure-Tone Average

A patient's **pure-tone average** (PTA) can be calculated by taking the average of the air conduction frequencies at 500, 1000, and 2000 Hz (Figure 1-3). This is an important calculation that can be used to compare changes in a patient's air thresholds

$$\frac{\text{Air Thresholds @ 500Hz + 1000Hz + 2000Hz}}{3} = \text{Pure-Tone Average}$$

FIGURE 1–3. Formula used to calculate the pure-tone average.

over time or to compare audiogram data from tests performed at different institutions. The PTA is typically similar (± 10 dB) to the speech thresholds (SRT or SDT).^{7,8}

If the PTA and the speech thresholds do not correlate well, it is important to consider the possibility of malingering or central auditory dysfunction. If the speech thresholds are much better than the PTA, **pseudohypoacusis** should be considered. If the PTA is significantly better than the speech thresholds, the possibility of central involvement should be considered. See Section 2: Speech Audiometry for more details regarding this expected correlation.

Step 1

ASSESS FOR HEARING LOSS

The first step in audiogram interpretation is to assess for the presence of hearing loss using the air conduction thresholds. An abnormal air conduction threshold alone does not specify which type of hearing loss exists (sensorineural, conductive, or mixed). Step 2 of the method will review how to differentiate the type of hearing loss noted on the audiogram.

Severity

If a patient's air conduction thresholds fall below 25 dB for adults and 20 dB for children, then some degree of hearing loss is present. Once you have characterized whether or not there is an impairment, you can begin to describe it. How severe is the loss? Is it symmetric? Severity of hearing loss is described using the standard terminology shown in Figure 1–4.

When an audiogram is completed, patients often may ask a clinician how much hearing they have lost. The classification system presented in Figure 1–4 can be used to describe the hearing loss severity in terms easy for them to understand. For example, if

It is important to note that hearing loss thresholds assessed by pure-tone audiometry may not always correspond with a patient's subjective impression of his or her hearing. Many extrinsic factors have an effect on an individual's actual level of communication impairment. It is important that clinicians explore other reasons for possible impairment, such as subjective tinnitus.

patients have an air conduction threshold of 30 dB at all frequencies, then one would describe their hearing loss as mild across all frequencies. Audiologists may also include a written summary report alongside the audiogram. In this report, they will describe the severity of the hearing loss using this classification system. Based on this handbook, one should be able to define the severity of the hearing loss without referencing an audiologist's report. However, if this report is available, cross-checking your interpretation may prove helpful.

Asymmetry

Asymmetry is judged by comparing the air conduction thresholds between the right and left ears. Initially, this may be best done by calculating each ear's pure-tone average. By using the PTA, the difference between the ears can be more easily compared. PTA differences ≥ 5 dB are considered asymmetric.

It is important to recognize the presence of asymmetry. One would generally expect the hearing in each ear to degrade at the same pace, especially with age-related changes or congenital etiologies. Loud noise-induced injury normally occurs to both ears equally, however, certain exposures are expected to be more unilateral in nature and may lead to hearing threshold asymmetry. These may not raise additional concern. For example, rifle or shotgun users may have their exposed, down-range ear more susceptible to damage (for right-handed shooters, this would be their left ear). Truck drivers may have increased noise exposure on their left side if always driving with the window down. A musician may note unilateral hearing loss if always

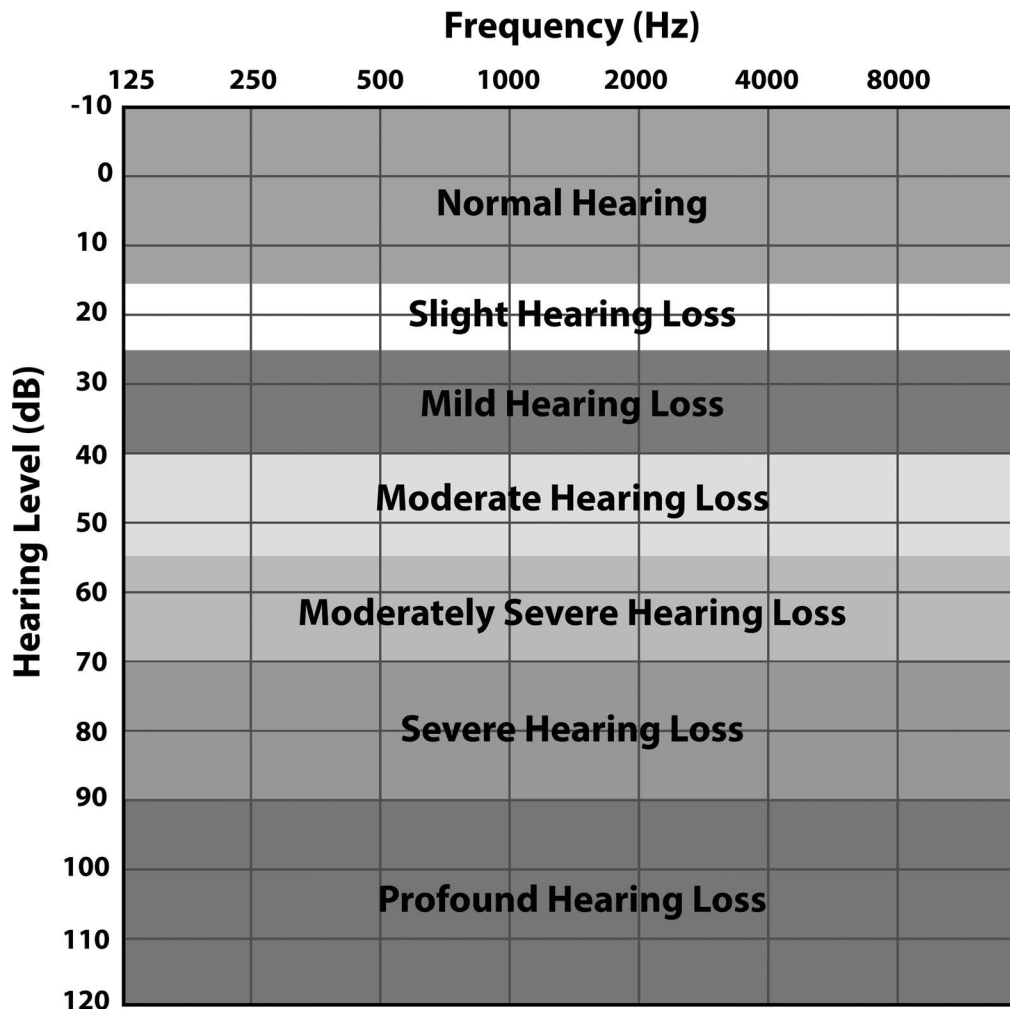


FIGURE 1–4. Classification scheme used to label severity of hearing loss in adults.

maintaining the same position next to the speaker system.

Regardless of suspected cause, the presence of asymmetry should always trigger a medical evaluation by an otolaryngologist. When asymmetric

hearing loss is present, further assessment is pursued, including a full otologic history, pertinent physical examination, and consideration of imaging.⁹ Further information regarding asymmetry will be reviewed in step 2.

Step 2

DETERMINE THE TYPE OF HEARING LOSS

Once a hearing loss has been identified and the severity has been measured, the next step is to determine whether it is a sensorineural, conductive, or mixed loss. This is done by examining the pure-tone audiogram bone and air conduction thresholds closely.

Sensorineural Hearing Loss

A **sensorineural hearing loss** is best determined by **bone conduction thresholds**. Air conduction thresholds were discussed under step 1 and are obtained via headphones or in-the-ear inserts. Bone conduction thresholds are usually measured after all air threshold testing is completed. This is done using a bone oscillator or vibratory apparatus placed overlying either mastoid.¹⁰ Similar to air thresholds, bone thresholds are measured as the lowest level (dB) at which 50% of the pure tones introduced via the oscillator are perceived by the patient for frequencies between 250 and 4000 Hz.⁴ The audiologist makes appropriate markings on the audiogram grid, represented by < for Right and > for Left, as shown in Figure 1–2.

Bone conduction testing is not specific to the side on which the oscillator is placed. For example, a right-sided oscillator is transmitting pure tones to both the right and left inner ear organs. If asymmetry exists, the better hearing cochlea will preferentially transmit the signal to the brain. In order to isolate the bone conduction thresholds by side, a masking noise must be used.

When bone conduction thresholds are found to be abnormal (ie, the threshold falls below 25 dB for adults and 20 dB for children), some degree of sensorineural hearing loss is present. Remember: air thresholds are nonspecific. Abnormal air thresholds can be caused by a problem anywhere from the ear canal to the brain since both the conduction and sensorineural components are being tested. Bone threshold abnormalities, however, isolate the nerve signaling pathway, indicating that hearing issue must be a damaged sensorineural component. This may include any part of the auditory pathway to the auditory cortex: **cochlea**, **vestibulocochlear nerve**, or brainstem relay centers. See Section 3 for further information on the auditory pathway. Sensorineural hearing loss can be caused by various conditions: **presbycusis**, noise-induced, **ototoxicity** from radiation therapy or medications, traumatic brain or temporal bone injuries, **Meniere's disease**, and various tumors. See Table 1–1 for a full list.

Sensorineural hearing loss is generally considered irreversible. One exception to this is **sudden sensorineural hearing loss**. Patients who experience sudden-onset hearing changes that persist over hours to days should be evaluated urgently with an audiogram. Clinicians need to examine these patients' bone conduction thresholds closely to determine if sensorineural asymmetry exists. If possible, these new thresholds should be compared to any prior available audiometric data. Urgent otolaryngology evaluation and treatment should follow if new or worsened asymmetry exists.^{11,12}

TABLE 1–1. Causes of Sensorineural Hearing Loss Divided Into Cochlear and Retrocochlear Pathologies

Causes of Sensorineural Hearing Loss	
Cochlear (Sensory Type)	Retrocochlear (Neural Type)
Autoimmune inner ear disease	Vestibulocochlear or eighth nerve lesions/tumors (vestibular schwannoma)
Cochlear trauma	Facial or seventh nerve lesions/tumors (facial schwannoma)
Congenital infections (rubella, toxoplasmosis, CMV, syphilis, etc.)	Central nervous system neoplasm
Congenital syndromes (Sitckler, Pendred, Usher, etc.)	Stroke (or other thromboembolic disease)
Genetically inherited (connexin 26)	Auditory neuropathy
Infection mediated	Auditory processing disorder
Meniere's disease	Central nervous system infection
Noise induced	Multiple sclerosis
Otosclerosis (cochlear involvement)	
Ototoxic agents	
Presbycusis	

Conductive Hearing Loss

A **conductive hearing loss** occurs when there is pathology in the external or middle ear that blocks sound from reaching the cochlea. Pure-tone audiograms do not directly contain measures of conductive hearing loss. The type of hearing loss is best determined by the presence of an **air-bone gap**. Once the audiologist has determined a patient's air and bone conduction thresholds (described in step 1), the air-bone gap will need to be calculated to determine what degree of hearing loss is present due to conductive pathology. This is simply done by examining each frequency individually and subtracting the bone conduction thresholds from the air conduction thresholds. For example, if a patient's bone conduction threshold at 2000 Hz is 10 dB and the air conduction threshold is 50 dB, the air-bone gap at 2000 Hz is reported as 40 dB. Each frequency may have a different air-bone gap and should be reported separately.

The various causes of conductive hearing loss are shown in Table 1–2, and include various conditions,

Often **immittance** testing will also be abnormal with conductive hearing loss. When it is not, one should suspect a potential underlying ossicular chain fixation or otosclerosis. This is discussed further in Section 3: Immittance Testing.

such as: cerumen impaction, otitis media, **cholesteatoma**, and **otosclerosis**. Many of these causes can be reversible, unlike the majority of sensorineural hearing loss etiologies. A full evaluation by an otolaryngologist should follow whenever conductive hearing loss is noted to see if it can be regained.

Mixed Hearing Loss

A **mixed hearing loss** occurs when there is both a sensorineural and a conductive component noted on the pure-tone audiogram. In this situation, there

TABLE 1–2. Causes of Conductive Hearing Loss Divided by Location of Pathology

Causes of Conductive Hearing Loss		
Outer Ear	Tympanic Membrane	Middle Ear
Otitis externa	Eustachian tube dysfunction (limiting normal tympanic membrane movement)	Cholesteatoma
Acquired canal atresia (surgical or traumatic)	Tympanic membrane mass (benign or malignant)	Chronic serous otitis media (middle ear effusion)
Mass obstructing canal (benign or malignant)	Tympanostomy tube in tympanic membrane	Congenital malformation of ossicles
Impacted cerumen	Tympanic membrane perforation	Hemotympanum
Congenital canal atresia (stenosis or absence)	Tympanosclerosis	Middle ear mass (benign or malignant)
Obstructive exostosis		Ossicular chain discontinuity
		Ossicular chain fixation
		Otitis media with effusion
		Otosclerosis (oval window involvement)

is impairment to both the conduction system in the outer or middle ear, as well as the inner ear nerve components. The resultant audiogram in this condition would have abnormal bone conduction thresholds as well as evidence of air-bone gaps simultaneously. For example, if a patient's bone conduction threshold at 2000 Hz is 30 dB and the air conduction threshold is 50 dB, there is a mild sensorineural

component to the hearing loss plus an additional air-bone gap of 20 dB signifying the conductive hearing loss component. Reversibility of this hearing loss would again depend on the severity of the conductive component.

Summary

Once the air and bone conduction thresholds have been obtained, one can fully assess the type(s) of hearing loss present:

1. If abnormal bone thresholds are noted, then a sensorineural hearing loss is present at those frequencies.
2. If air-bone gaps are measured with normal bone thresholds, then a conductive hearing loss is present at those frequencies.
3. If both abnormal bone thresholds and air-bone gaps are recorded, then a mixed hearing loss is present at those frequencies.

Some clinicians will use tuning forks to help evaluate whether a patient's hearing loss is conductive, sensorineural, or mixed, even before audiogram data are obtained. Multiple tuning forks exist to examine individual frequencies (256 Hz, 512 Hz, 1024 Hz, 2048 Hz). The most widely used clinical tests are the Weber and Rinne tests. See Appendix B: Bedside Hearing Tests for a full description of these in-clinic techniques.

Some institutions consider hearing loss to be asymmetric when a 10-dB difference exists between the right and left ears at two neighboring frequencies.^{13,14} Others will consider asymmetry when a 15-dB difference exists between three neighboring frequencies.¹⁵ A few studies have suggested that a >15-dB difference at 3000 Hz or 4000 Hz is the most indicative for an underlying pathologic process.^{16,17} When asymmetry is noted, neuroimaging is strongly considered to evaluate for primary central nervous system lesions, with contrast magnetic resonance imaging (MRI) being the most sensitive screening test for these lesions.

It is important to report the type of hearing loss noted both globally as well as for each specific frequency.

Asymmetry should again be revisited. While significant asymmetry is easily observed on an audiogram, there is not a clear consensus in the literature as to what defines clinically important asymmetry. As noted above, most clinicians are concerned with asymmetry as it relates to the bone conduction thresholds, as they denote that sensorineural hearing loss is present. Generally, conditions like presbycusis or noise-induced loss will lead to symmetric bone threshold degradation. If the bone conduction thresholds are significantly asymmetric between ears, unilateral conditions like **vestibular schwannoma** must be considered.

Masking

Masking is often employed when significant asymmetry is noted between each ear's thresholds (both air and bone). A sound stimulus presented to the test ear will eventually be detected by the opposite ear due to **crossover hearing**, if the **contralateral ear** has significantly better hearing than the test ear.^{1,4} For example, a person with unilateral profound hearing loss will not respond until the intensity of the signal is increased so high that the opposite (non-test) ear will detect the signal and the patient will respond. During testing, to avoid crossover hearing, a masking sound is presented to the non-test ear while measuring the test ear thresholds.¹⁸

When masking is employed, the better hearing ear is prevented from creating a false hearing threshold that underestimates the degree of loss in the worse hearing ear. Thus, masking effectively eliminates crossover hearing. Remember that bone thresholds are not ear specific. Without masking, the resultant bone thresholds represent only the threshold of the better hearing ear. When there is significant air conduction asymmetry, the resultant intensity needed to accurately test the poorer hearing ear can lead to crossover stimulation of the contralateral ear if masking is not used. Masking can thus be applied to bone conduction and air conduction thresholds alike.

To perform air conduction masking, the audiologist applies headphones or in-the-ear inserts that play narrowband noise centered at the test frequency in the better hearing (nontest) ear. The degree of masking needed to prevent crossover hearing is determined by the audiologist. For each tested frequency, the audiologist will adjust the amount of masking based off the type of headphone used and degree of hearing loss, taking into account the amount of **interaural attenuation** present (see below).

When bone conduction asymmetries are noted, masking will also be needed. As mentioned in the above example, a person with a unilateral profound hearing loss will detect a tone in the contralateral ear presented by bone conduction to the poorer ear. Bone conduction stimulates both cochleas almost equally. Once the audiologist effectively masks the nontest ear, the audiologist can isolate and measure the true threshold of the poorer hearing ear (test ear). Just like with air conduction masking, the audiologist will adjust the amount of masking needed for bone conduction testing, based on the use of the bone oscillator and degree of hearing loss, taking into account the amount of interaural attenuation present.

Successful masking can only occur when crossover hearing can be accounted for and avoided. To do so, an audiologist must first determine the interaural attenuation (IA) for each testing device (headphones, in-the-ear insert, and bone oscillator).¹⁹ Interaural attenuation is the amount of dB that are lost in the process of crossover hearing. As sound travels through the ear system, the acoustic energy will decrease and the amount of sound presented to the contralateral ear will be affected. Sound emitted from these devices will differ in the amount of interaural attenuation they create.^{1,4}

A **masking dilemma** can occur when there is a moderate to severe conductive hearing loss in both ears.² In this setting, it is impossible to adequately mask the hearing of the non-test ear without at the same time masking the hearing of the test ear. This is because the intensity of sound needed to mask the nontest ear is so loud, it crosses over to the tested ear. Understanding this dilemma avoids many potential pitfalls when trying to identify the true bone threshold in the poorer test ear.

- The minimum IA of headphones suggested to rule out crossover hearing is 40 dB.
- The minimum IA of insert earphones suggested is 75 to 85 dB at frequencies up to 1000 Hz and 50 to 65 dB above 1000 Hz.
- The minimum IA of bone conduction is typically 0 dB, but effective masking is introduced when there is an air-bone gap 10 dB or greater.

Pseudohypoacusis or Nonorganic Hearing Loss

An audiologic evaluation relies on accurate responses from the patient in order to trust the audiogram thresholds. However, some patients may give false responses on purpose. Rather than always accepting the rendered audiogram information as fact, one must know how to detect exaggerated or questionable results.²⁰ Audiologists have a few different techniques to trick a patient into revealing if the loss is organic or nonorganic.²¹

The most common clinical test used is the **Stenger test**.⁴ Any patient who presents with a unilateral or asymmetric hearing loss should be considered for a Stenger test, especially if there is anything suspicious in his or her history at presentation (attention-seeking behavior, medical-legal issues, direct/indirect financial gain, etc.). This test is performed as follows:

1. Two pure tones of the same frequency are presented to each ear simultaneously.
2. These tones only differ in their intensity.

3. The supposedly better hearing ear will get the same intensity of sound throughout the test.
4. The supposedly poorer hearing ear will get pure tones of gradually increasing intensity.

For example, a sound is played 10 dB above the voluntary threshold in the supposedly better hearing ear, and at the same time, a sound is played 10 dB lower in the poorer ear. The patient should acknowledge the signal in the better hearing ear. As the sound is increased in the poorer ear, a patient with a genuine hearing loss will continue to respond to the sound in the better hearing ear. A patient feigning impairment will stop responding to the sound in the good ear once the level in the worse ear is louder relative to the good ear.

If true pathology exists, the sound in the better hearing ear will be heard regardless of the increasing signal intensification on the poorer hearing ear. In cases of pseudohypoacusis, patients will stop responding once the worse ear begins to hear the pure tones, in order to keep the façade of poor hearing. This lack of an admission is what is referred to as a positive Stenger's test and will be reported on the audiogram as such.

Reliability

When evaluating for conditions like pseudohypoacusis, another important element to consider is the audiologist's opinion on the **reliability**, or repeatability, of the pure-tone results.² Some audiogram assessments will include a comment on the validity of the measured thresholds, especially if the audiologist judges the patient responses to be inconsistent, difficult to replicate, or a positive Stenger test is noted. In these situations, the audiologist may note that the audiometric assessment may be of questionable reliability. When the reliability is determined to be good, then the obtained results can be trusted as an accurate assessment of the patient's true hearing thresholds. If no mention of reliability is noted on an audiogram, one can assume the audiologist trusted the obtained results. Speech testing is also subject to this assessment as well. See Section 2: Speech Audiometry for further information.

Step 3

CHECK IF AUDIOGRAM FITS A CLASSIC PATTERN

At this point in the interpretation process, it is beneficial to globally review the pure-tone grid to see if its appearance matches any classic patterns. This can be especially helpful for those new to audiogram interpretation. Audiogram data elicited in certain disease states characteristically have significant commonalities that can be quickly recognized. Once one recognizes patterns in audiogram data, the learner can use these to help confirm suspected diagnoses and guide clinical interventions or counseling.

Pattern recognition can also help guide interpretation of the remainder of the audiogram data. These patterns can extend to the associated immittance and speech testing. For example, patients with otosclerosis will have a conductive hearing loss noted in the involved ear. Classically, there is an isolated increase in the bone conduction threshold noted at 2000 Hz called a **Carhart notch**. The associated **tympanometry** data should be normal

or near normal with a central peak at ambient pressure (type A or A_s tympanogram). Seeing a type A or A_s tympanogram in the setting of **ipsilateral** conductive hearing loss should raise suspicion that otosclerosis is the underlying disease causing the abnormal audiometric findings.²² More information on tympanometry will be discussed in Section 3.

The following audiogram grids are representative of these patterns. Some patterns represent specific disease states (Meniere's disease, otosclerosis, etc.), and others may suggest causality to the hearing loss pattern seen (noise-induced SNHL, presbycusis, etc.). It is important to emphasize that forming comparisons has significant limitations. One should complete a full audiogram interpretation regardless if a pattern is globally seen with the audiogram grid. Also, subtle pertinent findings in the speech or immittance testing may be missed if one focuses solely on making comparisons to classic audiogram representations.

Classic Audiogram Patterns

Audio 1: Normal Hearing (Bilateral)

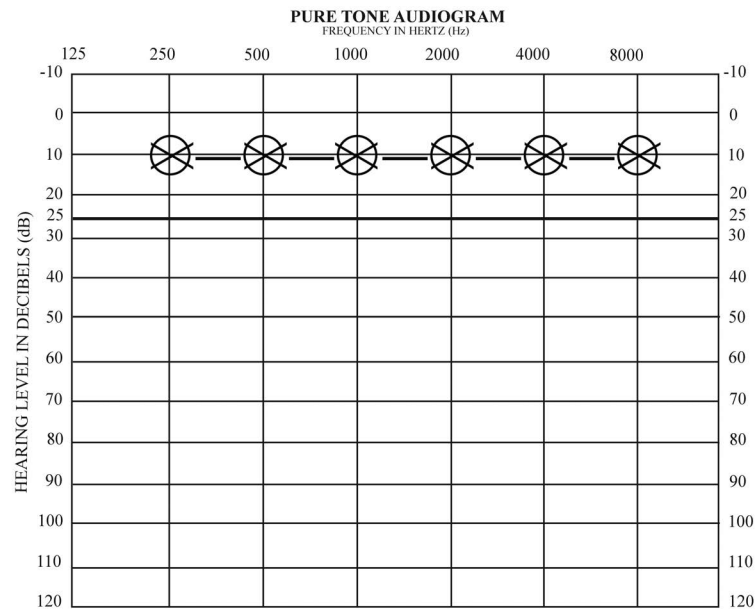


FIGURE 1–5. Normal hearing (bilateral).

Audio 2: Mild SNHL (Bilateral)

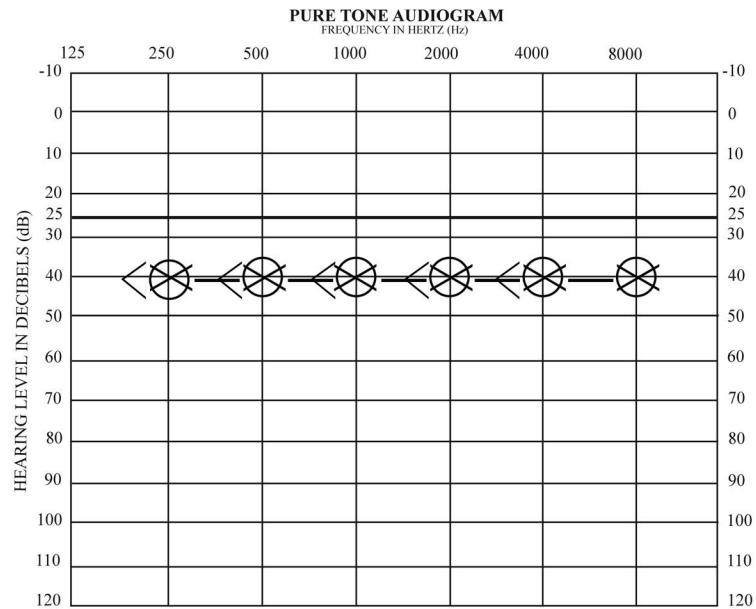


FIGURE 1–6. Mild SNHL (bilateral).

Audio 3: Moderate to Severe SNHL (Bilateral)

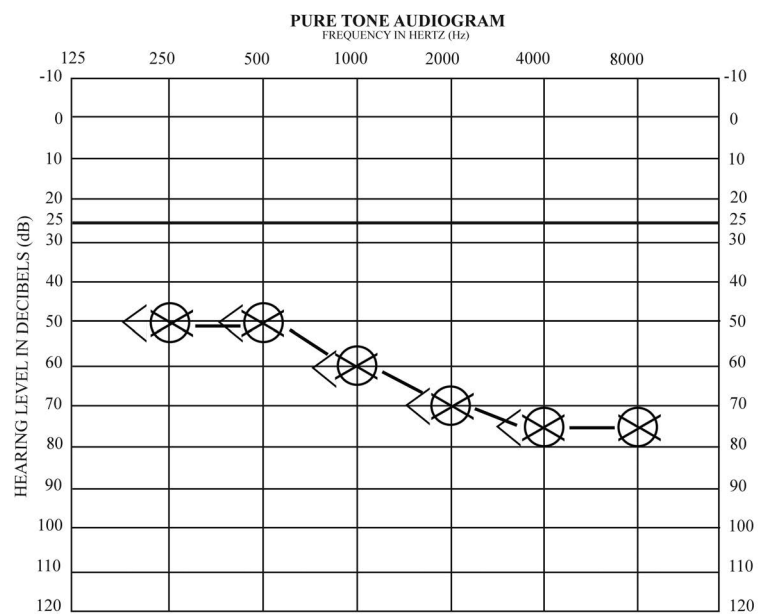


FIGURE 1–7. Moderate-to-severe SNHL (bilateral).

Audio 4: Mild CHL (Bilateral)

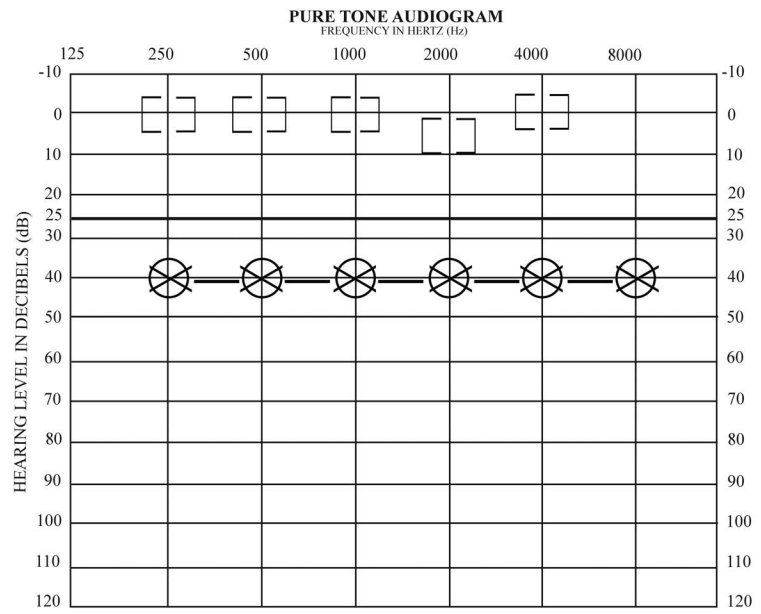


FIGURE 1–8. Mild CHL (bilateral).

Audio 5: Maximal CHL (Bilateral)

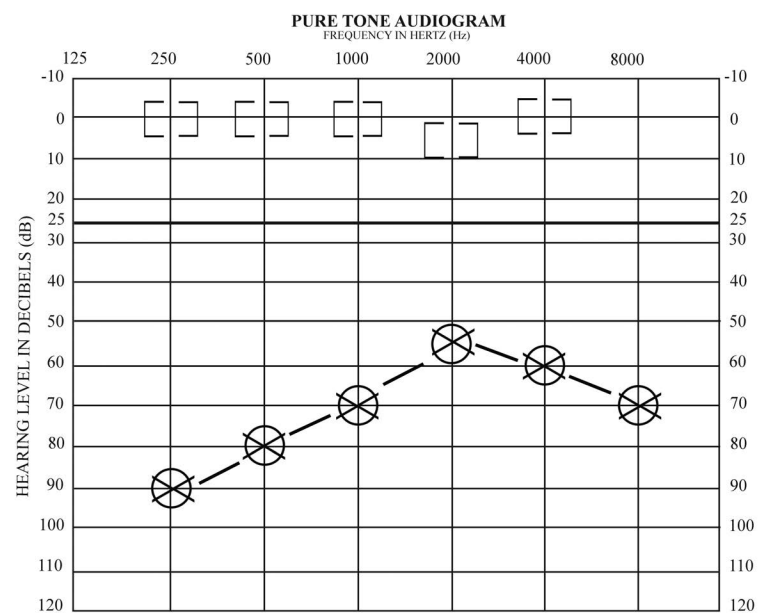


FIGURE 1–9. Maximal CHL (bilateral).

Audio 6: Moderate-to-Profound Mixed Hearing Loss (Bilateral)

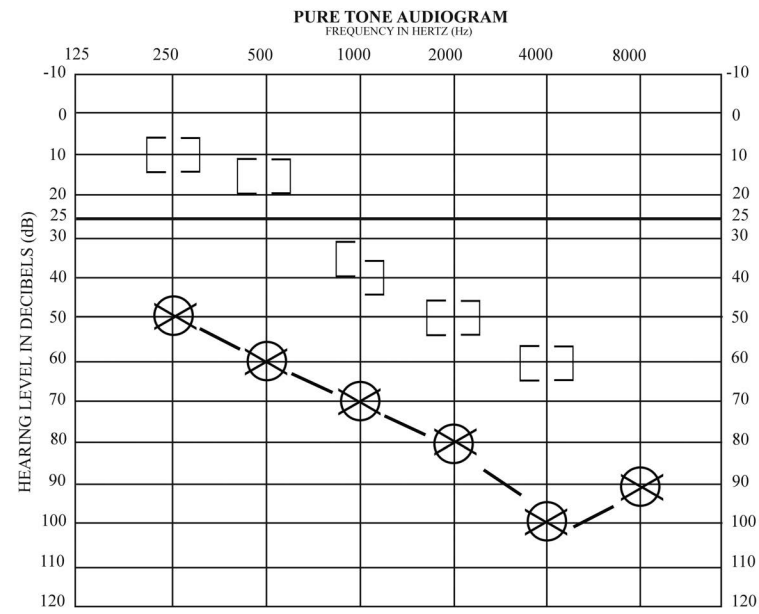


FIGURE 1–10. Moderate-to-profound mixed hearing loss (bilateral).

Audio 7: Presbycusis (Bilateral)

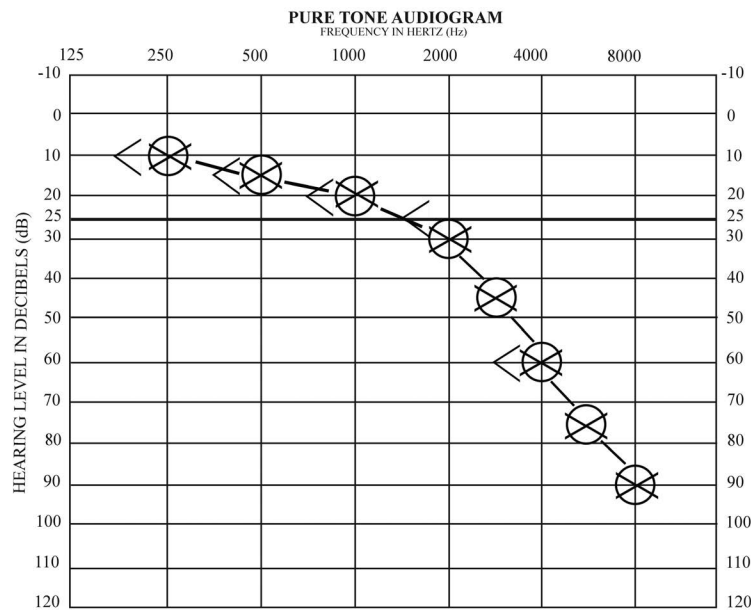


FIGURE 1–11. Presbycusis—down-sloping SNHL (bilateral).

Audio 8: Noise-Induced Hearing Loss (Bilateral)

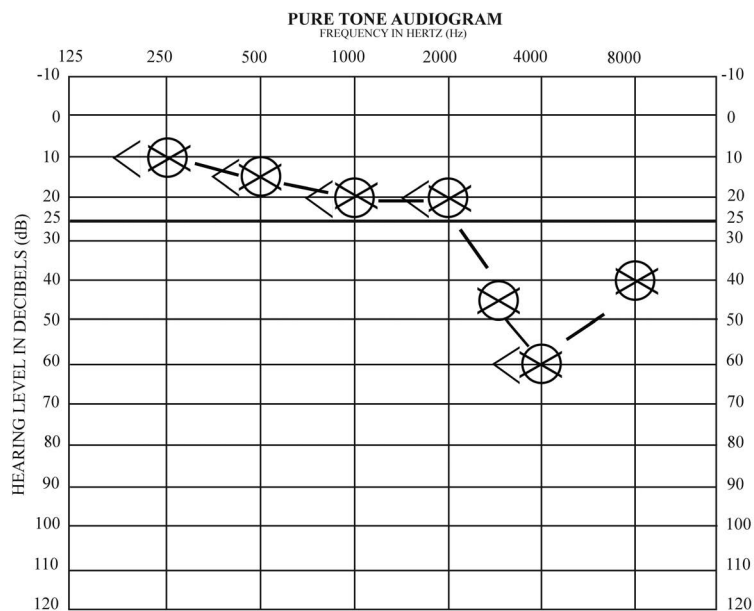


FIGURE 1–12. Noise-induced hearing loss—high-frequency sensorineural hearing loss with significant notching at 4000 Hz (bilateral).

Audio 9: Otosclerosis (Bilateral)

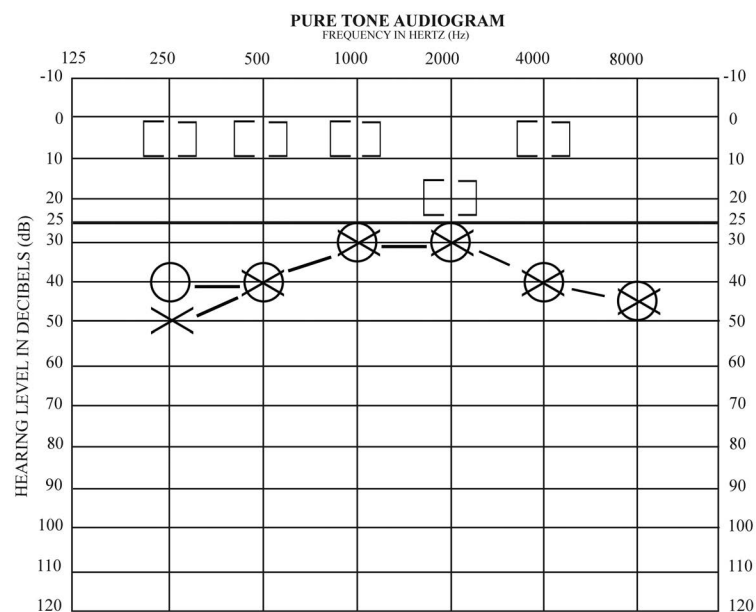


FIGURE 1–13. Otosclerosis—conductive hearing loss with Carhart notch at 2000 Hz (bilateral).

Audio 10: Meniere’s Disease (Unilateral)

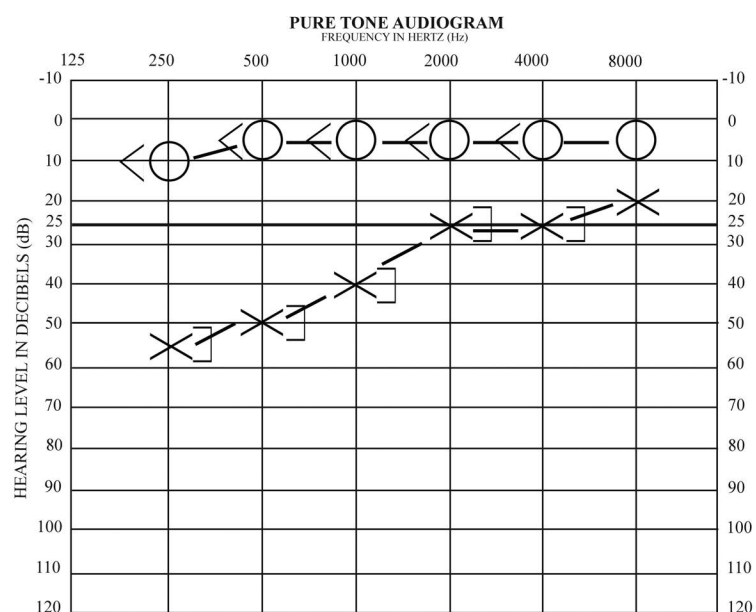


FIGURE 1–14. Meniere’s disease—low-frequency sensorineural hearing loss with up-sloping in higher frequencies (unilateral).

Audio 11: Congenital Sensorineural Hearing Loss (Bilateral)

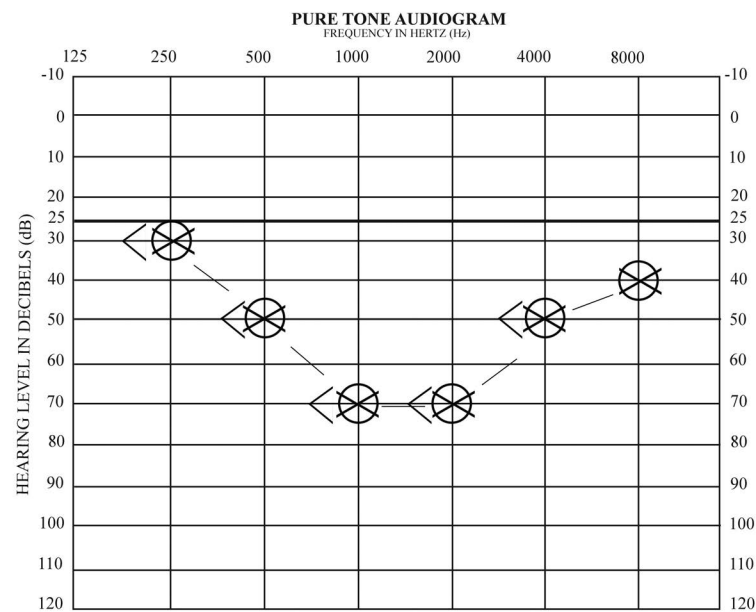


FIGURE 1–15. Congenital sensorineural hearing loss—mid-frequency involvement in a “cookie-bite” pattern (bilateral).

Audio 12: Anacusis (Bilateral)

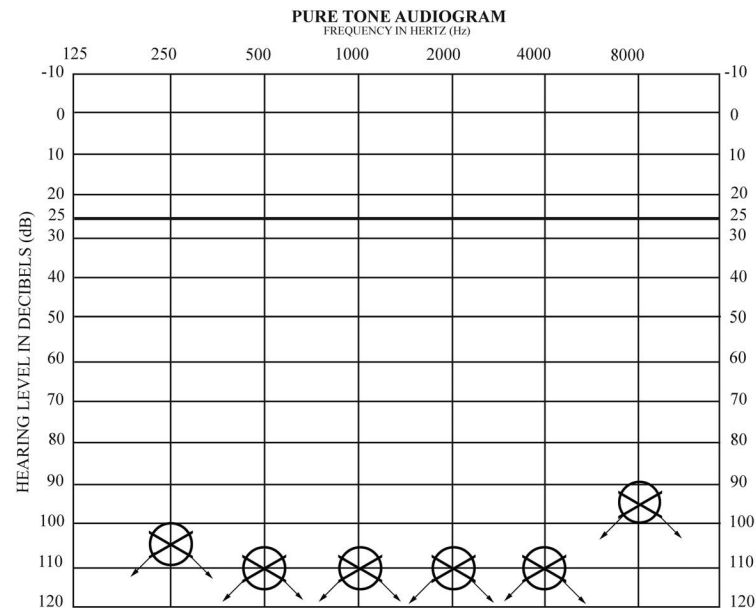


FIGURE 1–16. Anacusis—total sensorineural hearing loss (bilateral).

References

1. Gelfand SA. *Essentials of Audiology*. New York, NY: Thieme; 2009.
2. Stach B. *Clinical Audiology: An Introduction*. San Diego, CA: Nelson Education; 2008.
3. American Speech-Language-Hearing Association. *Guidelines for Manual Pure-Tone Threshold Audiometry*. American Speech-Language-Hearing Association; 2005.
4. Chasin M, English K, Hood LJ, Tillery KL. In: Katz J, ed. *Handbook of Clinical Audiology*. New York, NY: Wolters Kluwer; 2014.
5. Tharpe AM, Sewald R, eds. *Comprehensive Handbook of Pediatric Audiology*. San Diego, CA: Plural Publishing; 2016.
6. Davis H, Kranz FW. The international standard reference zero for pure-tone audiometers and its relation to the evaluation of impairment of hearing. *J Speech Lang Hear Res*. 1964;7(1):7–16.
7. Graham JT. Evaluation of methods for predicting speech reception threshold. *Arch Otolaryngol*. 1960;72(3):347–350.
8. Quiggle RR, Glorig A, Delk JH, Summerfield AB. Predicting hearing loss for speech from pure-tone audiograms. *Laryngoscope*. 1957;67(1):1–15.
9. Campbell K. *Essential Audiology for Physicians*. San Diego, CA: Singular; 1998.
10. Carhart R. Clinical application of bone conduction audiometry. *Arch Otolaryngol*. 1950;51(6):798–808.
11. Hughes GB, Freedman MA, Haberkamp TJ, Guay ME. Sudden sensorineural hearing loss. *Otolaryngol Clin North Am*. 1996;29(3):393–405.
12. Schreiber BE, Agrup C, Haskard DO, Luxon LM. Sudden sensorineural hearing loss. *Lancet*. 2010;375(9721):1203–1211.
13. Margolis RH, Saly GL. Asymmetric hearing loss: definition, validation, and prevalence. *Otol Neurotol*. 2008;29(4):422–431.
14. Urban SL, Benninger MS, Gibbens ND. Asymmetric sensorineural hearing loss in a community-based population. *Otolaryngol Head Neck Surg*. 1999;120(6):809–814.
15. Monsell EM. New and revised reporting guidelines from the Committee on Hearing and Equilibrium. *Otolaryngol Head Neck Surg*. 1995;113(3):176.
16. Saliba I, Bergeron M, Martineau G, Chagnon M. Rule 3,000: a more reliable precursor to perceive vestibular schwannoma on MRI in screened asymmetric sensorineural hearing loss. *Eur Arch Otorhinolaryngol*. 2011;268(2):207–212.
17. Schlauch RS, Levine S, Li Y, Haines S. Evaluating hearing threshold differences between ears as a screen for acoustic neuroma. *J Speech Lang Hear Res*. 1995;38(5):1168–1175.
18. Lidén G, Nilsson G, Anderson H. Masking in clinical audiometry. *Acta Otolaryngol*. 1959;50(1–2):125–136.
19. Zwislack J. Acoustic attenuation between the ears. *J Acoustic Soc Am*. 1953;25(4):752–759.
20. Qiu WW, Stucker FJ, Yin SS, Welsh LW. Current evaluation of pseudohypacusis: strategies and classification. *Ann Otol Rhinol Laryngol*. 1998;107(8):638–647.
21. Rintelmann WF, Schwan SA, Blakley BW. Pseudohypacusis. *Otolaryngol Clin North Am*. 1991;24(2):381–390.
22. Danesh AA, Shahnaz N, Hall JW. The audiology of otosclerosis. *Otolaryngol Clin North Am*. 2018;51(2):327–342.

Section 2

SPEECH AUDIOMETRY

Step 4

Compare Speech Thresholds to Pure-Tone Audiometry

Step 5

Assess Word Recognition Scores

Introduction

Speech audiometry is an essential part of the audiologic assessment. During the initial part of the audiogram, thresholds are obtained with pure tones, which quantify the degree of hearing loss (see Section 1: Pure-Tone Audiometry). After pure-tone testing is completed, speech audiometry is performed. The results of this testing reflect one’s ability to understand spoken words and can better relate the everyday usability of the hearing. Thus, speech testing is a more accurate representation of the person’s ability to communicate.

There are two main types of speech testing: (1) speech reception threshold (SRT) or speech detection threshold (SDT) and (2) word recognition testing (WRT).¹ These are generally reported in the box represented in Figure 2–1.

Several beneficial pieces of information are furnished from speech audiometry. As discussed in Section 1, speech results are used to cross-check the

SPEECH AUDIOMETRY			☐ RECORDED	☐ MLV
EAR	SRT	SDT	WRS	LEVEL
R				
L				

FIGURE 2–1. Standard speech audiometry table on audiogram form.

validity of the pure-tone audiometry.^{2,3} The speech results can support the pure-tone findings of conductive, sensorineural, or mixed hearing losses depending on the degree speech discrimination is affected. Speech testing also helps guide hearing aid selection and programming. Finally, the speech audiometry can help identify the site of a pathologic lesion affecting the hearing. Integrating speech testing in conjunction with pure-tone data is essential in providing thorough audiogram interpretation.

Step 4

COMPARE SPEECH THRESHOLDS TO PURE-TONE AUDIOMETRY

Speech Thresholds

An important part of speech audiometry is the **speech reception threshold (SRT)**. The SRT is the lowest level at which a listener can repeat bisyllabic words correctly 50% of the time. These bisyllabic words are called **spondee words**, meaning each syllable has the same weight. Examples include “baseball” and “hotdog.”^{1,4} See Table 2–1 for examples of spondee words used for speech testing.

The speech reception threshold should roughly correspond to the pure-tone average (PTA) calculated above in step 1. In this way, the SRT serves as an important cross-check for consistency or reliability in the obtained pure-tone results. For example, if the pure-tone average is 25 dB, one would expect the SRT to be at or near this level as well. If the SRT differs from the PTA by more than ± 10 to 12 dB, then the reliability of the pure-tone results should be questioned.^{4,5} This is discussed further below.

Comparing the PTA and SRT is most accurate when the hearing thresholds are relatively similar across the PTA frequencies (ie, the pure-tone curve is flat at 500, 1000, and 2000 Hz). When there is a steeply up or down-sloping loss, then the comparison is more difficult, as the PTA can be significantly reduced by the outlying threshold. A better option in this scenario would be to compare the SRT to the one or two best frequencies to ensure reliability of results.^{6,7} Other formulas have been developed to assist with this comparison, but they are beyond the scope of this text.⁸

The spondee words may be presented from a pre-recorded list on the audiometer or via a live voice into the inserts/headphones.¹ If a **monitored live voice (MLV)** is used, the audiologist will apply settings and filters from the audiometer to ensure equal intensity is placed on each syllable in the spondee words before reading the list to the patient. To ensure the patient is able to perform the task without significant difficulty, many audiologists will test the better hearing ear first.⁹ Also, essential to the accurate assessment of the SRT, a patient will be faced away from an audiologist during this testing to ensure lip reading does not falsely elevate the results.

Speech detection threshold (SDT) is very similar to the speech reception threshold, except the SDT differs in the manner in which the threshold was obtained. The SDT is level at which a listener can identify the presence of speech 50% of the time.⁴ An audiologist will present a single word over and over at increasing intensities until the listener reacts (head turn, hand raise, etc.).⁹ The listeners do not have to identify the material as speech, but they must indicate an awareness of the presence of sound. The SDT is used less often than the SRT, and it is used mostly in the evaluation of select patients, such as the profoundly deaf, neurologically impaired, developmentally delayed, or in very young patients.¹⁰ Similar to the SRT, the SDT should be consistent with the pure-tone thresholds and can be compared to the PTA to check for consistency with the testing.

Importantly, there is another significant difference between the SDT and SRT. In SRT, the patient must understand the speech and repeat it back to the tester in order for it to be registered as the threshold level. In SDT, speech signals only need to be identified, not

TABLE 2–1. Common Spondee Words Used in Speech Recognition Testing (Used with permission from: American Speech-Language-Hearing Association. [1988]. Determining threshold level for speech).

Spondee Word List			
airplane	drawbridge	hot dog	padlock
armchair	duck pond	ice cream	pancake
backbone	eardrum	inkwell	playground
baseball	earthquake	mousetrap	railroad
birthday	eyebrow	mushroom	stairway
blackboard	greyhound	northwest	sunset
cookbook	hardware	nutmeg	toothbrush
cowboy	headlight	oatmeal	whitewash
doormat	horseshoe	outside	woodwork

recognized and repeated, for the responses to be registered as the detection threshold. Thus, SRT may be a more accurate representation of the threshold level in which speech is heard and understood.

There are multiple additional phenomena that can influence the validity of the SDT/SRT results:

False-negative results (better than expected when compared to the PTA):

When significant deficiencies or discrepancies exist in speech thresholds compared to pure-tone thresholds, one should consider the presence of malingering (pseudohypoacusis) or **retrocochlear** pathology. As discussed in Section 1: Pure-Tone Audiometry, when major inconsistencies are noted, the audiologist may question the reliability of the obtained results. Stenger testing may also help clarify this further. If the Stenger test is negative, further testing of the auditory pathway should be considered to evaluate for retrocochlear pathology. See Section 3: Immittance Testing for more information regarding acoustic reflex testing and the evaluation of the auditory pathway.

1. Patients who undergo several audiogram assessments over a short period of time may begin to recognize the spondee lists. This may lead to falsely elevated speech scores and is called **practice effect**.^{11,12}

False-positive results (worse than expected when compared to the PTA):

1. Speech testing requires that the patient understands the testing language well. For patients who have the testing performed in their nonprimary language, one might see poorer speech results than what their true hearing state may actually produce if their primary language was used in the testing.
2. A patient's education, intelligence level, or cognitive state, especially if limited, may also produce a falsely positive speech assessment. Patients may fail to respond to the testing if they do not recognize the words being tested or have limited abilities to follow directions during testing.^{4,9}

Speech thresholds are also used to calculate the **most comfortable loudness level (MCL)**. The MCL represents the intensity level above threshold at which a person most comfortably hears speech,

which is typically estimated at 40 dB above the SRT. In contrast, the **uncomfortable loudness level (UCL)** is the intensity level at or above which sound would be painful.¹³ Both the MCL and UCL are useful when programming hearing aids in patients with

sensorineural loss and are often reported separately on the audiogram. A more thorough discussion of specific hearing aid evaluations is beyond the scope of this book. Advanced audiology texts include multiple chapters dedicated to this process.^{4,9,14}

Step 5

ASSESS WORD RECOGNITION SCORES

Word Recognition Testing

The second part of speech audiometry is **word recognition testing (WRT)** or **speech discrimination testing**. This testing attempts to quantify the ability of the patient to discriminate speech with their current hearing impairment. For this test, a list of **monosyllabic words** is presented to the patients near their most comfortable listening level (MCL) or around 40 dB above the speech reception threshold. Similar to the SRT, the audiologist can utilize recorded lists on the audiometer or present them via monitored live voice settings. The patient is asked to repeat them and the correct responses are tracked.

Typically, a prefabricated list of 50 words includes all the sounds of the English language. Due to time constraints, half-lists may often be administered in busy clinics. There are different lists that can be used during word recognition testing.^{1,4} The lists can be administered in quiet or in noise. A standard audiogram machine will provide prerecorded options that can be used to obtain the **word recognition scores (WRS)**. A list is selected based on the patient's age, intellectual abilities, language, and type of word testing needed. A few examples of the word lists used are found in Table 2-2. The audiologist will vary the lists used during successive audiogram tests to ensure practice effect does not confound the results.

Once the scores are calculated, audiologists will report the percentage correct on the audiogram form (see Figure 2-1, page 22). A categorical grade is generally assigned to the WRS to help describe the obtained results. One commonly used grading system is outlined in Figure 2-3. It should be noted, however, that there is little agreement on specific word

recognition score boundaries, and these grades may differ depending on where the audiometric testing was performed.⁹ Having a graded assessment of the obtained word recognition scores can help a clinician delineate the patient's ability to recognize speech. Importantly, further testing is often considered if the percentage is found to be reduced.

Word recognition testing can assist in the diagnosis of the type of hearing loss. For example, in a conductive hearing loss, word recognition scores are generally preserved at around 80% to 100%, whereas in a cochlear sensorineural hearing loss, the WRS is usually reduced. In the case of a retrocochlear sensorineural hearing loss, word recognition is generally very poor or abnormally low, even in the setting of a normal PTA. Cortical or central lesions will lead to this finding, with the extreme example being **Broca's aphasia**.¹⁵

One exception to this plateau phenomenon is noted with retrocochlear pathology. In a patient with suspected retrocochlear disease, one may see a **rollover** phenomenon in the word recognition scores.¹⁶ As the intensity level of the presented words increases beyond the usual plateau, increasingly louder intensities will lead to a dramatically decreased word recognition score. Expert audiologists will know to perform this additional testing in patients with suspected retrocochlear pathologies. One such test is acoustic reflex threshold measurements. Refer to Section 3: Immittance Testing for more information on this topic.

TABLE 2-2. Common Monosyllabic Word Lists Used During Word Recognition Testing

Examples of Monosyllabic Word Lists			
NU – 6 (List 1)		W-22 (List 2)	
laud	love	your	and
boat	sure	bin	young
pool	knock	way	cars
nag	choice	chest	tree
limb	hash	then	dumb
shout	lot	ease	that
sub	raid	smart	die
vine	hurt	gave	show
dime	moon	pew	hurt
goose	page	ice	own
whip	yes	odd	key
tough	reach	knee	oak
puff	king	move	new
keen	home	now	live
death	rag	jaw	off
sell	which	one	ill
take	week	hit	rooms
fall	size	send	ham
raise	mode	else	star
third	bean	tare	ear
gap	tip	does	thin
fat	chalk	too	flat
met	jail	cap	well
jar	burn	with	by
door	kite	air	ale

Source: Used with permission from Hirsh, I. J., Davis, H., Silverman, S. R., Reynolds, E. G., Eldert, E., & Benson, R. W. (1952). Development of materials for speech audiometry. *Journal of Speech and Hearing Disorders*, 17(3), 321–337 and adapted from Tillman, T. W., & Carhart, R. (1966). An expanded test for speech discrimination utilizing CNC monosyllabic words: Northwestern University Auditory Test No. 6. Northwestern University, Evanston, IL, Auditory Research Lab.

Note. Only one sample list is shown from each – multiple lists are available to prevent practice effect.

$$\frac{\text{Number of correct responses}}{\text{Total number of words presented}} \times 100 = \text{Word Recognition Score}$$

FIGURE 2–2. Formula used to calculate word recognition scores.

Score (%)	Category
90-100	Excellent
75-89	Good
60-74	Fair
50-59	Poor
<50	Very Poor

FIGURE 2–3. Grading system for word recognition scores.

As mentioned above, word recognition testing is performed at or near the MCL or around a 40-dB hearing level above the speech reception threshold. Testing below this intensity level may falsely lead to poorer scoring. Word recognition scoring should plateau once the intensity is greater than a ~30-dB hearing level above SRT. For this reason, word recognition testing is performed above that level normally. Once hitting the plateau, increasing the intensity level of the presented monosyllabic words

should not generate any further variation. If there is a variation in scoring, you may suspect malingering or retrocochlear pathology.

References

1. American Speech-Language-Hearing Association. *Determining Threshold Level for Speech*. 1988.
2. Graham JT. Evaluation of methods for predicting speech reception threshold. *Arch Otolaryngol*. 1960;72(3):347–350.
3. Quiggle RR, Glorig A, Delk JH, Summerfield AB. Predicting hearing loss for speech from pure-tone audiograms. *Laryngoscope*. 1957;67(1):1–15.
4. Gelfand SA. *Essentials of Audiology*. New York, NY: Thieme; 2009.
5. Chaiklin JB, Ventry IM. Spondee threshold measurement: a comparison of 2- and 5-dB methods. *J Speech Hear Disord*. 1964;29(1):47–59.
6. Carhart R, Porter LS. Audiometric configuration and prediction of threshold for spondees. *J Speech Lang Hear Res*. 1971;14(3):486–495.
7. Gelfand SA, Silman S. Functional components and resolved thresholds in patients with unilateral nonorganic hearing loss. *Br J Audiol*. 1993;27(1):29–34.
8. Carhart R. Observations on relations between thresholds for pure tones and for speech. *J Speech Lang Hear Res*. 1971;36(4):476–483.
9. Chasin M, English K, Hood LJ, Tillery KL. In: Katz J, ed. *Handbook of Clinical Audiology*. New York, NY: Wolters Kluwer; 2014.
10. Tharpe AM, Seewald R, eds. *Comprehensive Handbook of Pediatric Audiology*. San Diego, CA: Plural Publishing; 2016.
11. Walker C. Learning effects with a closed response set nonsense syllable test. *Aust J Audiol*. 1982;4:27–31.
12. Theodoridis GC, Schoeny ZG. Procedure learning effects in speech perception tests. *Audiology*. 1990;29(4):228–239.
13. Hawkins DB, Prosek RA, Walden BE, Montgomery AA. Binaural loudness summation in the hearing impaired. *J Speech Lang Hear Res*. 1987;30(1):37–43.
14. Stach B. *Clinical Audiology: An Introduction*. San Diego, CA: Nelson Education; 2008.
15. Johnson EW. Auditory findings in 200 cases of acoustic neuromas. *Arch Otolaryngol*. 1968;88(6):598–603.
16. Bess FH, Josey AF, Humes LE. Performance intensity functions in cochlear and eighth nerve disorders. *Am J Otol*. 1979;1(1):27–31.

Section 3

IMMITTANCE TESTING

Step 6

Determine Tympanometry Type

Step 7

Check Ear Canal Volume (ECV)

Step 8

Evaluate Acoustic Reflexes

Introduction

Audiogram interpretation concludes with a close assessment of **immittance testing** results. Immittance testing is a collection of tests including tympanometry, **static immittance**, and **acoustic reflex testing**. The primary purpose of this assessment is to determine the ability of sound to travel through the **tympanic membrane** (eardrum) and middle ear space. Additionally, acoustic reflexes help evaluate the acoustic pathways involved in sound signal transfer. By reviewing and understanding the measured reflexes, one can infer the potential location of a pathologic lesion causing the abnormal hearing thresholds. Furthermore, immittance testing provides valuable objective measurements that may help determine existing negative middle ear pressure, tympanic membrane immobility, eustachian tube dysfunction, and discontinuity or immobility of ossicular chain. As such, it should be a part of all comprehensive audiologic evaluations.¹

The term **immittance** is derived from two individual electrophysiologic measurements used to evaluate middle ear status: (1) impedance and (2) admittance. **Impedance** is a measure of the opposition to the flow of energy in a closed system (ie, how much sound energy is *impeded* from passing through the tympanic membrane). **Admittance** is a measure of the flow of energy through a closed system (ie, how much sound energy is *admitted* through the tympanic membrane). Immittance (Im – mittance) is a combination of these

two terms and is used to indicate the general principle of evaluating acoustic energy flow across the tympanic membrane.²

Since this testing is performed with electrophysiological techniques, no behavioral response is needed from the patient. Thus, immittance testing should be obtainable in nearly every patient, regardless of their age, and for this reason, it is a highly valuable piece of the audiometric assessment.^{3,4} To measure immittance, the external ear canal is sealed with an air-tight probe that is connected to a device that delivers a pure tone (usually at 226 Hz) at differing levels of air pressure. This testing triggers the tympanic membrane to move in the response to this air pressure. The reflected acoustic energy, changes in canal volume, and other measurements are received back into the probe and compared to the properties of the initial energy presented.⁵

A tympanic membrane with low impedance (and high admittance) more readily transfers the acoustic energy from the probe tone, whereas a tympanic membrane with high impedance (and low admittance) is more resistant to that energy flow. Changes to the impedance and admittance in the system, such as with **tympanosclerosis**, middle ear effusion, or tympanic membrane tumors, are measured and reported. Clinicians can use standard patterns of immittance testing (see Table 3–2) to compare the resultant patient-specific testing and help confirm or rule out suspected diagnoses.^{2,6–8}

Step 6

DETERMINE TYMPANOMETRY TYPE

Tympanometry

Tympanometry measures the capacity of the tympanic membrane to admit sound (admittance or **compliance**) and is therefore an important assessment of the conductive mechanism of the ear. Specifically, tympanometry measures the compliance of the tympanic membrane and underlying middle ear space to the same intensity sound presented at various air pressures. These pressures are presented in a closed system via an air-tight ear probe as described above. If a hermetic seal cannot be obtained for the testing, tympanometry is aborted and the audiologist makes a note of this issue.

Tympanometry is a dynamic test, meaning the tympanic membrane's compliance is measured at different air pressure levels, as the air pressure is systematically increased or decreased in this closed system.⁹ The test probe emits a long pure tone at 226 Hz (or 220 Hz) frequency with varying degrees of negative and positive pressures and then measures the compliance of this system.¹ A curve is generated from this testing as the air pressure is varied. The air pressure emitted is standardized in terms of mm H₂O (millimeters of water) or daPa (**decapascal**) and is the independent variable in this testing (ranging from -300 daPa to +200 daPa). The x-axis of a tympanogram represents the fluctuating air pressure as emitted from the measurement probe. Resultant compliance measurements obtained from the varied air pressures are recorded on the y-axis, and they represent the dependent variable in this testing. The recordings on the y-axis reflect the actual compensated static acoustic admittance of the tympanic membrane. Compliance, or admittance, is recorded in **millimhos** (mmhos) or in milliliters (mL).¹ It may be easiest to think about compliance measurements as equivalent to tympanic

membrane mobility, but this number actually describes the membrane's ability to admit sound. Further discussion of this measurement can be found under the compliance section below. Normative values for this testing are shown in Table 3-1.

Tympanograms are classically defined as type A, B, or C, based on the Jerger classification system described in 1970 (see Figure 3-1).^{10,11} This terminology has been universally adopted to describe the curves obtained with tympanometry testing.

Type A Tympanograms

A normal middle ear is most compliant at or near ambient pressure, and thus the peak in a normal tympanogram is centered between -100 and 100 mm H₂O (daPa) on the x-axis. When this occurs, the resultant curve is described as a type A tympanogram. The **peak compliance**, or **peak static acoustic admittance**, recorded on the y-axis of a type A tympanogram is also within a normal range (0.2 to 1.8 mmhos or mL).² This point is labeled Y_{tm} . See the section entitled "Middle Ear Pressure and Peak Compliance" below for further discussion on this topic.

There are subtypes of type A tympanograms depending on the measured compliance of the tympanic membrane. When there is a centrally located peak with reduced compliance, this is labeled type A_s (A_s for shallow). This reflects a middle ear impedance that is abnormally high, and much of the air pressure energy is reflected back into the probe with minimal displacement of the tympanic membrane.⁹

Type A_d tympanograms (A_d for deep) have a centrally located peak at ambient pressure as well as a measured hypercompliance of the tympanic membrane. Thus, the tympanic membrane is displaced significantly more with the standard ambient air pressure emitted from the probe.

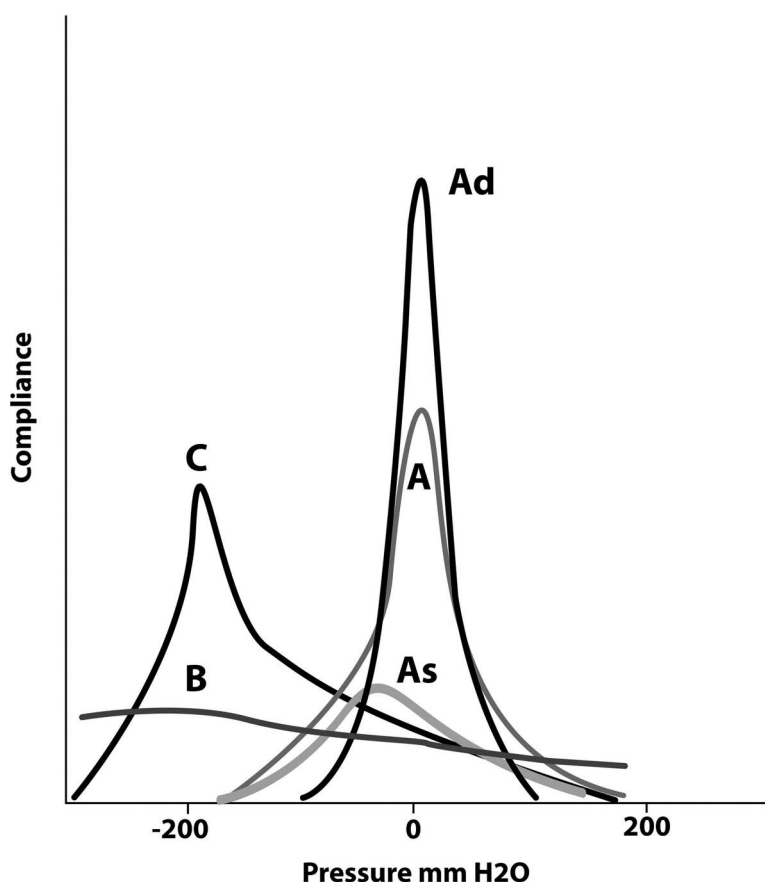


FIGURE 3-1. Tympanograms type A-C, A_s , and A_d . Adopted from: Jerger J. Clinical experience with impedance audiometry. *Arch Otolaryngol.* 1970;92(4):311–324. – Used with permission.

When a type A_s tympanogram is noted, diagnoses that reflect reduced compliance of the tympanic membrane should be considered, such as some middle ear effusions, ossicular chain fixation, or otosclerosis.

When a type A_d tympanogram is noted, diagnoses that reflect increased compliance of the tympanic membrane should be considered, such as a healing tympanic membrane perforation (**monomeric membrane**) or ossicular chain discontinuity.

Type B Tympanograms

A type B tympanogram has a flat line with no peak, or variability, in compliance. Hence, no significant variation is seen in the recorded compliance measurements of the tympanic membrane with the varying air pressures presented. In other words, no matter how much pressure is applied via the probe, the tympanic membrane does not move. Type B

tympanograms are most commonly associated with otitis media with effusion, since the fluid in the middle ear space prevents membrane mobility. Nevertheless, it can also be associated with a large cerumen impaction, tympanic membrane perforation, patent **tympanostomy tube**, or mass in the middle ear.^{12,13}

When a type B tympanogram is noted, it can be helpful to evaluate the canal volume measurements as well (see below). When the canal volume is within normal limits, one should suspect middle ear disease. When the canal volume is elevated above normal limits, then one should suspect a tympanic membrane perforation or patent tympanostomy tube. When the canal volume is smaller than expected, one would expect a cerumen impaction or ear canal occluding mass.

Type C tympanograms are considered synonymous with poor eustachian tube ventilation. When negative pressure measurements are noted to be prolonged over multiple months, clinicians will often intervene with targeted eustachian tube therapies designed to assist with return of middle ear ventilation. This may include topical intranasal steroids or antihistamines, **myringotomy** ± tympanostomy tubes to allow air pressure equalization across the tympanic membrane, or balloon dilation eustachian tuboplasty.

Type C Tympanograms

A type C tympanogram is seen when the peak in compliance along the y-axis continues within the normal range (ie, the measurement of compliance continues to lie between 0.2 and 1.8 mmhos). This compliance peak, however, is no longer centered over normal ambient pressures as seen with a type A tympanogram. Rather, the peak is shifted leftward toward negative pressures greater than –100 mm H₂O or daPa. Type C tympanograms are common in patients with **eustachian tube dysfunction** and represent inadequate equalization of the middle ear air into the eustachian (or pharyngotympanic) tube.¹⁴

Once the tympanometry data is collected and the Jerger type is determined, the audiologist will report this finding under the immittance section of the audiogram report. Generally, a table format is used to report this information along with middle ear pressures, physical volume testing, and compliance. An example of this table is found in Figure 3–2.

Middle Ear Pressure and Peak Compliance

As discussed above, air is emitted at varying pressures from –300 daPa to 300 daPa through the tympanometry probe in a closed air-tight system in order to obtain the compliance measurement for the tympanic membrane. An audiologist will report a **middle ear pressure**, or the emitted air pressure needed to reach the point of peak compliance. This is the pressure in which there is maximum acoustic energy admittance through the tympanic membrane into the middle ear space.⁵ Sometimes peak compliance is referred to as peak static acoustic admittance. As mentioned above, the normative values for peak compliance are around 0.2 to 1.8 mmhos for adults. The range can be more limited for children, generally 0.25 to 1.05 mmhos.¹⁵ See Table 3–1 for normal values for these measurements.

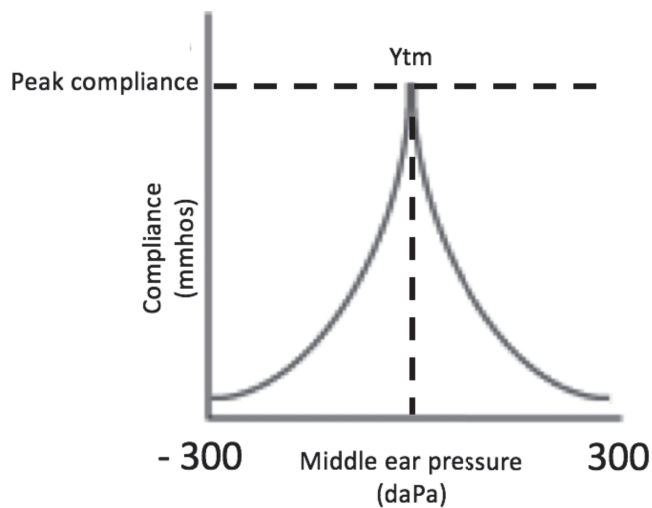
Peak compliance occurs when the ear canal pressure is equal on both sides of the eardrum or tympanic membrane, and the middle ear muscles are in a neutral state. This will be the point at which the tympanic membrane mobility is maximized. When this occurs, the x-axis value (middle ear pressure) and the y-value (compliance or peak compliance) are both recorded on the table shown in Figure 3–2. Graphically, the recorded compliance and middle ear pressure will coincide when the highest measurement is obtained on the peak on the tympanogram (or Y_{tm}). Thus, the value of the highest part of the peak on a type A tympanogram will be recorded for

IMMITTANCE TESTING		
Tympanometry		
	RIGHT	LEFT
TYMPANOGRAM TYPE		
MIDDLE EAR PRESSURE		
P.V.T		
COMPLIANCE		

FIGURE 3–2. Standardized table used to report immittance testing.

TABLE 3-1. Normative Values for Adult and Pediatric Immittance Testing

Normative Immittance Values		
Measure	Adult (>7 Years Old)	Pediatric (6 Months to 7 Years Old)
Middle ear pressure	-100 to 100 daPa	-100 to 100 daPa
Ear canal volume	0.6 to 1.5 mL	0.4 to 1.0 mL
Peak static acoustic compliance (Y_{tm})	0.2 to 1.8 mmhos	0.25 to 1.05 mmhos

**FIGURE 3-3.** Graphical depiction of the standard measurements taken during tympanometry testing and the data points they represent.

If eustachian tube dysfunction is suspected as a cause for reduced hearing or increased ear symptoms, a quick check of the middle ear pressure value along with the Jerger type can be very helpful. One would expect to see a Jerger type C tympanogram with a reported middle ear pressure >-100 daPa.

Caution is advised when interpreting tympanometry data. While the typical results are reported in Table 3-2, some pathologies can have multiple possible tympanometry outcomes. For example, some ears with otosclerosis may have a type A tympanogram rather than type A_s . Unexpected tympanogram results do not completely rule out a suspected diagnosis, highlighting the importance to incorporate all of the audiometric data, including the patient's history, in your clinical assessments.

compliance and should be centered at or around 0 daPa. For reference, Figure 3-3 shows these measurements labeled on a standard type A tympanogram.

It is extremely important at this point to take a moment to reflect back to the pure-tone audiometry data reviewed Section 1 (steps 1-3). A close comparison between the immittance data and the audiometry data should generally reveal consistent results and may help confirm certain diagnoses. Unexpected differences can also be revelatory for other diagnoses. A few examples are outlined in Table 3-2 to show how this can be done.

TABLE 3–2. Examples of Expected Pure-Tone Audiometry and Tympanometry Results for Selected Diagnoses

Diagnosis	Presence of Sensorineural Hearing Loss (SNHL)	Expected Pure-Tone Threshold Results	Expected Tympanogram Results
Normal	No	Normal air and bone conduction	Type A
Otitis media with effusion	No	Abnormal air conduction Normal bone conduction Presence of air-bone gap	Type B
Tympanic membrane perforation (– SNHL)	No	Abnormal air conduction Normal bone conduction Presence of air-bone gap	Type B
Tympanic membrane perforation (+ SNHL)	Yes	Abnormal air and bone conduction Presence of air-bone gap	Type B
Otosclerosis (– cochlear involvement)	No	Abnormal air conduction Normal bone conduction Presence of air-bone gap	Type A _s
Otosclerosis (+ cochlear involvement)	Yes	Abnormal air and bone conduction Presence of air-bone gap	Type A _s
Isolated tympanosclerosis	No	Normal air and bone conduction	Type A _s
Isolated ossicular chain discontinuity	No	Abnormal air conduction Normal bone conduction Presence of air-bone gap	Type A _d
Healing tympanic membrane perforation (monomer present)	No	Normal air and bone conduction	Type A _d
Tympanostomy tube	No	Normal or abnormal air conduction Normal bone conduction Presence of air-bone gap	Type B
Eustachian tube dysfunction without hearing complaints	No	Normal air and bone conduction	Type C

Step 7

CHECK EAR CANAL VOLUME (ECV)

Static Immittance

Another electrophysiologic measurement obtained during a standard audiogram is the **static immittance** or the ear canal physical volume test (PVT or ECV). This type of immittance correlates to the measured volume of the ear canal to the tympanic membrane when a 226-Hz probe is used.¹⁶ The measures obtained are generally expressed in milliliters (mL), but millimhos (mmhos) may also be used. By using the same probe and closed system technique, the volume of air needed during the tympanometry testing to obtain the compliance measurements can be derived from compliance measures and reported as the ear canal volume or ECV. Once this calculation is performed, the ECV measurements are reported alongside tympanometry type, middle ear pressures, and compliance, as shown in Figure 3–2 (see page 35).

The static immittance represents the volume of the ear canal only when a tympanic membrane is intact. This measurement can be affected by various outer ear obstructive pathologies or by a global narrowing of the ear canal itself. A wide range of canal volumes can be recorded and still be considered within normal limits. Normative values for an adult ear canal range from 0.6 to 1.5 mL. Pediatric measured volumes have a more limited range from 0.4 to 1.0 mL. When ear canal volumes are measured to be larger than 2.0 mL in children and 2.5 mL in adults, then pathology affecting the tympanic membrane is considered.¹⁶ A normal canal volume measurement has high clinical value, as it signifies the tympanic membrane is intact (see Table 3–1).

In order to calculate the ECV, tympanometry probes will obtain compliance measurements at the various middle ear pressures emitted from the probe. The probe is able to estimate the ear canal

If you suspect a tympanic membrane perforation is present, the ECV measurement can be very useful. One would expect to see an abnormally large volume reported (~4–7 cm³) when there is a hole in the tympanic membrane. Perforations can be caused from several different etiologies, but one must not forget that the ECV may also be increased in the setting of a patent tympanostomy tube.

volume based on the difference in the compliance measured at different pressures emitted. This is graphically summarized on a type A tympanogram in Figure 3–4. Please note that this calculation has been generalized to match the level of material presented in this text. For a more thorough discussion of how this measurement is obtained, please refer to a comprehensive audiometric science text.^{2,17}

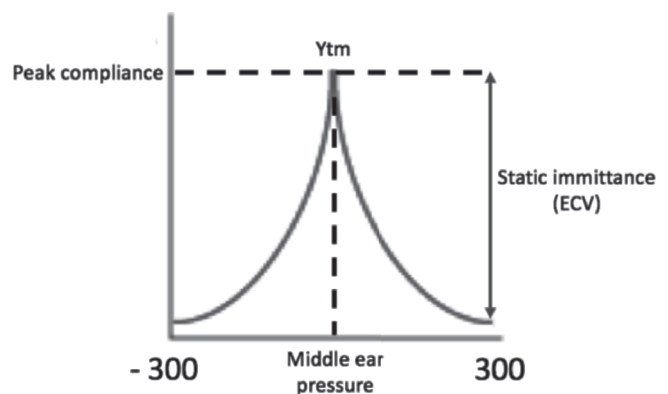


FIGURE 3–4. Graphical depiction of the static immittance (ECV).

Step 8

EVALUATE ACOUSTIC REFLEXES

Acoustic Reflex

Our neurosensory system has the ability to respond to stimuli quickly via reflex pathways. Reflexes serve to enhance the speed of the response as a defense mechanism to avoid injury. The auditory system utilizes the **acoustic reflex** to limit significant **autophony** when individuals are speaking, as well as protect its neurosensory organ, the cochlea, from high-intensity sounds. A sound stimulus will trigger an involuntary signal to stiffen the middle ear conduction system to avoid full acoustic energy transmission into the cochlea via the **oval window**. As a result, the cochlea will not receive the full stimulus intensity.⁵

The acoustic reflex pathway involves the facial (seventh) cranial nerve, vestibulocochlear (eighth) cranial nerve, and the **stapedius muscle**. Sometimes this reflex is called the stapedius reflex for this reason.¹⁷ The structures involved in this reflex are fully outlined in Figure 3–5. Sound energy transfers from the cochlea to the eighth nerve cochlear nucleus (afferent) into the brainstem at the **superior olivary nucleus** where the signal is transferred to the ipsilateral and contralaterally to the facial nuclei. From there, the facial nerve carries back a signal (efferent) to the stapedius muscle on both sides, stimulating immediate contraction.¹⁸ As a result, both stapes bones are pulled away from the oval window on the cochlea via the stiffening force of the stapedius muscle. In turn, the tympanic membrane is prevented from further full movement. Changes in movement can be registered on a tympanometry probe as compliance shifts in both the tested and contralateral ear. The information gathered via this reflex testing can be used to look for pathologic lesions involving any of the structures in the pathway.^{19,20}

The measurement of the **acoustic reflex threshold (ART)** refers to the lowest sound intensity tone that elicits stapedius contraction. In a normal hearing ear, sounds are presented to the tympanic membrane around 60 dB above their hearing thresholds. The intensity is increased until a response is elicited. One should expect then that sounds around 80 dB will trigger the acoustic reflex. Since an intense signal presented to one ear leads to stiffening and protection of both membranes, both can be evaluated simultaneously during the testing. The audiologist places the tympanometry probe into the ear being tested and places a secondary probe to measure the reflex in the contralateral ear. Results are reported by the ear that is stimulated by the signal and are recorded in a table as shown in Figure 3–6.

As shown in Figure 3–6, not all frequencies need to be measured, as acoustic reflex thresholds are not observed outside of 500-, 1000-, 2000-, and 4000-Hz frequencies.¹⁷ The audiologist will begin at 500 Hz by introducing sound stimuli around 70 dB and then increasing by 5 dB until the reflex is measured—hence the acoustic reflex threshold. Reflex thresholds are usually detected around 80 to 100 dB. The testing does not surpass 115 dB to avoid hurting

Acoustic reflex thresholds are reported with respect to the ear that receives the initial stimulus tone via the probe. The resultant reflex is then measured bilaterally as being ipsilateral or contralateral to the stimulated ear. For example, a “right contralateral reflex” is when the signal is played to the right ear and the measurement is taken from the left (contralateral) ear.

Remember that a stimulus presented to one ear usually results in bilateral reflex measurements, but one may only measure unilateral responses depending on if pathology is present or if only the ipsilateral ear was measured. When evaluating reflexes, both the ipsilateral (same side) and contralateral (opposite side) measurements may be attempted and reported. As a general rule, any stimulus signal that must traverse a damaged area prior to reaching the brainstem will have an effect on the measured reflex threshold on both sides.

may be able to deduce the site of a pathologic lesion (middle ear, cochlear, or retrocochlear), which may help confirm a suspected diagnosis. The existence or absence of reflexes can also provide further confirmation on a patient's hearing thresholds.²⁰

Interpretation of the reflexes requires a basic understanding of the acoustic reflex pathway (see Fig-

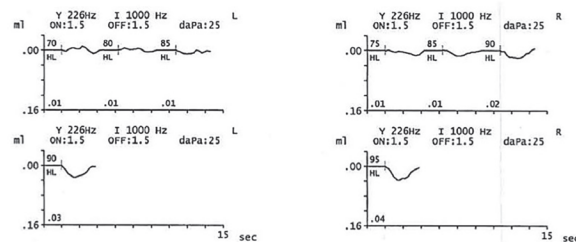
ure 3-5), in order to comprehend the measured results. An exhaustive review of this pathway will not be provided within this text. Further information regarding the specific structures that comprise the pathway as well as disease states that affect each area can be found in more comprehensive clinical texts. However, just like with audiometric data, one can learn to identify patterns to aid in reflex threshold interpretation.^{2,17} The following illustrations (comprising Figures 3-8A to 3-8E) summarize the expected reflex findings according to the location and nature of the pathology responsible for the hearing loss.

Outer and Middle Ear Pathology

Bilateral Conductive Hearing Loss

Outer or middle ear disease can affect the conduction of sound energy into the cochlea. The resultant alteration in conduction can prevent the initiation of the acoustic reflex. Furthermore, a secondary effect can occur if middle ear disease prevents the contraction of the stapedius or blocks the full reception of the reflex signal

NORMAL REFLEX THRESHOLDS



ABSENT REFLEX THRESHOLDS

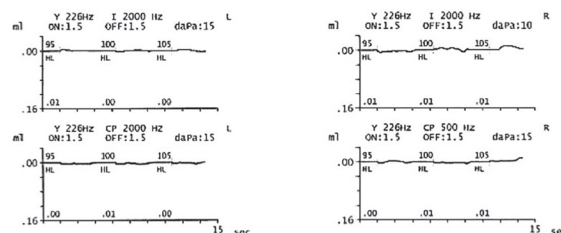


FIGURE 3-7. Example of an acoustic reflex graph that may be provided with audiogram data.

One notable exception exists when examining the reflexes of a patient with otosclerosis. Often the reflexes will be reduced or absent disproportionately to the degree of conductive hearing loss that is present. Understanding the otosclerosis disease process can help explain this phenomenon. As the stapes footplate seals to the oval window niche with otosclerotic plaques, stapes movement will be more limited. During acoustic reflex testing, the stapedius muscle contraction will be unable to elicit full stapes movement. This, in turn, leads to more limited tympanic membrane movement, which the probe registers as a reduced or absent acoustic reflex.

into the probe. Any stimulus presented to someone in this condition must overcome the air-bone gap to measure a reflex, which can sometimes exceed the maximum available stimulus level. The result, depending on amount of conductive loss, will be elevated or absent, as shown in Figure 3–8A. Both ipsilateral and contralateral measurements can be expected to be elevated or absent depending on the severity of the loss.

Unilateral Conductive Hearing Loss

In a unilateral conductive hearing loss situation, the results may vary depending on the severity of disease. For demonstration purposes, when a unilateral loss is discussed in this section, the right side will be shown as the affected ear. With mild conductive hearing loss, the resultant stapedial contraction may still be normal on the affected (right) side. However, as the conductive hearing loss becomes more severe, one should expect elevated or absent ipsilateral (right) and contralateral (left) reflexes when the right ear is stimulated. When the left side is stimulated, the reflexes would continue to be present on the ipsilateral side and either elevated or absent on the right side. No contralateral reflex is presumably seen if the right middle ear pathology is affecting the stapedial contraction.

Cochlear Pathology

Bilateral Sensorineural Hearing Loss

Similar to the bilateral conductive hearing loss shown above, bilateral cochlear hearing losses will

affect the acoustic reflexes when the impairment to the cochlea has progressed past a certain bone conduction threshold (assuming no air-bone gap or conductive hearing loss component exists). For cochlear hearing losses, bone conduction thresholds below 50 dB hearing level (HL) should have no effect on the reflexes and will likely be reported as present. As the hearing thresholds worsen in severity, the chances of an elevated or absent reflex increase, as shown in the bottom portion of Figure 3–8C.

Measurement of acoustic reflex thresholds can help identify a physiologic phenomenon known as **recruitment**. This finding is noted primarily with cochlear hearing losses. In a normal ear, as discussed above, reflex thresholds are typically tested around 60 dB above their hearing thresholds and a reflex around 80 dB is noted. With loudness recruitment, when cochlear pathology is present and the hearing thresholds are elevated, the acoustic reflex is triggered or recruited by sound levels only 30 to 40 dB above their thresholds.²¹

Unilateral Sensorineural Hearing Loss

The expected outcomes of the acoustic reflex in unilateral, right-sided sensorineural hearing loss of cochlear origin are shown in Figure 3–8D. Again, remember that as a general rule, any signal that must traverse a damaged area will affect the measured reflex threshold on both sides. Thus, whenever the stimulus is presented to the right ear, both reflexes measured in patients with significant cochlear pathology will be elevated or absent. Whenever the signal is presented to the normal left ear, the reflexes will be unaffected.

Retrocochlear Pathology

Acoustic reflex thresholds can be particularly helpful in situations where retrocochlear disease is

This paradoxical recruitment phenomenon can be clinically relevant as it is also observed outside of the testing environment. Patients with cochlear disorders may request others “to speak louder” and then immediately follow with the complaint to “stop shouting.”²² This response to sound perception becoming rapidly more intense with increasing sound level is recruitment.

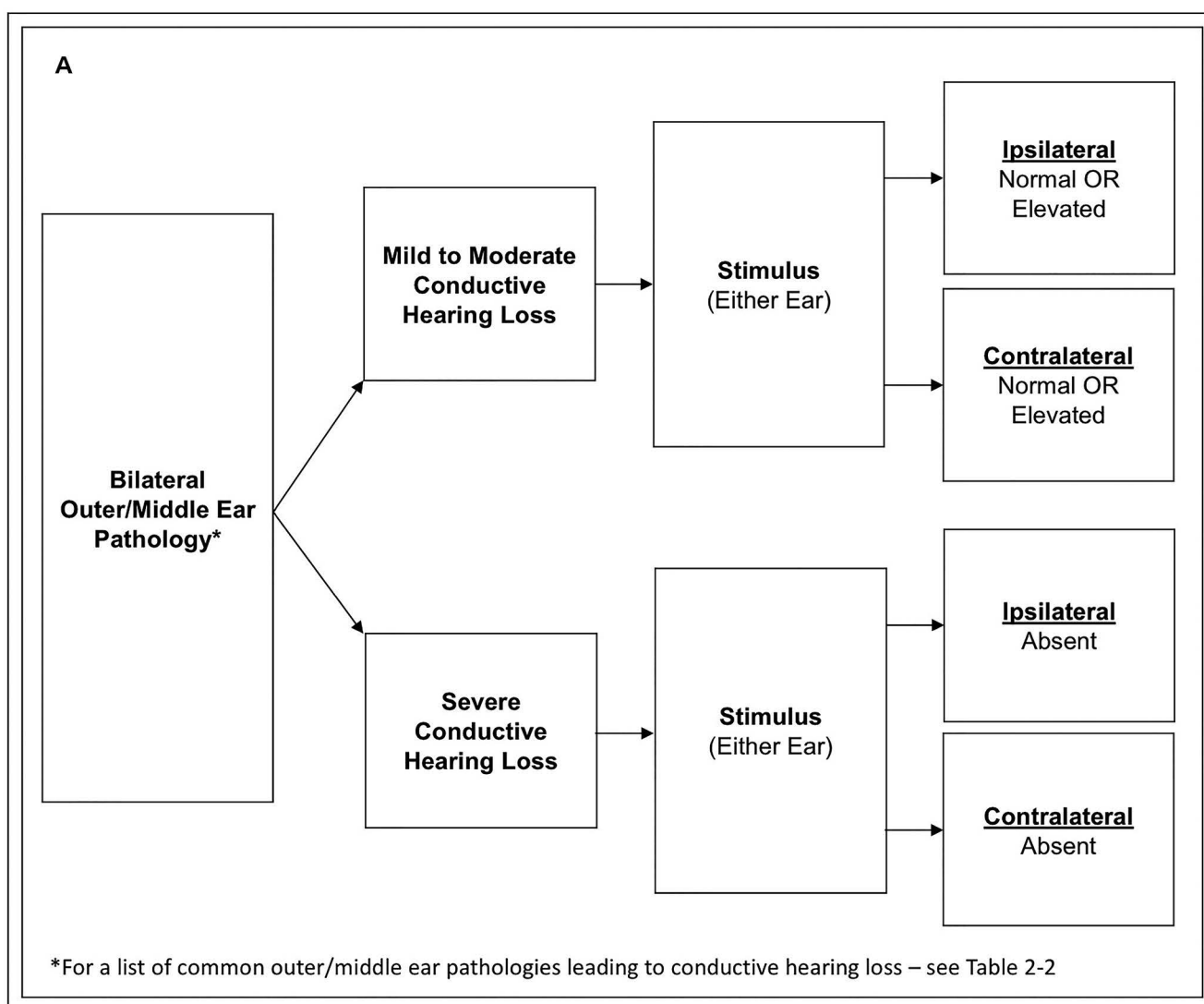


FIGURE 3–8. A. Expected acoustic reflexes seen with bilateral outer and middle ear pathologies leading to conductive hearing loss. **B.** Expected acoustic reflexes seen with right unilateral outer and middle ear pathology leading to conductive hearing loss. **C.** Expected acoustic reflexes seen with bilateral cochlear pathologies leading to sensorineural hearing loss. **D.** Expected acoustic reflexes seen with right unilateral cochlear pathology leading to sensorineural hearing loss. **E.** Expected acoustic reflexes seen with right unilateral vestibulocochlear nerve pathology leading to sensorineural hearing loss. (*continues*)

suspected to be the cause of the hearing loss. Retrocochlear refers to any structure in the acoustic reflex pathway that receives afferent stimulation from the cochlea. This relationship is demonstrated in Figure 3–9 (see page 49).

As discussed in Section 2: Speech Audiometry, aside from word recognition scores and the rollover phenomenon, reflex testing is the only other section

of the audiogram that measures retrocochlear disease. Otherwise, most data collected during audiometric testing identifies and measures conductive or cochlear pathologies only. While these regions comprise the preponderance of pathology that affects the hearing signaling pathway, other important retrocochlear structures along the pathway can also be affected. The most classic example of

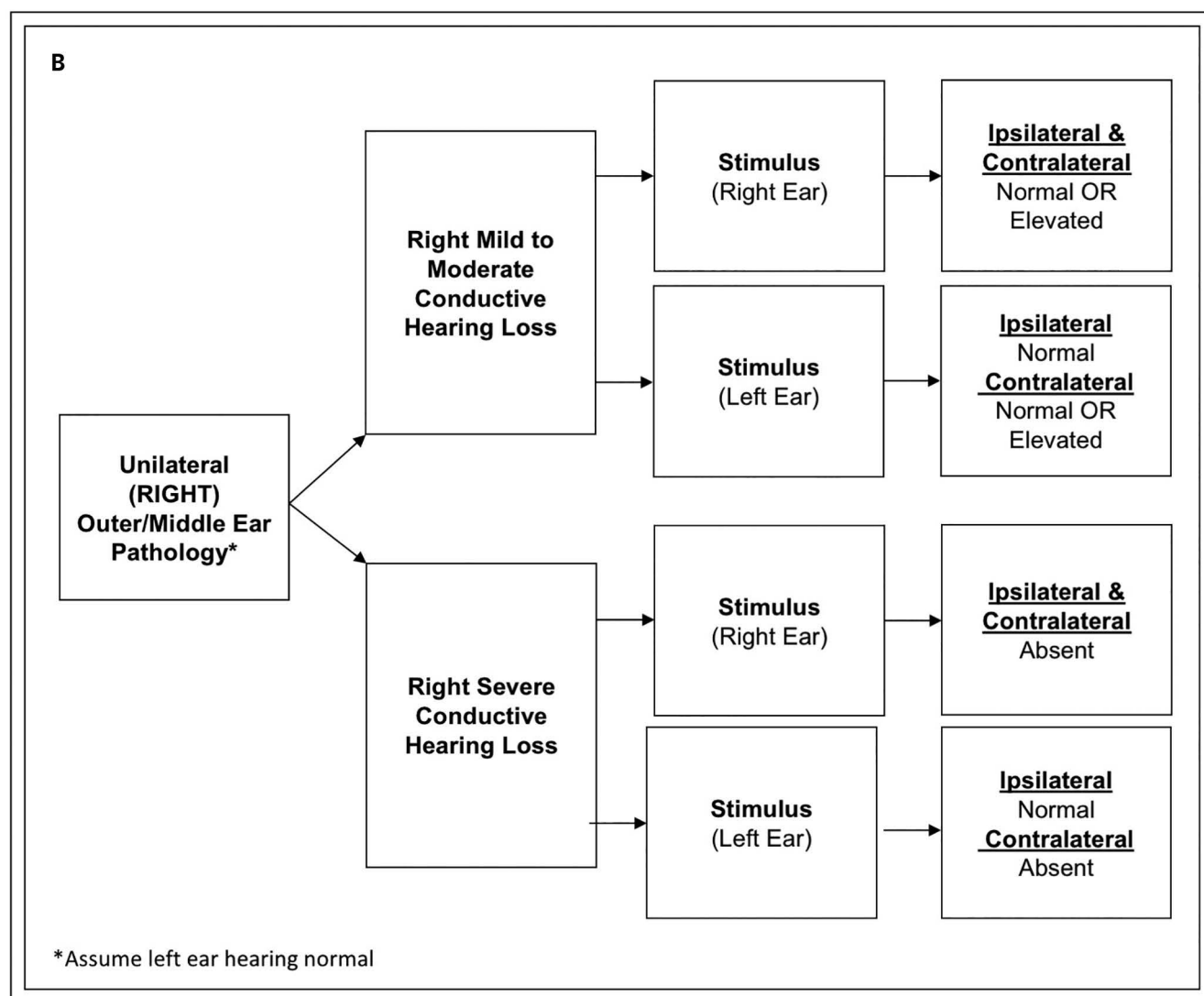


FIGURE 3-8. (continued)

a retrocochlear pathology is a vestibulocochlear (eighth) cranial nerve tumor (ie, **acoustic neuroma** or vestibular schwannoma). Lesions or defects of the neuronal projections in the brainstem or acoustic areas of the central nervous system (superior olivary nucleus) can also affect the measured reflex thresholds but are beyond the scope of this text.²

The vestibulocochlear (eighth) nerve is the most common site for retrocochlear pathology to occur. While the patient may present with a sensorineural hearing loss, the acoustic reflexes may show an entirely different pattern than what is expected with

While the reflexes may suggest retrocochlear pathology, they cannot be relied on as a screening test for diagnosis as false positives (abnormal reflexes with no pathology) and false negatives (normal reflexes with pathology) do occur. A good clinician will incorporate many factors (history, exam, pure-tone thresholds, the presence of unexplained sensorineural asymmetry, word recognition scores, and acoustic reflex testing) when considering this diagnosis.

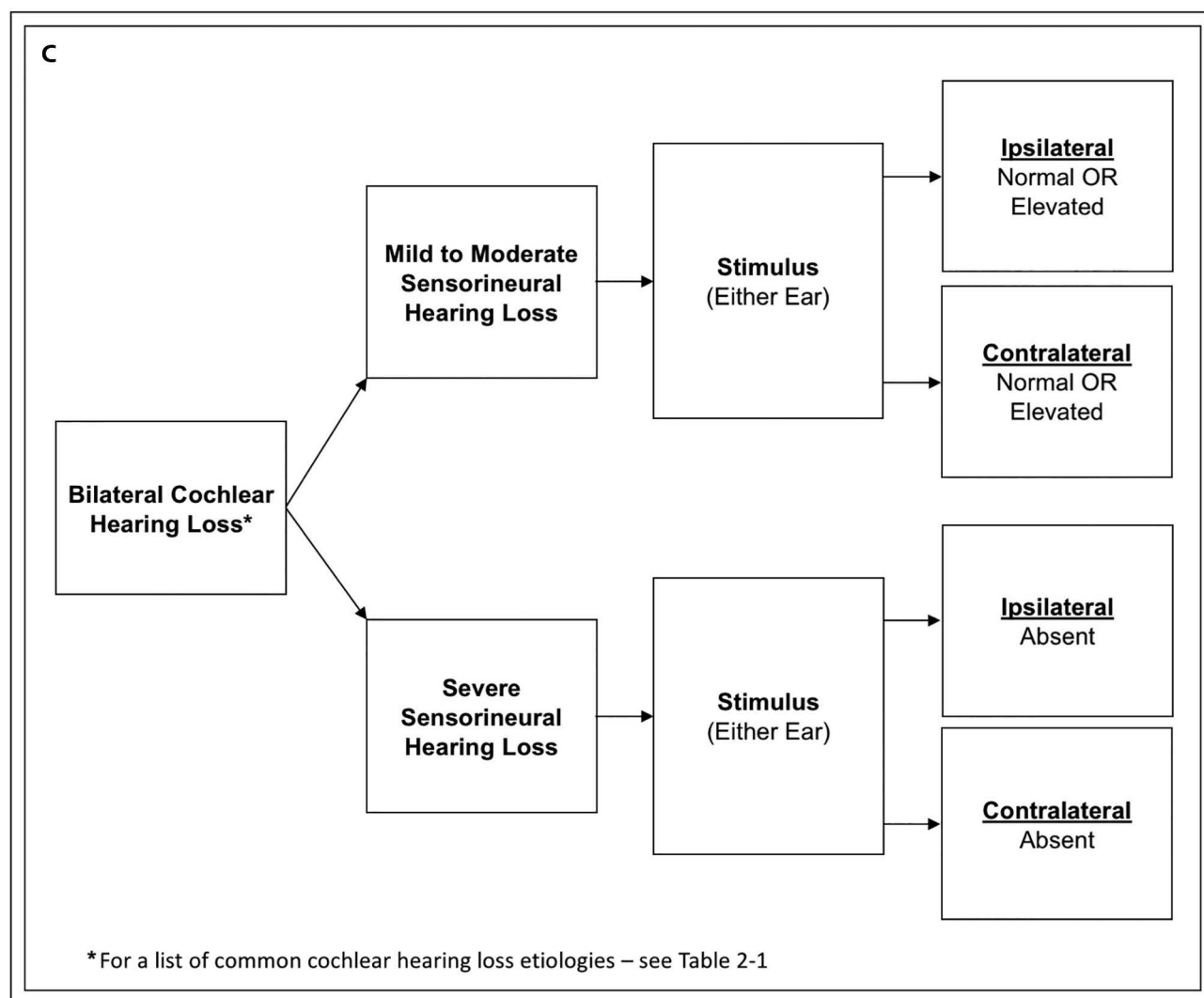


FIGURE 3-8. (continued)

cochlear pathologies discussed above. With vestibulocochlear nerve pathology, it is highly likely that reflexes will be absent or unusually elevated compared to their actual hearing threshold level (see Figure 3-8E). When this occurs, the suspicion of retrocochlear pathology should be raised.

The expected outcome of the acoustic reflex in unilateral, right-sided sensorineural hearing loss of retrocochlear origin caused by vestibulocochlear nerve pathology is shown in Figure 3-8E. When the right side is stimulated, both ipsilateral and contralateral reflexes will be absent, present but

elevated, or present with **acoustic decay**. Acoustic decay is a phenomenon that can be seen with a vestibulocochlear nerve pathology. Whenever a retrocochlear pathology is suspected, the audiologist may opt to perform additional reflex testing to monitor for acoustic decay. With acoustic reflex decay, the audiologist repeats the stimulus 5 seconds after initial testing at 10 dB above the acoustic reflex threshold already recorded. In vestibulocochlear nerve pathologies, the second reflex response decreases or decays by more than half of expected.^{23,24}

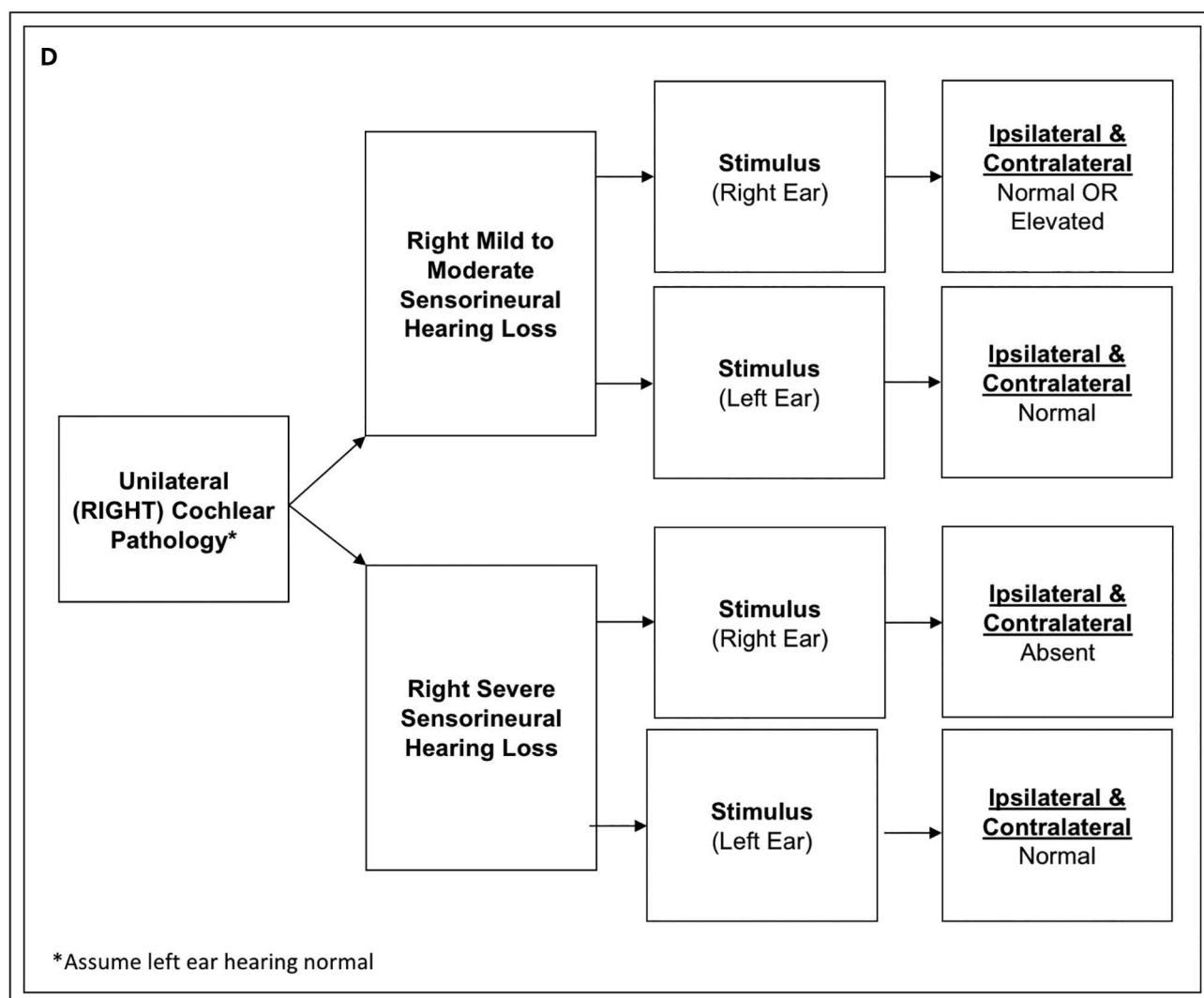


FIGURE 3-8. (continued)

It is worth noting that acoustic decay is not pathognomonic for eighth nerve lesions, as there is a high false-positive rate in patients with Meniere's disease (>20%).²⁵ This decay technique is also a sensitive measure of facial (seventh) nerve dysfunction. In a patient with seventh nerve pathology, like **Bell's palsy** or a facial schwannoma, the reflex decay may also be abnormal.^{2,17,26} Additional testing is necessary to interpret the finding of acoustic decay to determine the exact nature of the pathology.

Retrocochlear pathology affecting the more proximal portions of the reflex pathway will not be discussed in this text given the extreme variety of results that can be seen. It should be noted that given the decussation of nerve fibers in the brainstem and the multiple nuclei involved in the reflex pathway, the acoustic reflex threshold results depend on the size and location of the brainstem lesions. Once at the level of the brainstem, it is highly likely that other neurologic symptoms are present that will assist in making the diagnosis beyond what reflex testing will offer. More in-depth information on acoustic reflex testing can be found in dedicated neurotology and audiometric science texts.^{2,5,17}

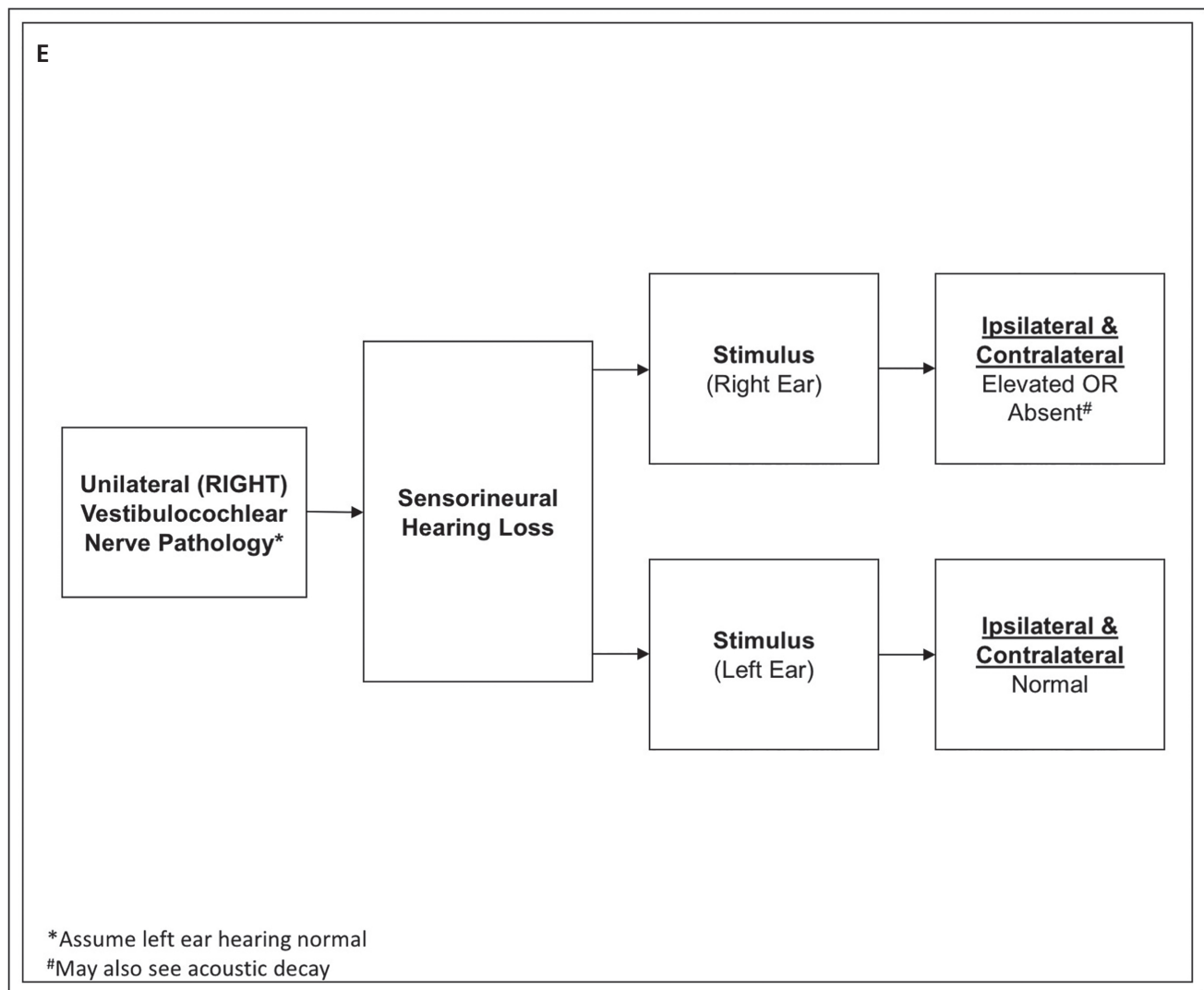


FIGURE 3–8. (continued)

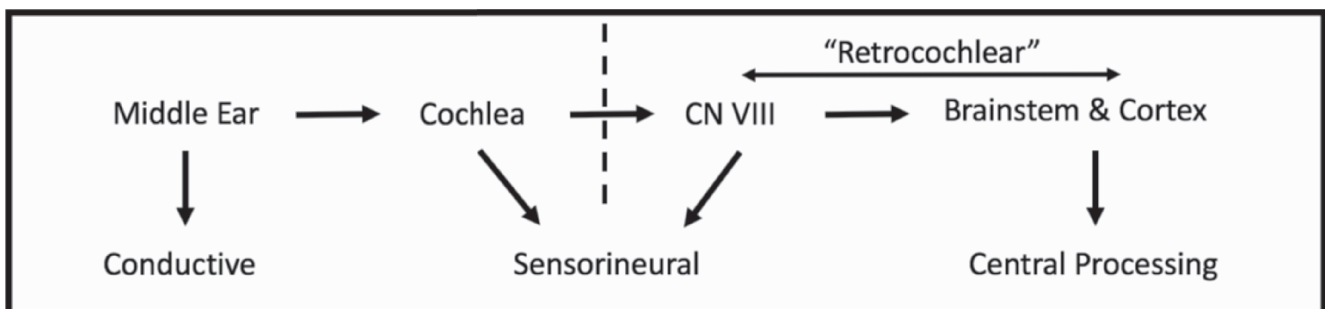


FIGURE 3–9. Areas of auditory system considered retrocochlear disease. Adopted from Davies, R.A. (2016). Audiometry and other hearing tests. In: *Handbook of Clinical Neurology*, 137, 157–176. Joseph M. Furman, Thomas Lempert. Used with permission.

References

1. American Speech-Language-Hearing Association. *Tympanometry*. American Speech-Language-Hearing Association; 1988.
2. Chasin M, English K, Hood LJ, Tillery KL. In: Katz J, ed. *Handbook of Clinical Audiology*. New York, NY: Wolters Kluwer; 2014.
3. Harker LA, Van Wagoner R. Application of impedance audiometry as a screening instrument. *Acta Otolaryngol*. 1974;77(1-6):198-201.
4. Paradise JL, Smith CG, Bluestone CD. Tympanometric detection of middle ear effusion in infants and young children. *Pediatrics*. 1976;58(2):198-210.
5. Stach B. *Clinical Audiology: An Introduction*. San Diego, CA: Nelson Education; 2008.
6. Alberti PWRM, Kristensen R. The clinical application of impedance audiometry: a preliminary appraisal of an electro-acoustic impedance bridge. *Laryngoscope*. 1970;80(5):735-746.
7. Eliachar I, Northern JL. Studies in tympanometry: validation of the present technique for determining intratympanic pressures through the intact eardrum. *Laryngoscope*. 1974;84(2):247-255.
8. Renvall U, Holmquist J. Tympanometry revealing middle ear pathology. *Ann Otol Rhinol Laryngol*. 1976;85:209-215.
9. Lilly DJ, Shanks JE. Acoustic immittance of an enclosed volume of air. In: *Hearing Assessment With the Acoustic Reflex*. 1981:145-160. New York, NY: Grune & Stratton.
10. Liden G, Peterson JL, Bjorkman G. Tympanometry a method for analysis of middle—ear function. *Acta Otolaryngol*. 1970;69(suppl 263):218-224.
11. Jerger J. Clinical experience with impedance audiometry. *Arch Otolaryngol*. 1970;92(4):311-324.
12. Orchik DJ, Dunn JW, McNutt L. Tympanometry as a predictor of middle ear effusion. *Arch Otolaryngol*. 1978;104(1):4-6.
13. Feldman AS. Tympanometry: application and interpretation. *Ann Otol Rhinol Laryngol*. 1976;85(2)(suppl):202-208.
14. Bluestone CD, Cantekin EI. Current clinical methods, indications and interpretation of eustachian tube function tests. *Ann Otol Rhinol Laryngol*. 1981;90(6):552-562.
15. Margolis RH, Hunter LL. Acoustic immittance measurements. In: *Audiology Diagnosis*. New York, NY: Thieme; 2000:381-342: Michael Valente and Ross J. Roeser, eds.
16. Van Camp KJ, Margolis RH, Wilson RH, Creten WL, Shanks JE. Principles of tympanometry. *ASHA Monographs*. 1986;24:1.
17. Gelfand SA. *Essentials of Audiology*. New York, NY: Thieme; 2009.
18. Borg E. On the neuronal organization of the acoustic middle ear reflex: a physiological and anatomical study. *Brain Res*. 1973;49(1):101-123.
19. Silman S, ed. *The Acoustic Reflex: Basic Principles and Clinical Applications*. Orlando, FL: Elsevier; 2012.
20. Feldman AS. Diagnostic application and interpretation of tympanometry and the acoustic reflex. *Audiology*. 1977;16(4):294-306.
21. Joris PX. Recruitment of neurons and loudness. *J Assoc Res Otolaryngol*. 2009;10(1):1-4.
22. Bacon SP, Oxenham AJ. Psychophysical manifestations of compression: hearing-impaired listeners. In: *Compression: From Cochlea to Cochlear Implants*. New York, NY: Springer; 2004:107-152: Sid P. Bacon, Richard R. Fay, and Arthur N. Popper, eds.
23. Anderson H, Barr B, Wedenberg E. Early diagnosis of vthith—nerve tumours by acoustic reflex tests. *Acta Otolaryngol*. 1970;69(suppl 263):232-237.
24. Jerger J, Harford E, Clemis J, Alford B. The acoustic reflex in eighth nerve disorders. *Arch Otolaryngol*. 1974;99(6):409-413.
25. Olsen WO, Noffsinger D, Kurdziel S. Acoustic reflex and reflex decay: occurrence in patients with cochlear and eighth nerve lesions. *Arch Otolaryngol*. 1975;101(10):622-625.
26. Turner RG, Shepard NT, Frazer GJ. Clinical performance of audiological and related diagnostic tests. *Ear Hear*. 1984;5(4):187-194.

Section 4

AUDIOGRAM WORKBOOK

Quick Reference Guide to the Method

Basic Audiograms

Intermediate Audiograms

Advanced Audiograms

Introduction

In this section, you will hone your audiogram interpretation skills using the method outlined in the text. We have constructed a wide variety of audiograms from many different hearing loss states and diseases. While not exhaustive, we hope that by practicing the examples here, you will build a strong foundation for interpretation.

The audiograms have been fabricated from real cases encountered by the authors. However, the data have been generalized so that patient-specific information is not represented. Patient ages and short clinical prompts have been furnished to help guide the context of each audiogram example. Every audiogram is accompanied by an official interpretation on the adjoining page. You will notice that each step in the method is represented in the interpretation provided. For some audiograms, portions of the testing may not be necessary or indicated, as in real clinical practice. Thus, on some examples, nonessential information may not be provided. This will be indicated in the interpretation guide.

To help you understand additional clinical context for the presented audiograms, supplementary clinical information may be supplied at the bottom of the interpretation. Also at the bottom of each interpretation, there will be a summary assessment with the resultant diagnosis derived from the presented audiometric data. Plan to cross-check this assessment with your suspected diagnosis to ensure they corroborate. Finally, each audiogram example

will list recommendations for further clinical evaluation and/or therapeutic interventions that can be considered after the assessment is realized. These recommendations are not meant to be exhaustive. They are included only for the reader to see what additional clinical measures may be considered once the audiogram is accomplished.

We have sorted the examples by difficulty to allow you to proceed to more advanced audiograms as you become comfortable with your interpretation skills. As such, we strongly encourage you to start from the beginning and progress stepwise. For more experienced clinicians, we urge you to use the intermediate and advanced examples to review key concepts or increase your interpretation proficiency.

Multiple blank interpretation templates have been provided at the end of the book to facilitate your practice. Fill these out prior to referencing the adjoining interpretations to help your skills develop. Also, on the next page, you will find an expanded reference guide to the 8-step interpretation method. Refer to this summary frequently to help you construct thorough interpretations of each audiogram. Finally, at the end of this section (Appendix A), a list of the audiogram examples organized by patient diagnosis has been provided. As you encounter cases or patients with similar diagnoses, this reference and the associated audiogram examples can help guide your clinical care.

We hope this section will become a wellspring of experience for you, as you increase your mastery of this skill. We wish you great success in growing and refining your abilities. Best of luck!

QUICK REFERENCE GUIDE TO THE METHOD

****Review patient's chief complaint and relevant history.****

- 1. Assess for hearing loss:** Is hearing loss present and to what degree?
 - a.** Measure severity of hearing loss (see Figure 1–4).
 - b.** Calculate the pure-tone average (PTA).
 - c.** Check for asymmetry.
- 2. Determine the type of hearing loss:** Is conductive, sensorineural, or mixed loss present?
 - a.** Calculate the air-bone gap.
 - b.** Determine if masking used.
 - c.** Check for any masking dilemmas.
 - d.** Was a Stenger test performed?
- 3. Check if audiogram fits a classic pattern**
 - a.** See Figures 1–5 through 1–16.
- 4. Compare speech thresholds to pure-tone audiometry**
 - a.** Determine if speech recognition threshold (SRT) or speech detection threshold (SDT) was performed.
 - b.** Check for consistency with the PTA.
- 5. Assess word recognition score (WRS) (see Figure 2–3)**
 - a.** Conductive hearing loss: Good WRS
 - b.** Cochlear hearing loss: Reduced or poor WRS
 - c.** Retrocochlear hearing loss: Poor WRS
 - d.** Check for rollover phenomenon.
- 6. Determine tympanometry type (see Figure 3–1)**
 - a.** Types:
 - A: Normal
 - Ad: Ossicular hypermobility, healed TM perforation
 - As: Ossicular fixation
 - B: Middle ear pathology, TM perforation, PET
 - C: Eustachian tube dysfunction, negative middle ear pressure
 - b.** Evaluate middle ear pressure and compliance.
 - c.** Ensure consistency with the pure-tone audiometry.
- 7. Check ear canal volume (ECV or PVT)**
 - a.** Normal ECV: 0.3 to 1.75
 - i.** TIP: Consider PET or perforated tympanic membrane with large ECVs.
- 8. Review acoustic reflexes (see Figure 3–8)**
 - a.** Normal acoustic reflexes from 80 to 105 dB
 - b.** Absent ipsilateral reflexes with middle ear, cochlear, retrocochlear pathology
 - c.** Contralateral reflexes may or may not be present depending on location of pathology.
 - d.** Check if acoustic decay testing done (if +, probable retrocochlear lesion).
 - e.** Confirm consistency with pure-tone data.

BASIC AUDIOGRAMS

Audiogram 1: Basic

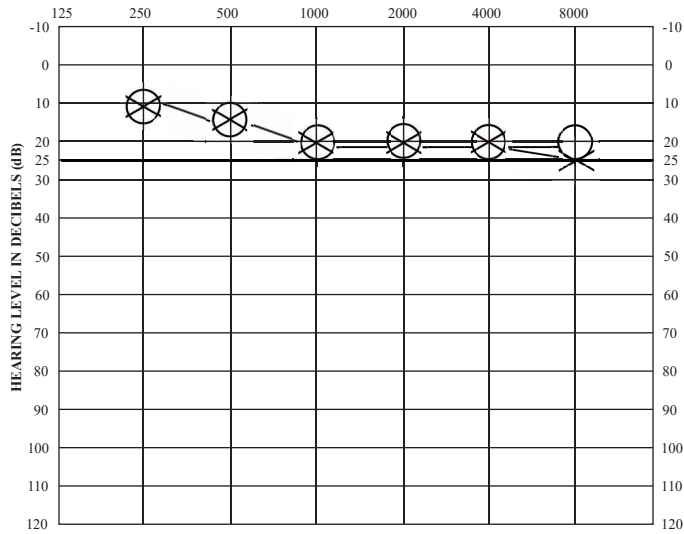
NAME: 1

DATE: _____

BY: _____

AGE: 58

PURE TONE AUDIOGRAM
FREQUENCY IN HERTZ (Hz)



SPEECH AUDIOMETRY



RECORDED



MLV

EAR	SRT	SDT	WRS	LEVEL
R	20		100%	60 dB HL
L	25		100%	65 dB HL

TRANSDUCER:

☐ HEADPHONES

☒ INSERTS

☐ SOUNDFIELD

TEST METHOD:

Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED			^	
SOUND FIELD		S		
AIDED		A		
OTHER TESTS MASKED	*			

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	- 50	- 45
P.V.T	1.2	1.3
COMPLIANCE	0.9	0.7

Reflex Threshold

	Contralateral				Ipsilateral		
	5K	1K	2K	4K	5K	1K	2K
RE							
LE							

REFLEX DECAY:

RT.

LT.

REFERRED BY: _____

HISTORY:

Patient c/o minimal difficulty hearing.

COMMENTS:

Audiogram 1: Basic—Interpretation

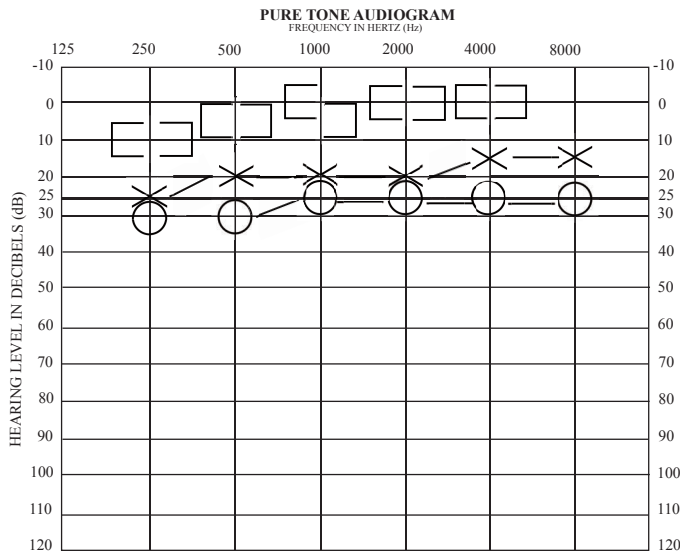
Audiogram: ____1____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Normal hearing PTA: RE, 18 dB; LE, 18 dB No asymmetry
Determine the type of hearing loss	Normal hearing (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Normal hearing, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally.
Determine tympanometry type (Figure 3–1)	Type A tympanograms, bilaterally
Check ear canal volume	Normal ECV
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Normal hearing sensitivity with normal tympanograms, bilaterally.

Recommendations: Follow up as needed.

Audiogram 2: Basic

NAME: 2 DATE: _____ BY: _____Age: 60

SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	30		92%	70 dB HL
L	25		100%	60 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
 TEST METHOD: ☐ SOUNDFIELD Std.

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING				
Tympanometry				
		RIGHT	LEFT	
TYMpanoGRAM TYPE		As	A	
MIDDLE EAR PRESSURE		0	0	
P.V.T		1.8	1.9	
COMPLIANCE		0.2	0.9	

Reflex Threshold									
	Contralateral					Ipsilateral			
		.5K	1K	2K	4K		.5K	1K	2K
RE									
LE									

REFLEX DECAY: RT. _____

LT. _____

REFERRED BY: _____

HISTORY: Patient presents with a history of frequent ear infections as a young child. He is here today for his annual audiologic evaluation. Denies tinnitus, otalgia, vertigo, and excessive noise exposure.

COMMENTS: _____

Otoscopy: calcification (tympanosclerosis) noted on TM, AU

Audiogram 2: Basic—Interpretation

Audiogram: 2

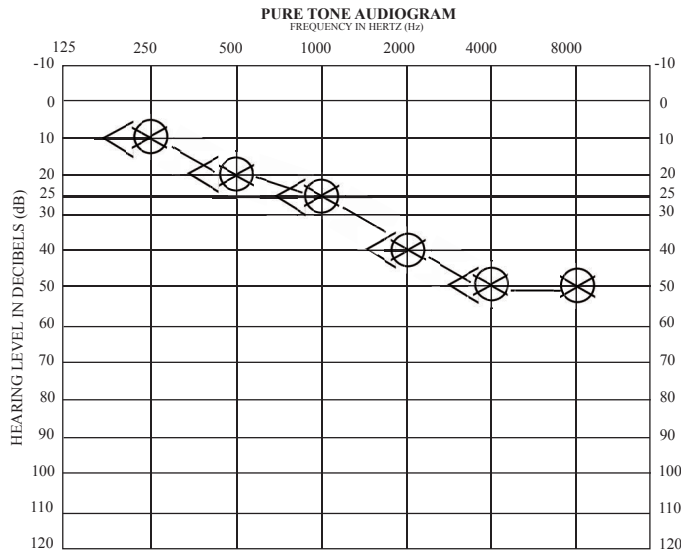
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Mild HL Left Ear: Normal hearing PTA: RE, 27 dB; LE, 20 dB No significant asymmetry
Determine the type of hearing loss	Conductive (bilateral) - Hearing is within normal limits for the left ear however there is an air-bone gap present.
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Bilateral conductive HL
Compare speech thresholds to pure tones	SRT is in agreement with PTs
Assess word recognition score (Figure 2–3)	Right Ear: Excellent Left Ear: Excellent
Determine tympanometry type (Figure 3–1)	Right Ear: Type As tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Right Ear: Normal ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	DNT

Assessment: Mild conductive hearing loss in the right ear and essentially normal hearing in the left ear, with 15 to 20 dB air-bone gaps throughout the frequency range. Air-bone gaps are likely due to tympanosclerosis.

Recommendations: No intervention from ENT standpoint. Observation and follow-up as needed.

Audiogram 3: Basic

NAME: 3 DATE: _____ BY: _____

Age: 72

SPEECH AUDIOMETRY
☒ RECORDED

☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	20		92%	60 dB HL
L	25		92%	65 dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD

TEST METHOD: Std.

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING
Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	15	10
P.V.T	0.9	0.9
COMPLIANCE	0.5	0.5

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
LT. _____

REFERRED BY: _____

HISTORY: Patient c/o hearing difficulty and difficulty hearing on telephone. Denies aural drainage, tinnitus, and vertigo.

COMMENTS: _____

Audiogram 3: Basic—Interpretation

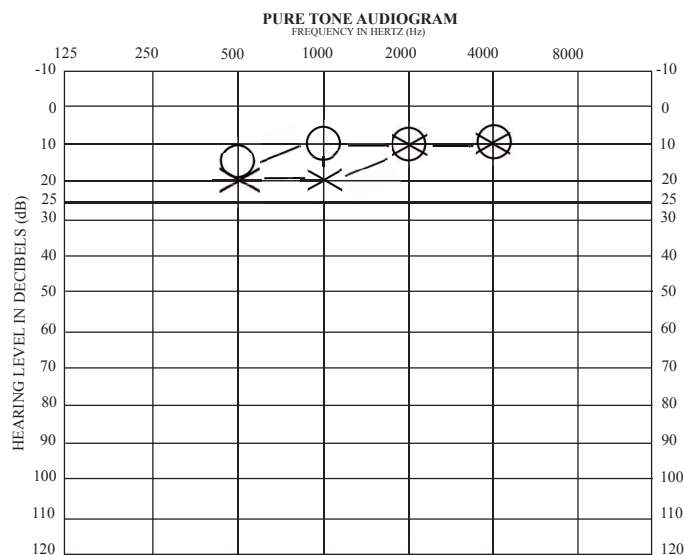
Audiogram: ____3____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Normal hearing sloping to moderate HL PTA: RE, 28 dB; LE, 28 dB No asymmetry
Determine the type of hearing loss	Sensorineural (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Sensorineural HL, bilaterally.
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally.
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Right Ear: Normal ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Normal hearing sloping to a moderate sensorineural hearing loss, bilaterally.

Recommendations: Consider amplification for both ears. ENT referral for medical clearance.

Audiogram 4: Basic

NAME: 4 DATE: _____ BY: _____Age: 3

SPEECH AUDIOMETRY ☐ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R		10		
L		15		
SF		20		

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☒ SOUNDFIELD
 TEST METHOD: VRA/CPA

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD		S		
AIDED		A		
OTHER TESTS MASKED		*		

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	B	C
MIDDLE EAR PRESSURE	---	-145
P.V.T	3.8	0.5
COMPLIANCE	---	0.3

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: ENTHISTORY: Hx of pressure equalization (PE) tubes, AUCOMMENTS: Patient's parent denies aural drainage and hearing concerns. Patient is congested today.

Otосcopy:

Right Ear - PET visualizedLeft Ear - PET extruded in ear canal

Audiogram 4: Basic—Interpretation

Audiogram: 4

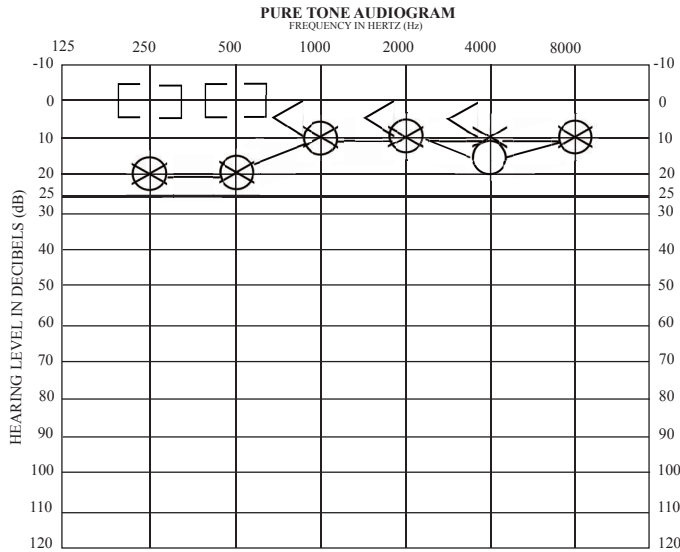
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Normal hearing, bilaterally PTA: RE, 12 dB; LE, 17 dB No significant asymmetry
Determine the type of hearing loss	Undetermined
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Normal hearing, bilaterally
Compare speech thresholds to pure tones	SDT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	N/A
Determine tympanometry type (Figure 3–1)	Right Ear: Type B tympanogram Left Ear: Type C tympanogram
Check ear canal volume	Right Ear: Large ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Hearing within normal limits, bilaterally, with a patent PET in the right ear and an extruded PET in the left ear. Left ear tympanogram suggests negative middle ear pressure and is consistent with audiometric results and reports of congestion.

Recommendations: Follow up with ENT. ENT may consider observation for left ear given normal hearing or replace left PET given type C tympanogram.

Audiogram 5: Basic

NAME: 5 DATE: _____ BY: _____
 Age: 4



SPEECH AUDIOMETRY				
			☐ RECORDED	☒ MLV
EAR	SRT	SDT	WRS	LEVEL
R	15		100%	50 dB HL
L	15		100%	50 dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD
 TEST METHOD: CPA

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	B	B
MIDDLE EAR PRESSURE	---	---
P.V.T	0.5	0.5
COMPLIANCE	---	---

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE						NR	
LE						NR	

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: 4-year-old patient presents with a history of chronic otitis media in both ears. Patient's mother reports patient is congested today.

COMMENTS: _____

Otoscopy:

Right Ear - clear canal

Left Ear - clear canal

Audiogram 5: Basic—Interpretation

Audiogram: 5

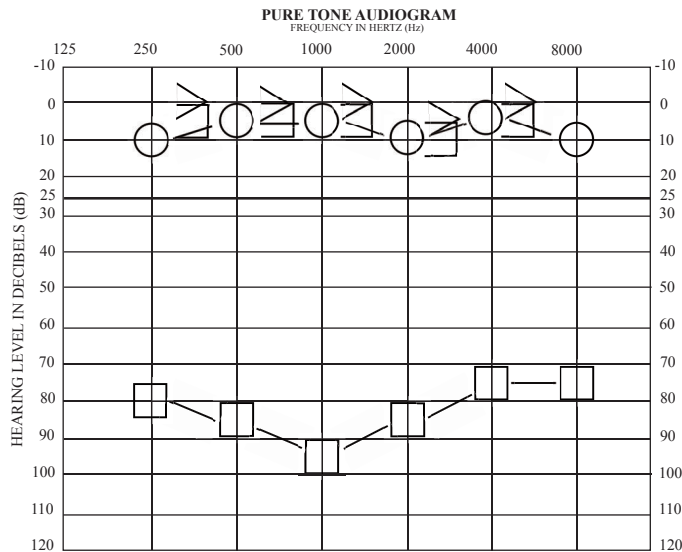
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Normal hearing with 20- to 25-dB air-bone gap at 250 and 500 Hz, bilaterally PTA: RE – 13 dB, LE – 13 dB No asymmetry
Determine the type of hearing loss	Conductive HL (low frequencies)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Conductive component
Compare speech thresholds to pure tones	SRTs are in agreement with PTs.
Assess word recognition score (Figure 2–3)	Excellent, bilaterally
Determine tympanometry type (Figure 3–1)	Type B tympanograms, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Ipsilateral reflexes are absent at 1000 Hz, bilaterally.

Assessment: Normal hearing sensitivity with 20 to 25 dB air-bone gap at 250 and 500 Hz, bilaterally. Low-frequency conductive component is consistent with abnormal tympanograms, suggesting middle ear pathology in both ears.

Recommendations: Follow up with ENT as scheduled. ENT to discuss with parent regarding patient's speech and hearing abilities. May opt to observe to see if self-resolution occurs. However, if patient presents with chronic infections and/or speech is affected, then PETs are likely to be recommended.

Audiogram 6: Basic

 NAME: 6 DATE: _____ BY: _____

 Age: 13

SPEECH AUDIOMETRY
☒ **RECORDED**
☐ **MLV**

EAR	SRT	SDT	WRS	LEVEL
R	0		100%	50 dB HL
L	80*		86%	110* dB HL

 TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD

 TEST METHOD: Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	B
MIDDLE EAR PRESSURE	0	---
P.V.T	0.8	6.5
COMPLIANCE	0.9	---

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

 REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

 HISTORY: Patient presents with a history of left-sided conductive hearing loss due to frequent PET placements and tympanic membrane perforation. Patient currently wears amplification for the left ear. Patient denies otalgia and dizziness.

COMMENTS: _____

Otoscopy:

Right Ear: clear canal

Left Ear: TM perforation

Audiogram 6: Basic—Interpretation

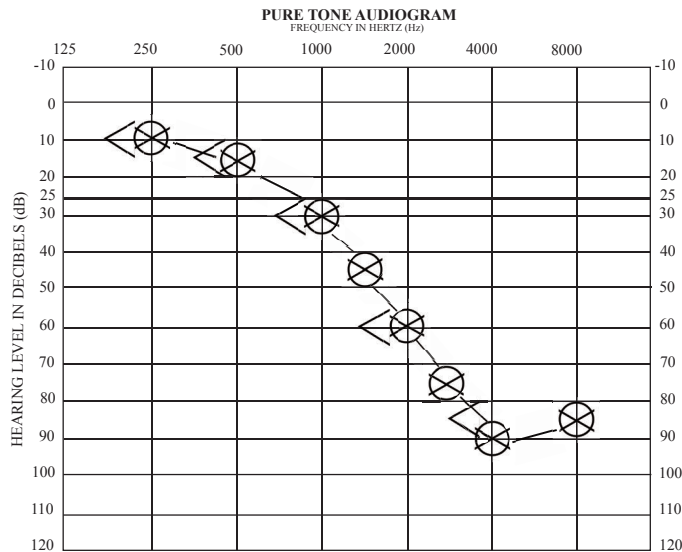
Audiogram: ____6____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Normal hearing Left Ear: Severe to profound HL PTA: RE, 6 dB; LE, 88 dB Significant asymmetry
Determine the type of hearing loss	Right Ear: Normal hearing Left Ear: Conductive
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral conductive HL
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Excellent Left Ear: Good
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type B tympanogram
Check ear canal volume	Right Ear: Normal ECV Left Ear: Large ECV
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Normal hearing sensitivity with a normal tympanogram in the right ear and a severe to profound conductive hearing loss with a flat tympanogram with a large ECV in the left ear (also known as a maximally conductive hearing loss), consistent with TM perforation.

Recommendations: Continue full-time hearing aid use. Return for annual audiologic evaluation. ENT may opt to offer closure of TM given normal status of right ear. Otherwise, observation and revisiting surgery option in a few years.

Audiogram 7: Basic

NAME: 7 DATE: _____ BY: _____Age: 80

SPEECH AUDIOMETRY

☒ RECORDED☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	40		88%	80 dB HL
L	40		88%	80 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
 TEST METHOD: ☐ SOUNDFIELD Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	1.0	1.1
COMPLIANCE	0.5	0.6

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE						NR	NR
LE						NR	NR

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient's spouse c/o increased difficulty hearing female voices and in background noise. Patient reports intermittent tinnitus. Denies otalgia, vertigo, and excessive noise exposure.

COMMENTS: _____

Audiogram 7: Basic—Interpretation

Audiogram: ____7____

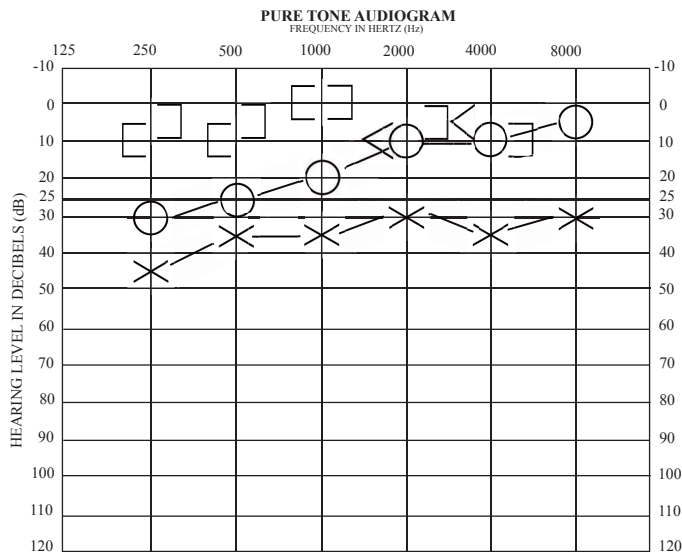
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Normal hearing sloping to a severe high-frequency HL, bilaterally PTA: RE, 35 dB; LE, 35 dB No asymmetry
Determine the type of hearing loss	Sensorineural (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	High-frequency sensorineural hearing loss, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is good, bilaterally.
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Ipsilateral: Right Ear—No response Left Ear—No response

Assessment: Normal sloping to severe high-frequency sensorineural hearing loss with normal tympanograms, bilaterally.

Recommendations: Consider amplification. Return to audiology clinic annually for monitoring purposes.

Audiogram 8: Basic

NAME: 8 DATE: _____ BY: _____
 Age: 6



SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	20		100%	50 dB HL
L	40		100%	70* dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD ☐ CPA

TEST METHOD: _____

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	B	B
MIDDLE EAR PRESSURE	---	---
P.V.T	6.0	0.5
COMPLIANCE	---	---

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents to the clinic with a history of pressure equalization tubes (PET).

COMMENTS: _____

Otосcopy:

Right Ear: PET visualized

Left Ear: PET visualized with cerumen in ear canal

Audiogram 8: Basic—Interpretation

Audiogram: 8

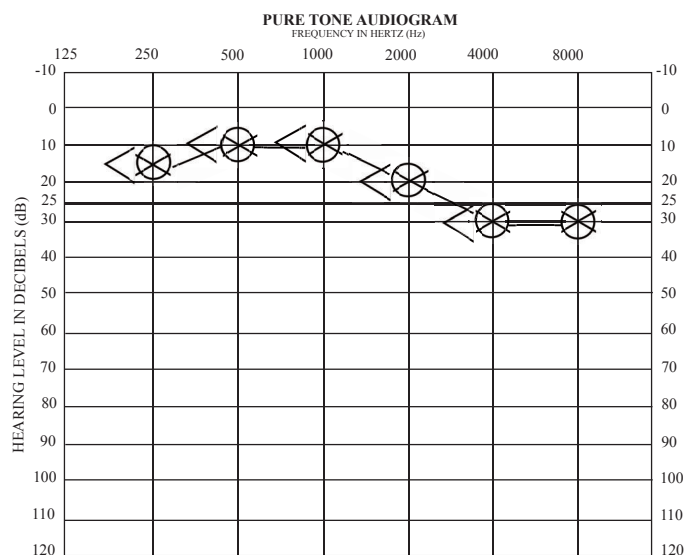
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Mild low-frequency HL rising to normal hearing Left Ear: Mild HL PTA: RE, 18 dB; LE, 33 dB Asymmetry
Determine the type of hearing loss	Conductive (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Conductive HL, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Excellent Left Ear: Excellent
Determine tympanometry type (Figure 3–1)	Right Ear: Type B tympanogram Left Ear: Type B tympanogram
Check ear canal volume	Right Ear: Large ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Testing indicated a low-frequency conductive hearing loss rising to normal hearing in the right ear and a mild conductive hearing loss in the left ear. Results are consistent with a patent PET in the right ear and probable middle ear pathology or obstructed PET in the left ear.

Recommendations: Follow up with ENT regarding probable middle ear pathology in left ear, for example, consider replacement of PET. Return to audiology clinic after medical management.

Audiogram 9: Basic

NAME: 9 DATE: _____ BY: _____

Age: 40


SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	25		92%	50 dB HL
L	25		92%	50 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD Std.

TEST METHOD: _____

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED	*			

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	-30	-60
P.V.T	0.6	0.6
COMPLIANCE	0.4	0.4

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE					85	90	90
LE					95	90	95

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents history of an ear infection 6 months ago. C/o minimal difficulty hearing. Intermittent tinnitus.
Denies otalgia, vertigo, and excessive noise exposure.

COMMENTS: _____

Audiogram 9: Basic—Interpretation

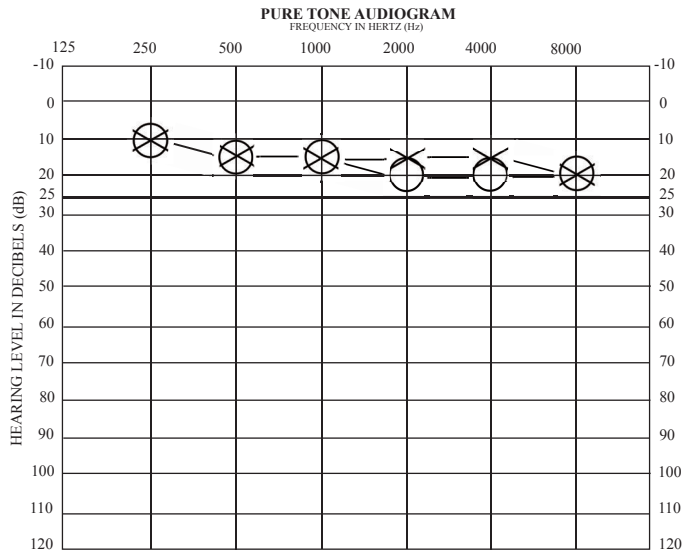
Audiogram: ____9____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Normal sloping to mild high-frequency HL, bilaterally PTA: RE, 13 dB; LE, 13 dB No asymmetry
Determine the type of hearing loss	Sensorineural (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Sensorineural HL, bilaterally.
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is good, bilaterally.
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Ipsilateral: Right Ear—Present at 500, 1000, & 2000 Hz Left Ear—Present at 500, 1000, & 2000 Hz

Assessment: Normal hearing sloping to a mild high-frequency sensorineural hearing loss with normal tympanograms, bilaterally.

Recommendations: Amplification is not recommended at this time and no ENT referral is necessary. Return to audiology clinic annually for monitoring purposes.

Audiogram 10: Basic

NAME: 10 DATE: _____ BY: _____Age: 80

SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	15		100%	45
L	15		100%	45

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD

TEST METHOD: Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↗	
UNSPECIFIED			△	
SOUND FIELD		S		
AIDED		A		
OTHER TESTS MASKED	*			

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	25	15
P.V.T	0.8	0.9
COMPLIANCE	0.7	0.9

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE		105	95			85	90
LE		95	90			85	90

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: NEW PATIENT: Patient c/o tinnitus

COMMENTS: _____

Audiogram 10: Basic—Interpretation

Audiogram: ____10____

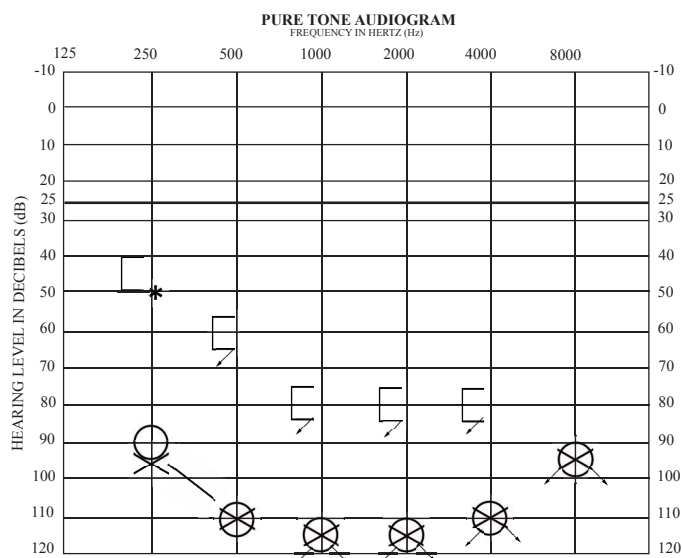
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Normal hearing, bilaterally PTA: RE, 16 dB; LE, 15 dB No asymmetry
Determine the type of hearing loss	Normal hearing (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Normal hearing, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally.
Determine tympanometry type (Figure 3–1)	Type A tympanograms, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Both ipsilateral and contralateral reflexes are present at 1000 and 2000 Hz, bilaterally.

Assessment: Normal hearing sensitivity with normal tympanograms, bilaterally.

Recommendations: Consider referral to ENT or expert audiologist to discuss sound therapy for tinnitus complaints.

Audiogram 11: Basic

NAME: 11 DATE: _____ BY: _____
Age: 20



SPEECH AUDIOMETRY			☐ RECORDED	☐ MLV
EAR	SRT	SDT	WRS	LEVEL
R		NR		
L		NR		

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
TEST METHOD: ☐ SOUNDFIELD Std.

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED			△	
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED	*			

IMMITTANCE TESTING		
Tympanometry		
	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	1.0	0.9
COMPLIANCE	0.3	0.5

Reflex Threshold							
	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: _____
RT. _____

LT. _____

REFERRED BY:

HISTORY: 20-year-old patient with known bilateral profound hearing loss here for her annual audiology exam. Patient wears hearing aids and is fluent in American Sign Language (ASL).

COMMENTS: _____

Audiogram 11: Basic—Interpretation

Audiogram: ____11____

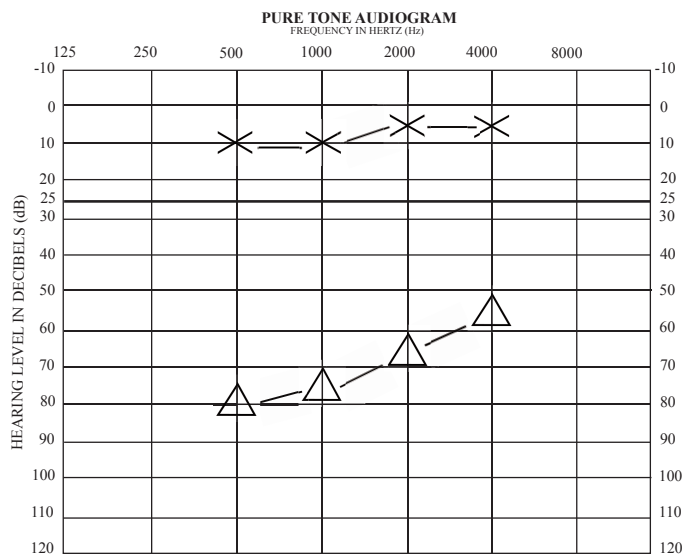
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Profound HL, bilaterally
Determine the type of hearing loss	Sensorineural (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Sensorineural HL, bilaterally
Compare speech thresholds to pure tones	SDT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	DNT
Determine tympanometry type (Figure 3–1)	Type A tympanograms, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	DNT

Assessment: Testing reveals a bilateral profound sensorineural hearing loss.

Recommendations: Return for annual audiologic evaluations. Continue full-time hearing aid use. Patient could be recommended for a cochlear implant evaluation. However, it is important to be respectful of Deaf culture beliefs and understand the patient may not be interested in a cochlear implant for those reasons.

Audiogram 12: Basic

NAME: 12 DATE: _____ BY: _____
 Age: 6



SPEECH AUDIOMETRY ☐ RECORDED ☒ MLV

EAR	SRT	SDT	WRS	LEVEL
R	65		92%	95* dB HL
L	5		100%	50 dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD
 TEST METHOD: CPA

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED	*			

IMMITTANCE TESTING						
Tympanometry						
				RIGHT	LEFT	
TYMPANOGRAM TYPE					A	
MIDDLE EAR PRESSURE					10	
P.V.T					0.8	
COMPLIANCE					0.4	

Reflex Threshold							
	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____

LT. _____

REFERRED BY: _____

HISTORY: Patient presents with right-sided microtia. He is seen today for his annual audiologic evaluation. Patient currently wears a Bone-Anchored Hearing System (BAHS) with softband. Previous hearing test indicates normal hearing in the left ear and a maximal conductive hearing loss in the right ear, consistent with microtia.

COMMENTS: _____

Audiogram 12: Basic—Interpretation

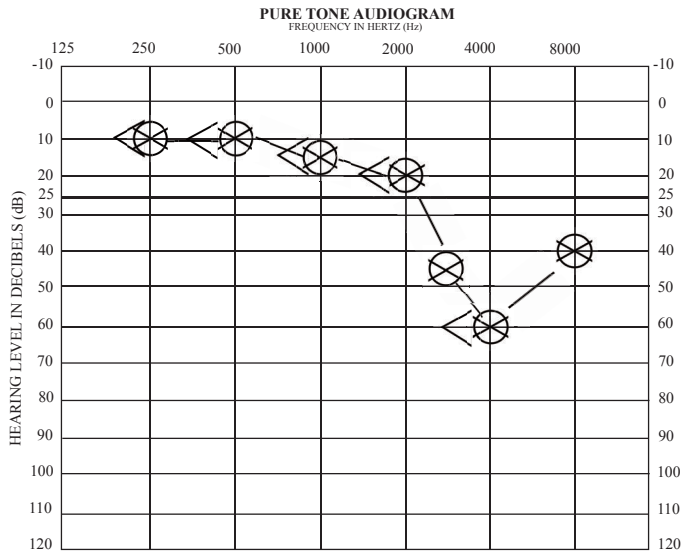
Audiogram: ____12____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Severe to moderate HL Left Ear: Normal hearing PTA: RE, 73 dB; LE, 8 dB Significant asymmetry
Determine the type of hearing loss	Undetermined Previous test indicates unilateral conductive HL; consistent with microtia
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral HL
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally.
Determine tympanometry type (Figure 3–1)	Right Ear: DNT Left Ear: Type A tympanogram
Check ear canal volume	Right Ear: DNT Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Severe rising to moderate hearing loss in the right ear and normal hearing sensitivity in the left ear with a normal tympanogram. (Until masked bone conduction can be completed, it is unknown if HL is purely conductive or mixed.) Previous testing suggests hearing loss to be conductive.

Recommendations: Return to audiology clinic for annual audiologic evaluation. Follow up with managing audiologist for hearing aid care. Consider surgical BAHS consultation and implant placement (children are eligible for surgical BAHS starting at age 5).

Audiogram 13: Basic

NAME: 13 DATE: _____ BY: _____Age: 32**SPEECH AUDIOMETRY**☒ RECORDED☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	25		96%	65 dB HL
L	25		92%	65 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD Std.

TEST METHOD: _____

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	1.2	1.2
COMPLIANCE	0.5	0.5

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE					90	NR	NR
LE					90	NR	NR

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents to the clinic with a history of active shooting and military history with heavy artillery.
Patient denies aural drainage, tinnitus, and vertigo.

COMMENTS: _____

Audiogram 13: Basic—Interpretation

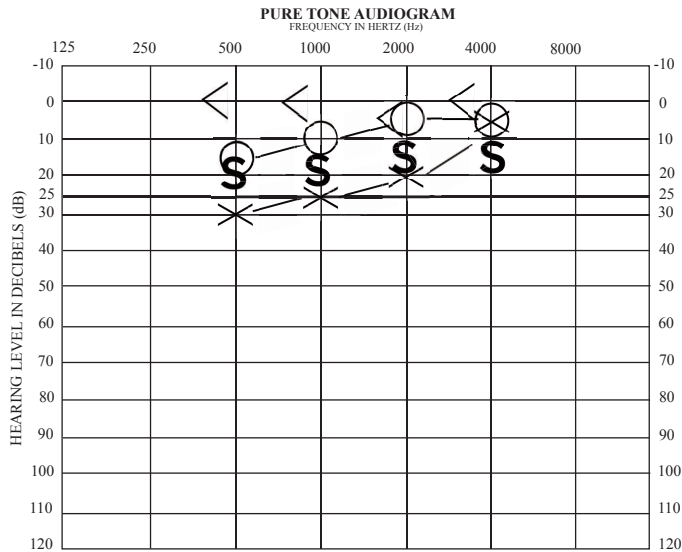
Audiogram: ____13____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Normal sloping to moderately severe HL PTA: RE, 15 dB; LE, 15 dB No asymmetry
Determine the type of hearing loss	Sensorineural (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Sensorineural HL, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Excellent Left Ear: Excellent
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Right Ear: Normal ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	Right Ipsilateral: No response at 1 and 2 K Left Ipsilateral: No response at 1 and 2 K

Assessment: Testing indicates normal hearing sloping to a moderately severe sensorineural hearing loss with a noise notch noted at 4000 Hz, bilaterally.

Recommendations: Follow up with ENT. Consider amplification options if experiencing hearing difficulties. Recommend hearing protection devices for the future. Annual audiologic evaluations for monitoring purposes.

Audiogram 14: Basic

NAME: 14 DATE: _____ BY: _____Age: 3

SPEECH AUDIOMETRY ☐ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R		10		
L		15		
SF		15		

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☒ SOUNDFIELD
 TEST METHOD: CPA

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	B	
MIDDLE EAR PRESSURE	---	
P.V.T	4.8	DNT
COMPLIANCE	---	

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: 3-year-old patient with a hx of pressure equalization (PE) tubes in both ears. Patient is here today for his PET check.
He currently has a left-sided ear infection, which is being treated with antibiotics.

COMMENTS: _____

Otoscopy

Right Ear: PET visualized

Left Ear: Otorrhea

Audiogram 14: Basic—Interpretation

Audiogram: 14

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Normal hearing Left Ear: Mild rising to normal hearing PTA: RE, 10 dB; LE, 25 dB Asymmetric
Determine the type of hearing loss	Conductive (unilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral conductive HL
Compare speech thresholds to pure tones	SDT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	N/A
Determine tympanometry type (Figure 3–1)	Right Ear: Type B tympanogram Left Ear: Did not test due to excessive drainage
Check ear canal volume	Right Ear: Large ECV Left Ear: DNT
Review acoustic reflexes (Figure 3–8)	N/A

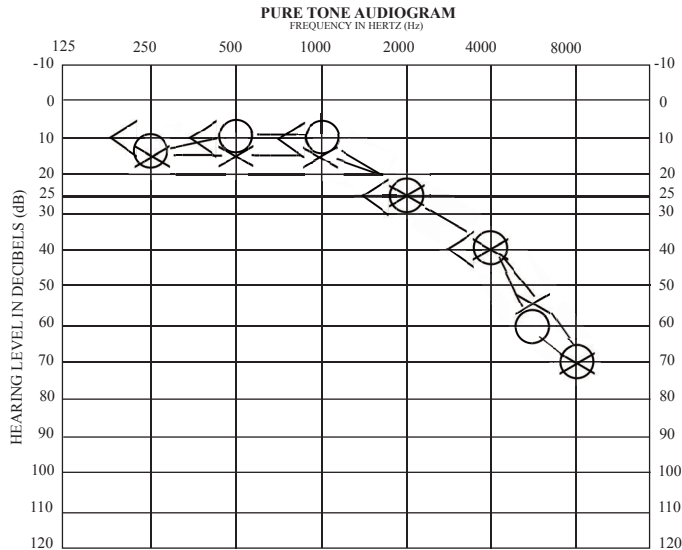
Assessment: Sound-field responses indicate normal hearing in the better hearing ear. Right ear indicates normal hearing with a patent PET. The left ear indicates a mild likely conductive HL rising to normal hearing, consistent with middle ear pathology.

Recommendations: Follow up with ENT for repeat microscopic examination and debridement to determine status of tube in left ear. Patient to continue antibiotics to assist with resolution. Return to audiology clinic after medical management.

Audiogram 15: Basic

NAME: 15 DATE: BY:

Age: 85



SPEECH AUDIOMETRY				
EAR	SRT	SDT	WRS	LEVEL
R	25		84%	65 dB HL
L	25		88%	65 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD

TEST METHOD: Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD		S		
AIDED		A		
OTHER TESTS MASKED		*		

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	1.5	1.3
COMPLIANCE	0.8	0.7

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE						85	
LE						85	

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient c/o hearing difficulties in background noise. Denies excessive noise exposure, tinnitus, dizziness, and otalgia. No previous hearing aid use.

COMMENTS: _____

Audiogram 15: Basic—Interpretation

Audiogram: ____15____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Moderately severe high-frequency hearing loss, bilaterally PTA: RE, 15 dB; LE, 18 dB No asymmetry
Determine the type of hearing loss	Sensorineural (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	High-frequency sensorineural HL, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Good Left Ear: Good
Determine tympanometry type (Figure 3–1)	Type A tympanograms, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Ipsilateral reflexes were present, bilaterally.

Assessment: Normal hearing sloping to a moderately severe high-frequency sensorineural hearing loss with normal tympanograms, bilaterally. Hearing loss likely due to presbycusis.

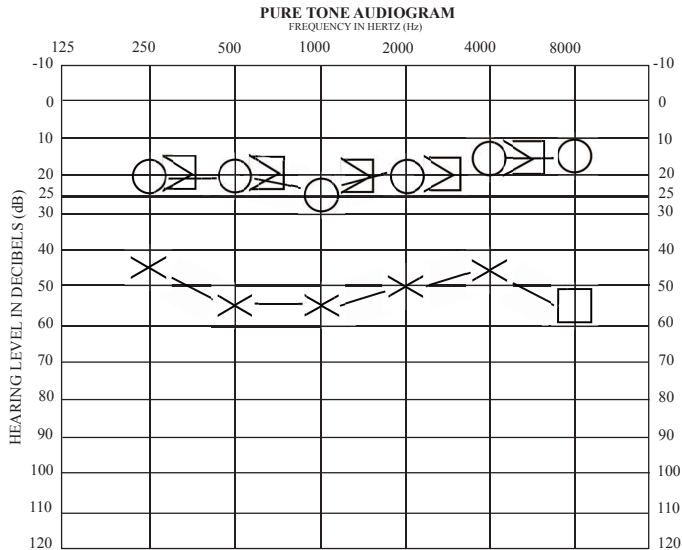
Recommendations: Consider bilateral amplification; referral to ENT for medical clearance.



INTERMEDIATE AUDIOGRAMS



Audiogram 16: Intermediate

NAME: 16 DATE: _____ BY: _____Age: 56

SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	20		100%	60
L	50		100%	90

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD

TEST METHOD: Std.

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	B
MIDDLE EAR PRESSURE	0	---
P.V.T	1.0	7.0
COMPLIANCE	0.4	---

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: ENT

HISTORY: Patient presents with a history of left-sided tympanic membrane perforation. Patient denies otalgia, otorrhea, and dizziness. Reports difficulty hearing, especially in the left ear.

COMMENTS: _____

Audiogram 16: Intermediate—Interpretation

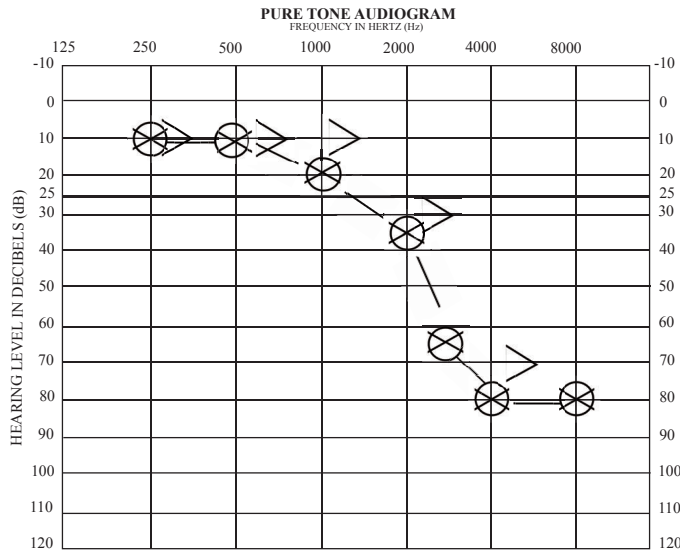
Audiogram: ____16____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right: Normal hearing Left: Mild to moderate HL PTA: RE, 22 dB; LE, 53 dB Asymmetric
Determine the type of hearing loss	Right: Normal Left: Conductive
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral conductive HL
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally.
Determine tympanometry type (Figure 3–1)	Right: Type A tympanogram Left: Type B tympanogram
Check ear canal volume	Right: Normal ECV Left: Large ECV
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Normal hearing with a type A normal tympanogram in the right ear and a mild to moderate conductive hearing loss with a type B tympanogram with a large ECV in the left ear. Left ear results are consistent with TM perforation.

Recommendations: ENT referral to discuss options to observe with use of amplification to assist affected ear versus attempt closure with tympanoplasty. Return to audiology clinic after medical management for repeat hearing test.

Audiogram 17: Intermediate

NAME: 17 DATE: _____ BY: _____Age: 60

SPEECH AUDIOMETRY ☐ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	30		80%	70 dB HL
L	30		84%	70 dB HL

DPOAE:

Right Ear: Present 1 - 1.5 Absent 2 - 8 kHz

Left Ear: Present 1 - 1.5 Absent 2 - 8 kHz

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELDTEST METHOD: Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	1.1	1.0
COMPLIANCE	0.5	0.5

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE					90	105	NR
LE					90	100	NR

REFLEX DECAY: RT. _____
LT. _____

REFERRED BY: _____

HISTORY: 60-year-old patient presents with a identified colon cancer and is receiving ongoing chemotherapy treatment (cisplatin). Patient and his caregiver deny aural drainage and ear infections. They reported some dizziness and tinnitus.

COMMENTS: _____

Audiogram 17: Intermediate—Interpretation

Audiogram: ____17____

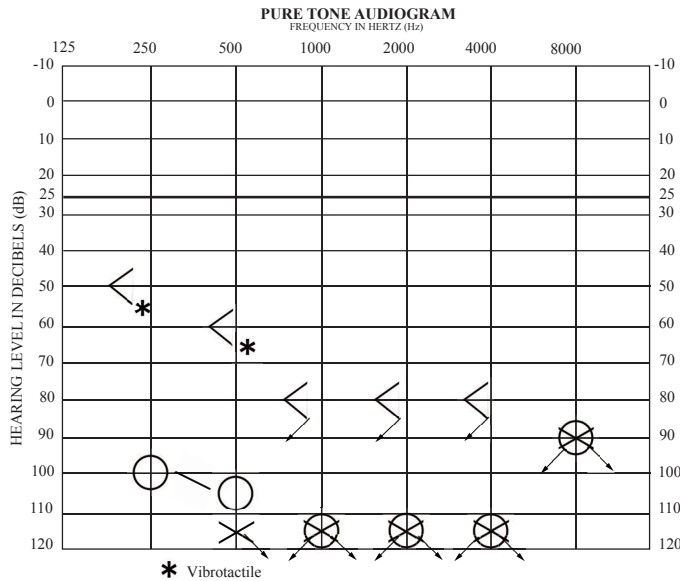
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Normal hearing sloping to a severe HL, bilaterally PTA: RE, 22 dB; LE, 22 dB Symmetric
Determine the type of hearing loss	Sensorineural (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Sensorineural HL, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is good, bilaterally.
Determine tympanometry type (Figure 3–1)	Type A tympanograms, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Absent ipsilateral acoustic reflexes at 2000 Hz in both ears; consistent with pure-tone findings

Additional Testing: DPOAEs indicate present responses from 1 to 1.5 and absent from 2 to 8 kHz in both ears. DPOAE results are consistent with pure-tone findings.

Assessment: Normal hearing precipitously sloping to a severe sensorineural hearing loss with normal tympanograms, bilaterally.

Recommendations: Return to audiology clinic per oncology protocol. During ongoing chemotherapy, patient should consider amplification for both ears to assist with communication.

Audiogram 18: Intermediate

NAME: 18 DATE: _____ BY: _____Age: 3

SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R		105	0/10	110 dB HL
L		110	0/10	110 dB HL

DPOAEs absent 1 - 8 kHz, bilaterally.

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
 TEST METHOD: ☐ SOUNDFIELD ☒ Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED	*			

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	-5
P.V.T	0.5	0.4
COMPLIANCE	0.3	0.3

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE						NR	
LE						NR	

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: 3-year-old patient acquired bacterial meningitis and is here for his audiogram. He passed his newborn hearing screening in both ears. Per parent report he is no longer responding to his name.

COMMENTS: _____

Audiogram 18: Intermediate—Interpretation

Audiogram: ____18____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Bilateral profound HL PTA: RE, 105 dB, LE, 105 dB No significant asymmetry
Determine the type of hearing loss	Sensorineural (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Anacusis
Compare speech thresholds to pure tones	SDT in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is poor in both ears.
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	No ipsilateral response in both ears at 1000 Hz

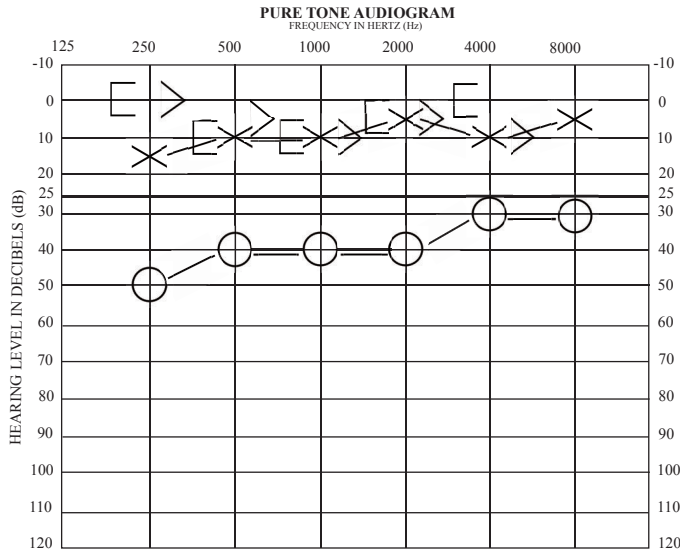
Assessment: Testing reveals a profound sensorineural hearing loss, bilaterally. Absent DPOAE and acoustic reflexes confirm this diagnosis.

Recommendations: Urgent referral to ENT and cochlear implant audiologist to consider cochlear implant. Likely to be a candidate for simultaneous bilateral implantation.

Audiogram 19: Intermediate

NAME: 19 DATE: BY:

Age: 15



SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	40		100%	80 dB HL
L	10		100%	50 dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD
 TEST METHOD: Std.

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	C	A
MIDDLE EAR PRESSURE	-360	0
P.V.T	0.5	0.7
COMPLIANCE	0.7	0.7

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE					NR	NR	NR
LE					85	85	85

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with a history of chronic otitis media and eustachian tube dysfunction. Patient c/o decreased hearing in the right ear.

COMMENTS: _____

Otoscopy:

Right Ear: clear canal

Left Ear: clear canal

Audiogram 19: Intermediate—Interpretation

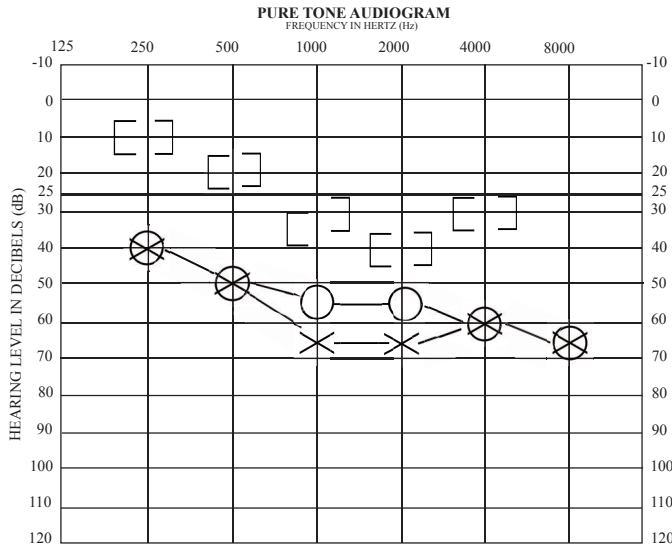
Audiogram: ____19____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Moderate to mild HL Left Ear: Normal hearing PTA: RE, 40 dB; LE, 8 dB Asymmetric
Determine the type of hearing loss	Conductive (unilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral conductive hearing loss
Compare speech thresholds to pure tones	SRTs are in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally.
Determine tympanometry type (Figure 3–1)	Right Ear: Type C tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Ipsilateral Right Ear: Absent .5, 1, 2 K Ipsilateral Left Ear: Present .5, 1, 2 K

Assessment: Right ear indicates a moderate to mild conductive hearing loss with a type C tympanogram, suggesting negative middle ear pressure. The left ear indicates normal hearing sensitivity and a normal tympanogram. Overall test results confirm probable middle ear pathology in the right ear, likely secondary to eustachian tube dysfunction.

Recommendations: Follow up with ENT for repeat microscopic examination, debridement, and evaluation for cholesteatoma.

Audiogram 20: Intermediate

NAME: 20 DATE: _____ BY: _____Age: 35

SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	50		92%	90 dB HL
L	55		96%	95 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD
 TEST METHOD: Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMpanogram TYPE	A	A
MIDDLE EAR PRESSURE	5	0
P.V.T	0.9	0.9
COMPLIANCE	0.2	0.2

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE					NR	NR	
LE					NR	NR	

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient is a 35 year old female currently pregnant and reports gradual decrease in hearing over the last few years. Patient denies aural drainage, vertigo, and aural fullness.

COMMENTS: _____

Audiogram 20: Intermediate—Interpretation

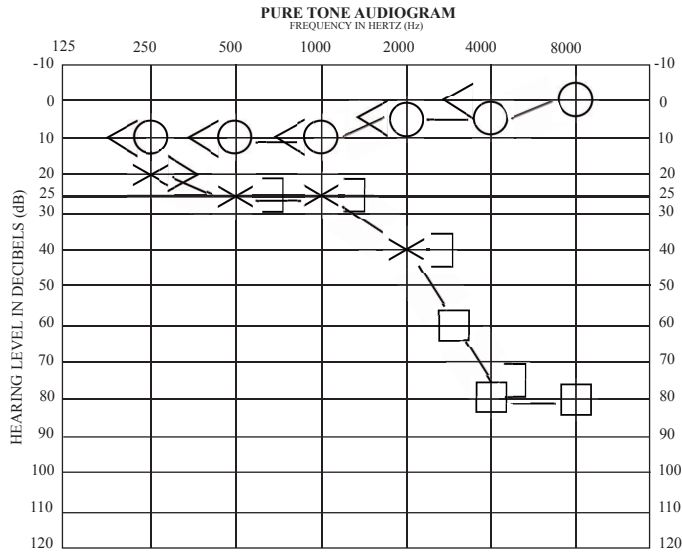
Audiogram: 20

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Mild sloping to moderately severe HL, bilaterally. PTA: RE, 53 dB; LE, 60 dB No significant asymmetry
Determine the type of hearing loss	Mixed (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Mixed hearing loss, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally.
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Absent ipsilateral responses at 500 and 1000 Hz, bilaterally

Assessment: Testing indicates a mild sloping to moderately severe mixed hearing loss with normal tympanograms, bilaterally.

Recommendations: Follow up with ENT to discuss diagnosis of otosclerosis. Consider amplification if surgical treatment is not an option at this time. In-office tuning fork examination from ENT may help guide recommendations.

Audiogram 21: Intermediate

NAME: 21 DATE: _____ BY: _____Age: 38

SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	10		100%	40 dB HL
L	35		80%	85* dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD

TEST METHOD: Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED			△	
SOUND FIELD		S		
AIDED		A		
OTHER TESTS MASKED	*			

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	20	0
P.V.T	1.1	1.3
COMPLIANCE	0.6	0.7

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE	90	90	90		80	80	80
LE	80	80	80		90	90	105

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient c/o decreased hearing in the left ear. Patient denies dizziness. He reports constant high-pitched tinnitus. The hearing has gradually decreased over time. He is a full-time interstate truck driver.

COMMENTS: _____

Audiogram 21: Intermediate—Interpretation

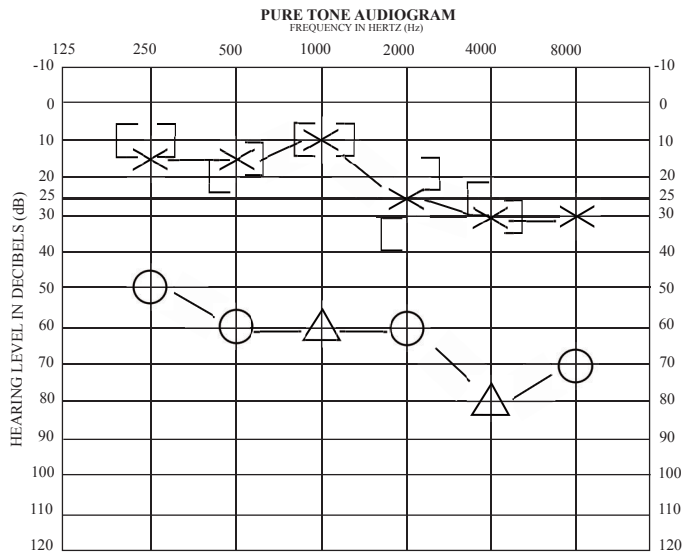
Audiogram: 21

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Normal hearing Left Ear: Normal sloping to severe high-frequency hearing loss PTA: RE, 8 dB; LE, 30 dB Significant asymmetry
Determine the type of hearing loss	Sensorineural (unilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral asymmetric SNHL
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Excellent Left Ear: Good
Determine tympanometry type (Figure 3–1)	Type A tympanograms, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Ipsilateral and contralateral reflexes were present at all measured frequencies in both ears.

Assessment: Testing indicates normal hearing sensitivity in the right ear and normal hearing from 250 to 1000 Hz sloping to a severe high-frequency sensorineural hearing loss in the left ear. Tympanograms and ARTs are normal in both ears, suggesting normal middle ear function.

Recommendations: Referral to ENT to consider MRI to rule out acoustic neuroma. However, patient informed audiologist that he often drives with his driver's side window down for hours at a time; therefore, hearing loss is likely due to excessive noise exposure. Counsel patient on hearing protection options available. Consider amplification for left ear.

Audiogram 22: Intermediate

NAME: 22 DATE: _____ BY: _____Age: 42

SPEECH AUDIOMETRY

☒ RECORDED☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	60*		92%	100* dB HL
L	25		100%	65 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD

TEST METHOD: Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD		S		
AIDED		A		
OTHER TESTS MASKED		*		

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	Ad	A
MIDDLE EAR PRESSURE	0	0
P.V.T	0.8	0.8
COMPLIANCE	4.2	0.5

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE	NR	NR	NR		NR	NR	NR
LE	NR	NR	NR		85	85	95

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents accidental right-sided ear injury. Denies otorrhea, tinnitus, and vertigo. Reported ear pain in the right ear.

COMMENTS: _____

Audiogram 22: Intermediate—Interpretation

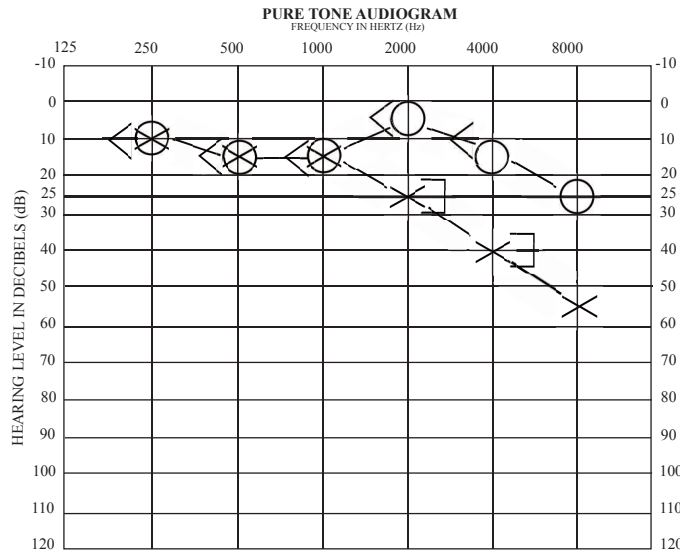
Audiogram: 22

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Moderate to severe HL Left Ear: Normal to mild HL PTA: RE, 60 dB; LE, 16 dB Asymmetry
Determine the type of hearing loss	Right Ear: Mixed Left Ear: Sensorineural
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Mixed HL
Compare speech thresholds to pure tones	SRT in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent in both ears.
Determine tympanometry type (Figure 3–1)	Right Ear: Type Ad tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Right Ear: Ipsilateral and contralateral responses are absent. Left Ear: Ipsilateral responses are present and contralateral responses are absent.

Assessment: Testing reveals a moderate to severe mixed hearing loss in the right ear and normal hearing from 250 to 2000 Hz sloping to a mild sensorineural hearing loss in the left ear. Right hearing loss is consistent with middle ear injury/ossicular chain discontinuity.

Recommendations: Referral to ENT; it appears patient may have suffered an ossicular disruption. Middle ear exploration and ossicular chain reconstruction should be considered. Consider amplification if surgical treatment is not an option.

Audiogram 23: Intermediate

NAME: 23 DATE: _____ BY: _____Age: 53

SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	20		100%	50 dB HL
L	20		40%	85 dB HL *

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
 TEST METHOD: ☐ SOUNDFIELD Std. _____

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED	*			

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	-50	-45
P.V.T	1.0	1.2
COMPLIANCE	0.6	0.7

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE		NR	NR	NR	90	95	95
LE		NR	NR	NR	95	95	NR

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient c/o gradual decreased hearing, aural fullness, and pulsing tinnitus in her left ear. She denies vertigo and excessive noise exposure.

COMMENTS: _____

Audiogram 23: Intermediate—Interpretation

Audiogram: 23

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Normal hearing Left Ear: Normal hearing sloping to moderate high-frequency HL PTA: RE, 11 dB; LE, 18 dB Asymmetry
Determine the type of hearing loss	Sensorineural (unilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	High-frequency unilateral sensorineural HL
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Excellent Left Ear: Poor
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Absent contralateral responses in both ears

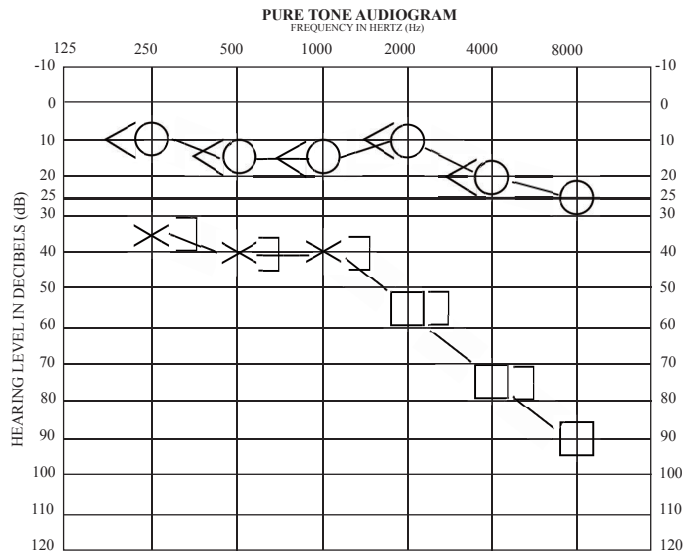
Assessment: Testing indicates normal hearing in the right ear and normal hearing sloping to a moderate sensorineural hearing loss in the left ear. Patient's symptoms and hearing loss may suggest an acoustic neuroma.

Recommendations: Follow up with ENT to rule out retrocochlear lesion (acoustic neuroma, lipomas, etc.) via MRI.

Audiogram 24: Intermediate

NAME: 24 DATE: BY:

Age: 53



SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	15		100%	50 dB HL
L	50		40%	90* dB HL
L			28%	100* dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
 TEST METHOD: ☐ SOUNDFIELD Std.

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	1.3	1.0
COMPLIANCE	0.5	0.5

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE	NR	NR	NR		90	90	95
LE	NR	NR	NR		NR		

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient c/o continued decrease in hearing, aural fullness, and tinnitus in her left ear. Previous audio indicated normal hearing in the right ear and normal hearing sloping to a moderate SNHL in the left ear. She reports her hearing has

COMMENTS: worsened. She did not previously seek treatment as recommended.

(Rollover method used)

Audiogram 24: Intermediate—Interpretation

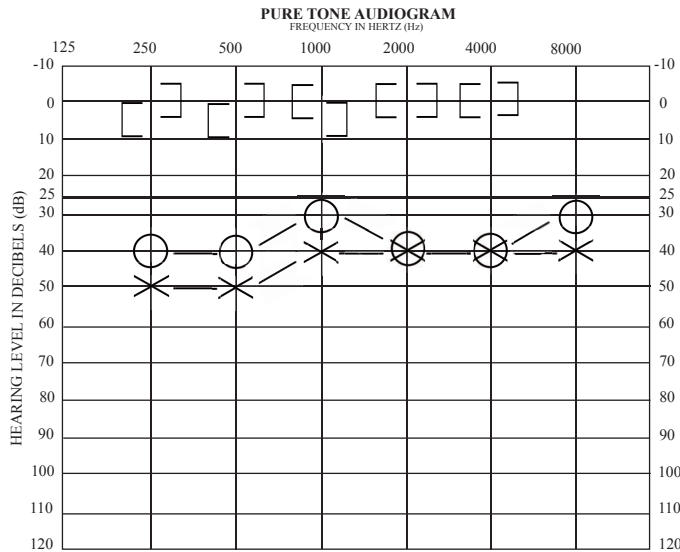
Audiogram: 24

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Normal hearing Left Ear: Mild sloping to profound high-frequency HL PTA: RE, 13 dB; LE, 45 dB Asymmetry
Determine the type of hearing loss	Sensorineural (unilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	High-frequency unilateral sensorineural HL
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–5)	Right Ear: Excellent Left Ear: Very Poor Rollover method used
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Present ipsilateral responses and absent contralateral responses in the right ear Absent ipsilateral and contralateral responses in the left ear

Assessment: Testing indicates normal hearing in the right ear and a mild sloping to profound high-frequency SNHL in the left ear. Poorer thresholds noted overall in left ear. Due to progressive unilateral high-frequency SNHL, poor WRS in the left ear, and patient's symptoms, this may suggest an acoustic neuroma.

Recommendations: Follow up with ENT for medical treatment and to rule out possible acoustic neuroma or other retrocochlear lesions. Follow up with ENT to rule out retrocochlear lesion via MRI.

Audiogram 25: Intermediate

NAME: 25 DATE: _____ BY: _____Age: 42

SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	40		100%	70 dB HL
L	45		100%	70 dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD
 TEST METHOD: Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	B	B
MIDDLE EAR PRESSURE	---	---
P.V.T	1.0	1.0
COMPLIANCE	---	---

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with otorrhea in both ears and c/o difficulty hearing. Patient denies ear pain, vertigo, and tinnitus.COMMENTS: Otoscopy: Aural drainage, bilaterally.

Audiogram 25: Intermediate—Interpretation

Audiogram: 25

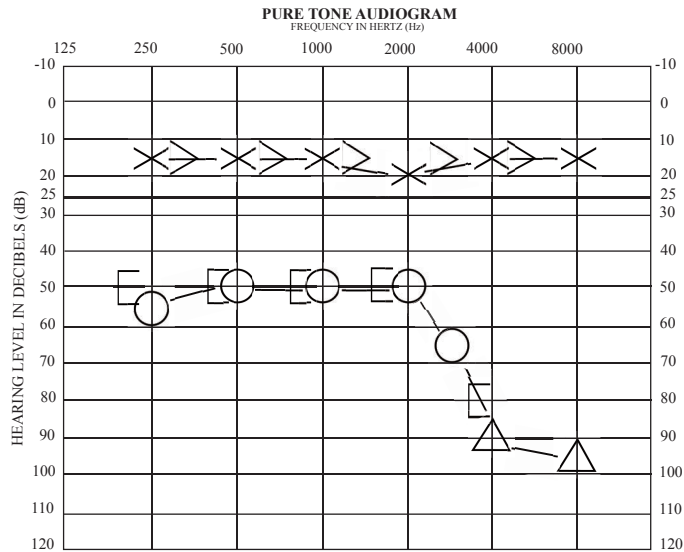
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Mild HL Left Ear: Moderate to mild HL PTA: RE, 36 dB; LE, 43 dB Slight asymmetry
Determine the type of hearing loss	Conductive (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Conductive HL, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally.
Determine tympanometry type (Figure 3–1)	Type B tympanogram, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Results indicate a mild conductive hearing loss in the right ear and a moderate rising to mild conductive hearing loss in the left ear with type B tympanograms, bilaterally. Results are consistent with bilateral otitis externa.

Recommendations: ENT referral for medical management of otitis externa with topical therapies and serial debridements. Will need to assess status of middle ear once canal infection is cleared. Return to audiology clinic after medical management for repeat audiogram.

Audiogram 26: Intermediate

NAME: 26 DATE: _____ BY: _____
 Age: 55



SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	60		80%	100* dB HL
L	15		96%	50 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD Std. _____
 TEST METHOD: _____

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMpanogram TYPE	A	A
MIDDLE EAR PRESSURE	5	0
P.V.T	0.9	0.9
COMPLIANCE	4.2	0.6

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE					NR	NR	
LE					85	90	

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with vertigo, tinnitus, ataxia, and facial paralysis. He had a stroke 3 months ago. Patient reports decreased hearing on right side.

COMMENTS: _____

Audiogram 26: Intermediate—Interpretation

Audiogram: 26

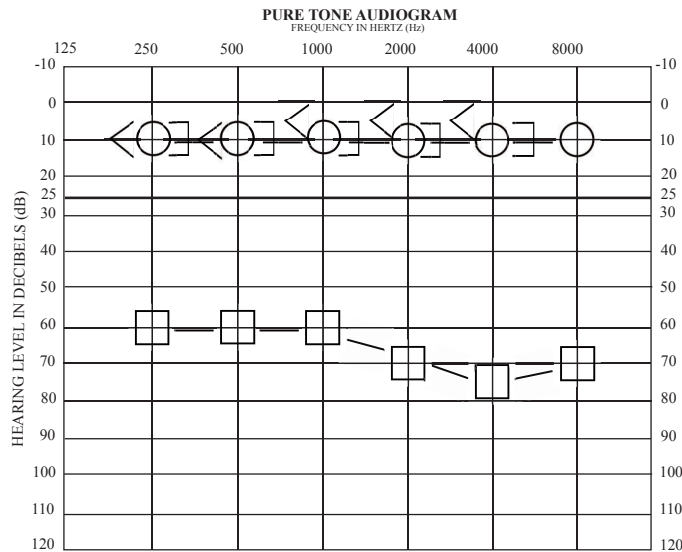
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Moderate sloping to profound high-frequency HL Left Ear: Normal hearing PTA: RE, 50 dB; LE, 18 dB Asymmetric
Determine the type of hearing loss	Right Ear: Sensorineural Left Ear: Normal hearing
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Sensorineural HL, unilaterally
Compare speech thresholds to pure tones	SRT in agreement with PTs
Assess word recognition score (Figure 2–3)	Right Ear: Good Left Ear: Excellent
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Right Ear: Normal ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	Right ipsilateral responses are absent and left ipsilateral responses are present.

Additional case information: Patient presents with sudden hearing loss in the right ear. Patient recently suffered a stroke 3 months ago. Presents with vertigo, tinnitus, ataxia, and facial paralysis.

Assessment: Testing reveals a moderate sloping to profound high-frequency SNHL in the right ear and normal hearing in the left ear.

Recommendations: ENT referral.

Audiogram 27: Intermediate

NAME: 27 DATE: _____ BY: _____Age: 55

SPEECH AUDIOMETRY				
EAR	SRT	SDT	WRS	LEVEL
R	10		100%	50 dB HL
L	60		100%	90 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
 TEST METHOD: ☐ SOUNDFIELD Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	B
MIDDLE EAR PRESSURE	0	---
P.V.T	1.2	4.0
COMPLIANCE	0.5	---

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE						85	
LE						NR	

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with history of left-sided canal wall down and is here for his post-op audiogram. Patient denies aural drainage.

COMMENTS: _____

Audiogram 27: Intermediate—Interpretation

Audiogram: 27

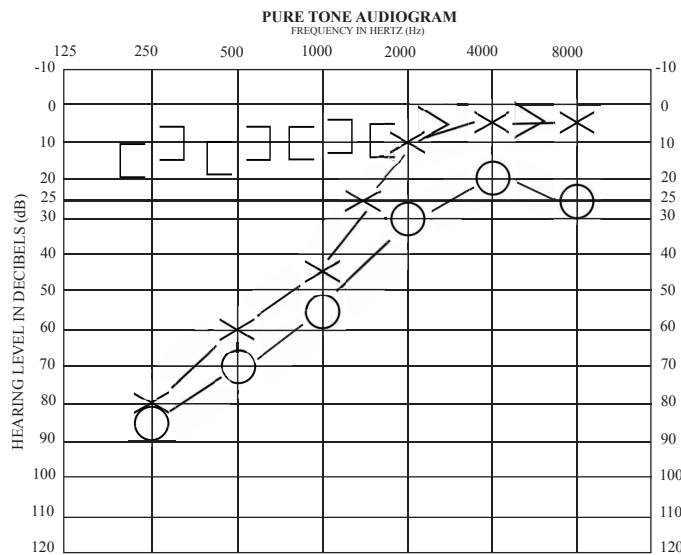
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Normal hearing Left Ear: Moderately severe to severe HL PTA: RE, 10 dB; LE, 63 dB Significant asymmetry
Determine the type of hearing loss	Conductive (unilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Conductive HL, unilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally.
Determine tympanometry type (Figure 3–1)	Right: Type A tympanogram Left: Type B tympanogram
Check ear canal volume	Right: Normal ECV Left: Large ECV
Review acoustic reflexes (Figure 3–8)	Right: Present ipsilateral response at 1000 Hz Left: Absent ipsilateral response at 1000 Hz

Assessment: Results indicate normal hearing in the right ear and a moderately severe sloping to severe conductive hearing loss in the left ear with a type B tympanogram with a large ear canal volume, consistent with canal wall down tympanomastoidectomy.

Recommendations: ENT follow-up as scheduled for observation and serial debridement. Consider amplification for left ear. Annual audiologic evaluations.

Audiogram 28: Intermediate

NAME: 28 DATE: _____ BY: _____
 Age: 17



SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	50		92%	85*
L	35		100%	75*

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD
 TEST METHOD: _____ Std.

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	B	B
MIDDLE EAR PRESSURE	---	---
P.V.T	3.0	5.0
COMPLIANCE	---	---

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with history of PETs in both ears. Reports otalgia one month ago.

COMMENTS: Otoscopy: TM perforation, bilaterally

Audiogram 28: Intermediate—Interpretation

Audiogram: 28

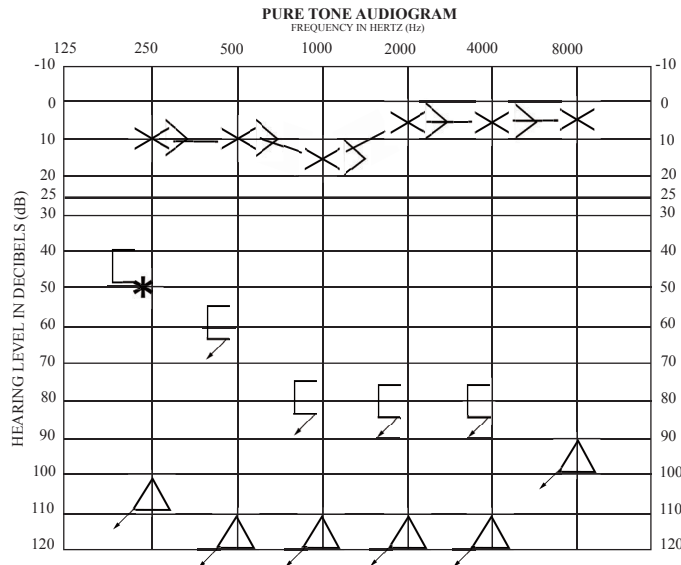
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Severe low frequency HL rising to normal hearing, bilaterally PTA: RE, 52 dB; LE, 38 dB No significant asymmetry
Determine the type of hearing loss	Conductive (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Conductive HL, bilaterally
Compare speech thresholds to pure tones	SDT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally
Determine tympanometry type (Figure 3–1)	Right Ear: Type B tympanogram Left Ear: Type B tympanogram
Check ear canal volume	Large ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Testing reveals a severe low-frequency conductive hearing loss rising to normal hearing with type B tympanograms with large ear canal volumes, bilaterally. Results are consistent with bilateral TM perforations.

Recommendations: Follow up with ENT for medical treatment. Patient likely to be a candidate for closure tympanoplasty, starting with the worse hearing (right) ear. Return to audiology clinic after medical management for repeat testing.

Audiogram 29: Intermediate

NAME: 29 DATE: BY:
 Age: 16



SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	NR		0/10	110 dB HL
L	5		100%	50 dB HL

DPOAEs

Right Ear: Absent 1 - 8 kHz

Left ear: Present 1 - 8 kHz

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD
 TEST METHOD: Std.

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMpanoGRAM TYPE	A	A
MIDDLE EAR PRESSURE	-15	0
P.V.T	0.8	0.8
COMPLIANCE	0.2	0.5

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE	NR	NR	NR		NR	NR	NR
LE	95	95	100		85	85	90

REFLEX DECAY: RT. _____

LT. _____

REFERRED BY: _____

HISTORY: Patient presents to the clinic with a history of head trauma. Denies aural drainage, tinnitus, and vertigo.

COMMENTS: _____

Negative Stenger

Audiogram 29: Intermediate—Interpretation

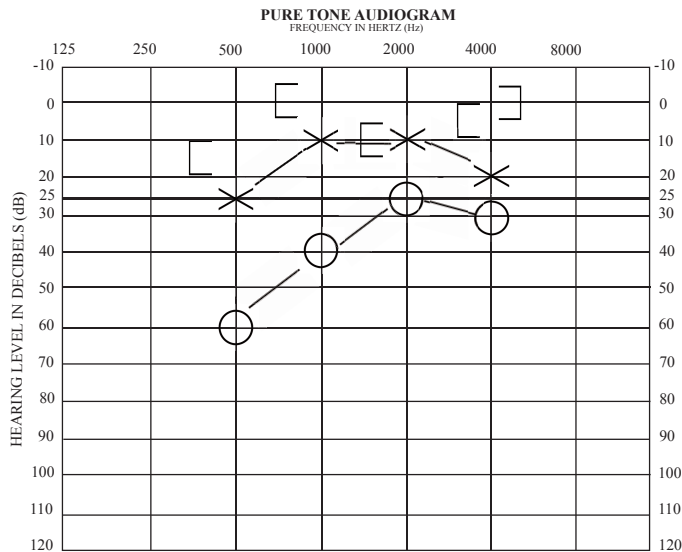
Audiogram: ____29____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Profound HL Left Ear: Normal hearing PTA: RE, 110 dB; LE, 10 dB Significant asymmetry
Determine the type of hearing loss	Sensorineural (unilateral) Negative Stenger
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral sensorineural HL
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Poor Left Ear: Excellent
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Right Ear: Normal ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	Right ipsi/contra: No response Left ipsi/contra: Present

Assessment: Testing indicates a profound sensorineural hearing loss in the right ear and normal hearing in the left ear. Additional testing indicated absent DPOAE responses in the right ear and present DPOAE responses in the left ear. Results are consistent with patient history of head trauma.

Recommendations: Follow up with ENT. For patients with single-sided deafness, a few options exist. A provider should discuss the use of a CROS system, a BAHS, or a cochlear implant.

Audiogram 30: Intermediate

NAME: 30 DATE: _____ BY: _____Age: 3

SPEECH AUDIOMETRY ☐ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R		40		
L		15		

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD CPA - Good to Fair

TEST METHOD: _____

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD		S		
AIDED		A		
OTHER TESTS MASKED		*		

IMMITTANCE TESTING		
Tympanometry		
	RIGHT	LEFT
TYMPANOGRAM TYPE	B	B
MIDDLE EAR PRESSURE	---	---
P.V.T	0.6	2.2
COMPLIANCE	---	---

Reflex Threshold							
	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
LT. _____

REFERRED BY: ENTHISTORY: - Patient presents with history of cleft lip and palate, pressure equalization tubes (PETs) in both ears, and conductive hearing loss. He is here today for his post-op audiogram.COMMENTS: Otoscopy
PETs visualized, AUPatient fatigued to left bone conduction.

Audiogram 30: Intermediate—Interpretation

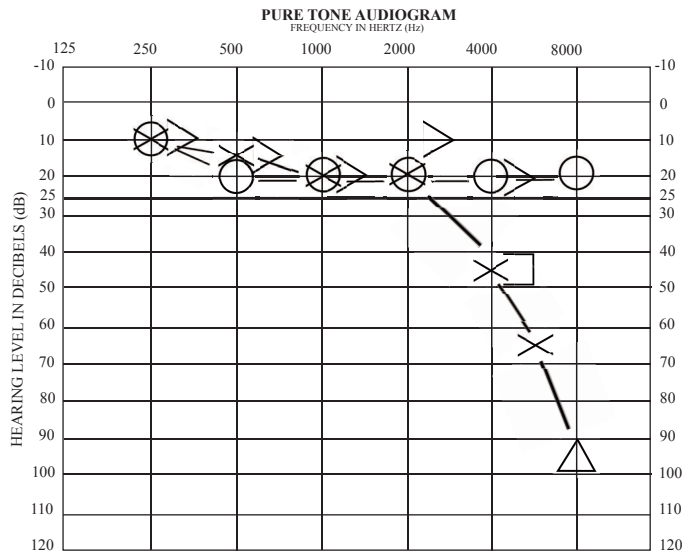
Audiogram: 30

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Moderately severe rising to mild HL Left Ear: Slight low-frequency HL rising to normal hearing PTA: RE, 42 dB; LE, 15 dB Asymmetry
Determine the type of hearing loss	Conductive (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Conductive HL, bilaterally
Compare speech thresholds to pure tones	SDT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	N/A
Determine tympanometry type (Figure 3–1)	Right Ear: Type B tympanogram Left Ear: Type B tympanogram
Check ear canal volume	Right Ear: Normal ECV Left Ear: Large ECV
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Right ear indicates a moderate rising to mild conductive hearing loss with a type B tympanogram and normal ECV, suggesting a plugged PET. The left ear indicates a slight hearing loss rising to normal hearing in the left ear with a type B tympanogram and large ECV, consistent with a patent PET. Slight HL likely conductive, however patient fatigued to the task.

Recommendations: Follow up with ENT to discuss status of right ear and consider replacement of tube. Return to audiology clinic after medical management.

Audiogram 31: Intermediate

NAME: 31 DATE: _____ BY: _____Age: 65

SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	20		100%	60 dB HL
L	25		92%	60 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD

TEST METHOD: Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED				*

IMMITTANCE TESTING		
Tympanometry		
	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	-45
P.V.T	1.2	1.3
COMPLIANCE	0.4	0.5

Reflex Threshold									
	Contralateral					Ipsilateral			
		.5K	1K	2K	4K		.5K	1K	2K
RE		100	95				85	90	
LE		85	100				85	95	

REFLEX DECAY:	RT. _____
	LT. _____

REFERRED BY: _____

HISTORY: Patient here for routine audiologic examination. Patient denies otorrhea, vertigo, tinnitus, excessive noise exposure, and otalgia. Reports sometimes has difficulty hearing in background noise.

COMMENTS: _____

Audiogram 31: Intermediate—Interpretation

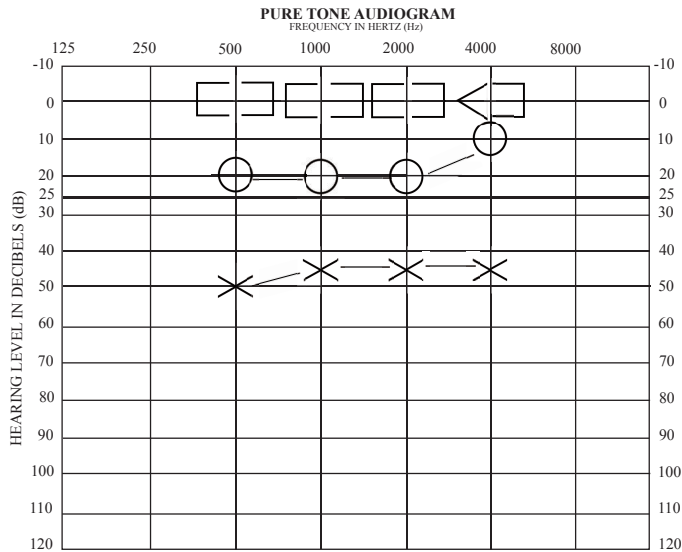
Audiogram: ____31____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Normal hearing Left Ear: Normal 250 to 2000 Hz sloping to profound high-frequency HL PTA: RE, 20 dB; LE, 18 dB Asymmetric
Determine the type of hearing loss	Sensorineural (unilateral) Negative Stenger at 8000 Hz
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral SNHL
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Excellent Left Ear: Excellent
Determine tympanometry type (Figure 3–1)	Type A tympanogram, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Present ipsilateral and contralateral reflexes in both ears

Assessment: The right ear reveals normal hearing sensitivity with a normal tympanogram. The left ear reveals normal hearing from 250 Hz to 2000 Hz sloping to a profound high-frequency sensorineural hearing loss. Although there are no otologic symptoms suggesting retrocochlear pathology, a referral to ENT is still required due to the unilateral asymmetric SNHL.

Recommendations: ENT referral for evaluation and consideration of MRI scan to rule out retrocochlear lesion.

Audiogram 32: Intermediate

NAME: 32 DATE: _____ BY: _____Age: 4

SPEECH AUDIOMETRY				
EAR	SRT	SDT	WRS	LEVEL
R	15		DNT	
L	40*		DNT	

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD

TEST METHOD: CPA

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMpanogram TYPE	B	B
MIDDLE EAR PRESSURE	---	---
P.V.T	7.0	0.4
COMPLIANCE	---	---

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with history of pressure equalization tubes (PET) in both ears. Today his parents report a recent ear infection, aural drainage, and patient c/o difficulty hearing in left ear.

COMMENTS: _____

Otосcopy:

Right Ear - PET visualized

Left Ear - PET extruded

Audiogram 32: Intermediate—Interpretation

Audiogram: 32

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Normal hearing Left Ear: Moderate HL PTA: RE, 20 dB; LE, 47 dB Significant asymmetry
Determine the type of hearing loss	Conductive (unilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral conductive HL
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Did not test – Young child with speech delay
Determine tympanometry type (Figure 3–1)	Type B tympanograms, bilaterally
Check ear canal volume	Right Ear: Large ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	N/A

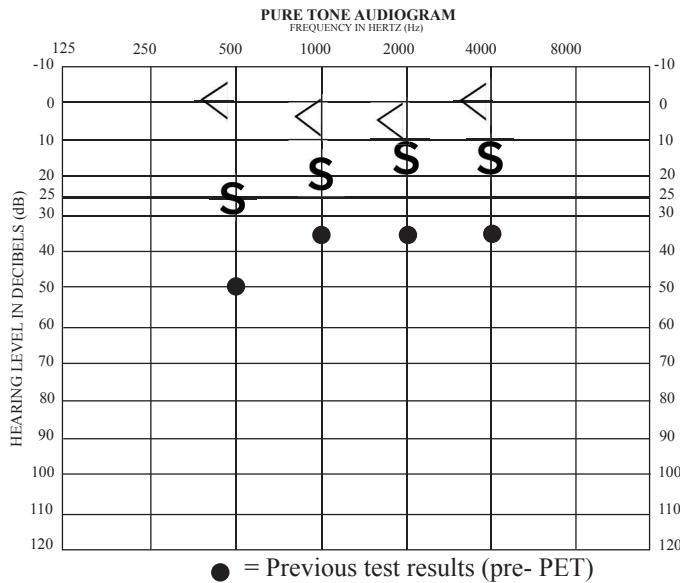
Assessment: The right ear reveals normal hearing with 20 dB air-bone gaps at 500 to 2000 Hz, consistent with a patent PET. The left ear reveals a moderate conductive hearing loss, which is suggestive of probable middle ear pathology and confirms extruded PET.

Recommendations: ENT referral to discuss repeat PET placement for the left ear. Return to clinic after medical management for repeat audiogram.

Audiogram 33: Intermediate

NAME: 33 DATE: BY:

Age: 2



SPEECH AUDIOMETRY ☐ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R		10		
L		10		
SF				

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☒ SOUNDFIELD
 TEST METHOD: VRA

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	B	B
MIDDLE EAR PRESSURE	---	---
P.V.T	7.0	6.8
COMPLIANCE	---	---

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents for his post-op PET placement audiogram. Parent denies aural drainage. Parent reported an improvement in patient's speech and hearing abilities.

COMMENTS: _____

Pure tones were attempted with headphones, however the patient fatigued to the task.

Audiogram 33: Intermediate—Interpretation

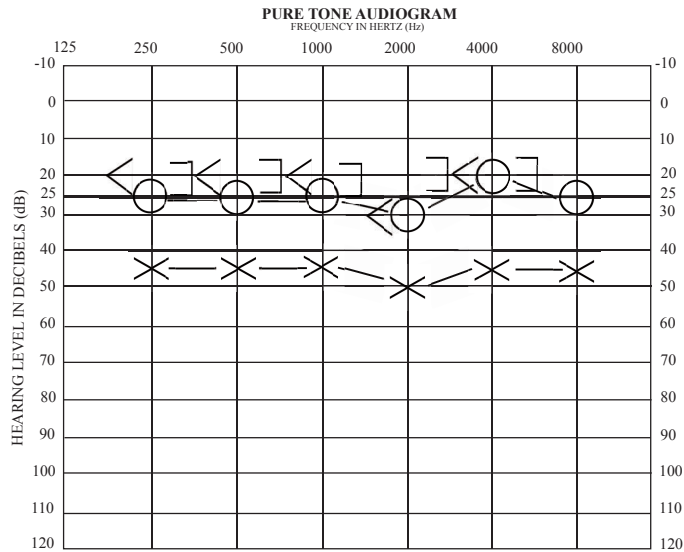
Audiogram: ____33____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Sound field indicates a slight low-frequency HL rising to normal hearing in the better hearing ear.
Determine the type of hearing loss	Normal hearing in the better hearing ear with conductive component in the low frequencies, likely due to patent PETs.
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	N/A
Compare speech thresholds to pure tones	SDT in agreement with PTs (SDT was completed using headphones)
Assess word recognition score (Figure 2–3)	N/A
Determine tympanometry type (Figure 3–1)	Right Ear: Type B tympanogram Left Ear: Type B tympanogram
Check ear canal volume	Large ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Testing reveals a slight conductive hearing loss rising to normal hearing in the better hearing ear (via sound field) with patent PETs in both ears. Improvement in thresholds noted overall compared to previous test results.

Recommendations: Follow-up audiogram and PET check with ENT in 6 months.

Audiogram 34: Intermediate

NAME: 34 DATE: _____ BY: _____Age: 49**SPEECH AUDIOMETRY**☒ RECORDED☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	25		100%	55 dB HL
L	40		100%	70 dB HL

TRANSDUCER:

☒ HEADPHONES
☐ SOUNDFIELD☐ INSERTS

TEST METHOD:

Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	B
MIDDLE EAR PRESSURE	0	0
P.V.T	1.0	1.0
COMPLIANCE	0.5	0.4

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY:

RT. _____

LT. _____

REFERRED BY: _____

HISTORY: Patient c/o feeling "plugged up" in left ear. Patient denies tinnitus and vertigo.

COMMENTS: _____

Audiogram 34: Intermediate—Interpretation

Audiogram: 34

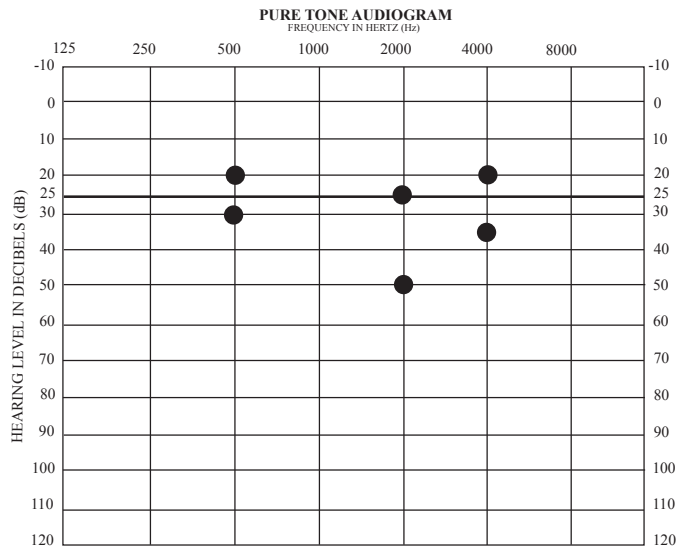
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Normal hearing with mild HL at 2000 Hz Left Ear: Mild to moderate HL PTA: RE, 26 dB; LE, 46 dB Asymmetry
Determine the type of hearing loss	Right Ear: Sensorineural at 2000 Hz Left Ear: Conductive
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral conductive HL
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally.
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type B tympanogram
Check ear canal volume	Right Ear: Normal ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Testing reveals normal hearing with a mild sensorineural hearing loss at 2000 Hz in the right ear and a mild to moderate conductive hearing loss in the left ear. Audiologic results are consistent with tympanograms, suggesting left CHL is likely due to probable middle ear pathology.

Recommendations: Referral to ENT. ENT to evaluate with microscopic exam for middle ear disease (effusion, mass, etc.).

Audiogram 35: Intermediate

NAME: 35 DATE: _____ BY: _____
 Age: 2



SPEECH AUDIOMETRY ☐ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R				
L				
SF		20	(Name, "BaBa")	

DPOAEs - Present 1 - 8 kHz, bilaterally

TRANSDUCER: ☐ HEADPHONES ☐ INSERTS
☒ SOUNDFIELD **VRA/CPA**

TEST METHOD: _____

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED	*			

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	0.5	0.5
COMPLIANCE	0.3	0.3

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE						P	
LE						P	

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with a history of developmental delay, speech delay, and prematurity. Patient passed their newborn hearing screening in both ears. Parents deny aural drainage, ear infections, and family history of congenital hearing loss.

COMMENTS: _____

Audiogram 35: Intermediate—Interpretation

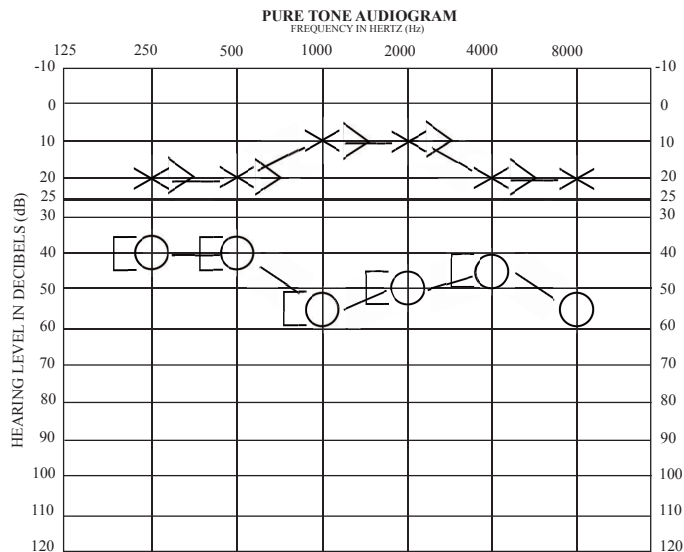
Audiogram: ____35____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Scattered one-time responses in the normal to mild hearing loss range.
Determine the type of hearing loss	Undetermined
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	N/A
Compare speech thresholds to pure tones	SDT obtained at 20 dB in the better hearing ear via sound-field testing.
Assess word recognition score (Figure 2–3)	N/A
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	ART screening indicated present response in both ears.

Assessment: DPOAEs were present 1 to 8 kHz, bilaterally. Behavioral and electrophysiological test results suggest near-normal to normal hearing, bilaterally. Repeat hearing testing is recommended to obtain ear-specific and frequency-specific information.

Recommendations: Return to audiology clinic in 4 to 6 months for repeat hearing test.

Audiogram 36: Intermediate

NAME: 36 DATE: _____ BY: _____Age: 50

SPEECH AUDIOMETRY				
		☒ RECORDED	☐ MLV	
EAR	SRT	SDT	WRS	LEVEL
R	50		68%	95* dB HL
L	15		88%	60* dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD Std.

TEST METHOD: _____

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD		S		
AIDED		A		
OTHER TESTS MASKED		*		

IMMITTANCE TESTING

Tympanometry		
	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	1.0	1.0
COMPLIANCE	0.5	0.5

Reflex Threshold							
	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE						100	
LE						85	

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient c/o decreased hearing in the right ear. Denies otalgia, dizziness, and aural fullness. Reports occasional tinnitus.

COMMENTS: _____

Audiogram 36: Intermediate—Interpretation

Audiogram: ____36____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Mild to moderate HL Left Ear: Normal hearing PTA: RE, 48 dB; LE, 13 dB Significant asymmetry
Determine the type of hearing loss	Sensorineural (unilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral SNHL
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Fair Left Ear: Good
Determine tympanometry type (Figure 3–1)	Type A tympanograms, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Ipsilateral reflexes were present in the right and left ears at 1000 Hz.

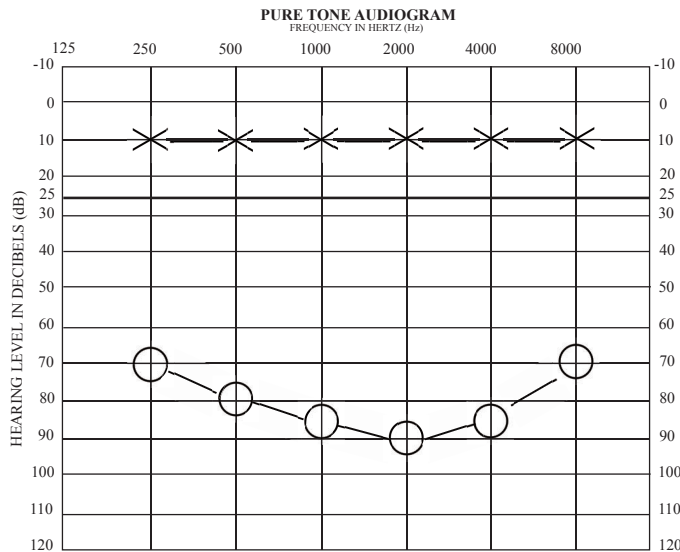
Assessment: The right ear reveals a mild to moderate sensorineural hearing loss with a normal tympanogram. The left ear reveals normal hearing sensitivity with a normal tympanogram.

Recommendations: ENT referral. Likely to recommend MRI to evaluate for cerebellopontine angle tumor or lesions. Consider hearing aid evaluation to discuss treatment options for the right ear.

Audiogram 37: Intermediate

NAME: 37 DATE: BY:

Age: 13



SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	50		100%	70 dB HL
L	20		100%	70 dB HL

DPOAEs: Present 1 - 8 kHz, bilaterally

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD
 TEST METHOD: Std.

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	0.7	0.6
COMPLIANCE	0.5	0.6

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE						85	
LE						85	

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Referred on school screening in right ear

COMMENTS: - Patient readily understands soft speech

Positive Stenger

- Many false positives

- Half-word responses to SRTs

Audiogram 37: Intermediate—Interpretation

Audiogram: ____37____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Severe HL Left Ear: Normal hearing PTA: RE, 85 dB; LE, 10 dB Significant asymmetry
Determine the type of hearing loss	N/A
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral HL
Compare speech thresholds to pure tones	SRT is not in agreement with PTA for the right ear
Assess word recognition score (Figure 2–3)	Excellent, bilaterally
Determine tympanometry type (Figure 3–1)	Type A tympanogram, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Present ipsilateral responses at 1000 Hz in both ears

PEARL: PTA must agree with SRT within 10–12 dB. Differences of 10 dB or more suggests malingering.

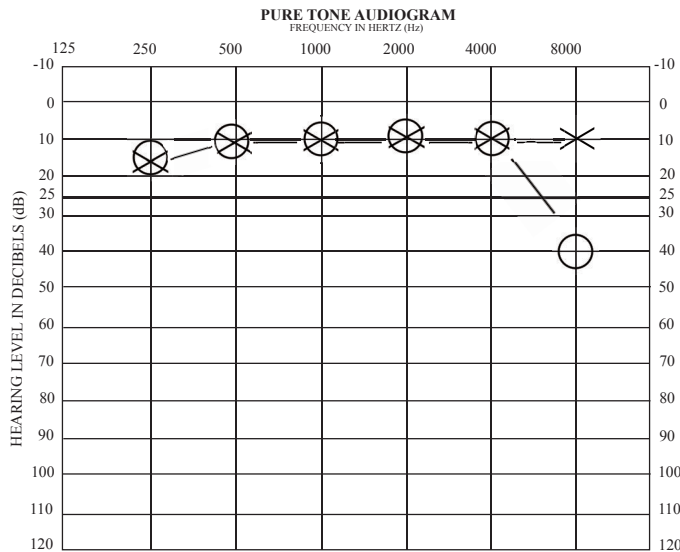
Assessment: Normal hearing in the left ear and a positive Stenger in the right ear with present DPOAE and ARTs. Results suggest patient is malingering for the right ear.

Recommendations: Return to audiology clinic for repeat hearing test. If patient unable to cooperate, consider auditory brainstem response (ABR) test. Parents were counseled on results.

Audiogram 38: Intermediate

NAME: 38 DATE: BY:

Age: 50



SPEECH AUDIOMETRY ☐ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	10		100%	40 dB HL
L	10		100%	40 dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD

TEST METHOD: Std.

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE		
MIDDLE EAR PRESSURE		
P.V.T		
COMPLIANCE		

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with no hearing complaints.

COMMENTS: _____

Audiogram 38: Intermediate—Interpretation

Audiogram: ____38____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Normal hearing, bilaterally, with the exception of a mild HL at 8000 Hz in the right ear PTA: RE, 10 dB, LE, 10 dB Asymmetry at 8000 Hz
Determine the type of hearing loss	Normal with exception of 8 kHz
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	N/A
Compare speech thresholds to pure tones	SDT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Excellent, bilaterally
Determine tympanometry type (Figure 3–1)	N/A
Check ear canal volume	N/A
Review acoustic reflexes (Figure 3–8)	N/A

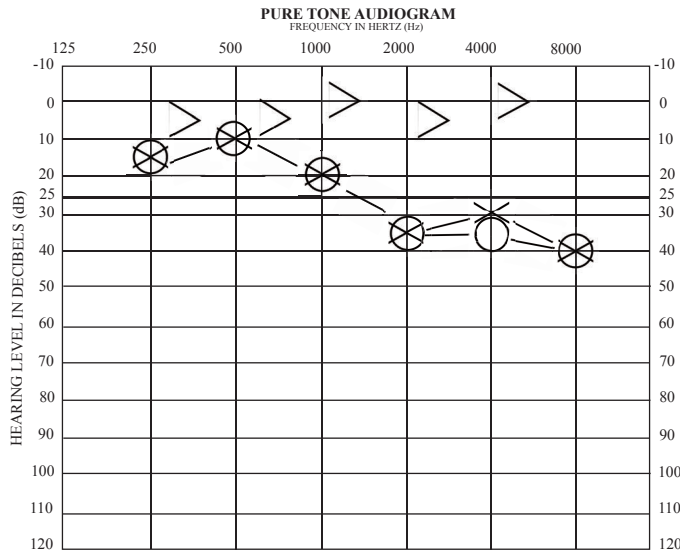
Assessment: Normal hearing sensitivity, bilaterally. Audiologist rechecked 8000 Hz using insert earphones and noted an improvement. As frequencies get higher, the wavelength gets shorter. The distance between the eardrum and the earphone is ~1½ inches in an adult ear. At a certain point, usually 8 kHz, the direct and reflected waves can partially cancel, resulting in elevated thresholds.

Recommendations: No further testing needed.

Audiogram 39: Intermediate

NAME: 39 DATE: BY:

Age: 92



SPEECH AUDIOMETRY ☐ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	10		100%	40 dB HL
L	10		100%	40 dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD
 TEST METHOD: Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED				*

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	1.1	1.5
COMPLIANCE	0.5	0.5

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE					80	80	80
LE					80	80	80

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with no hearing complaints.

COMMENTS: _____

Audiogram 39: Intermediate—Interpretation

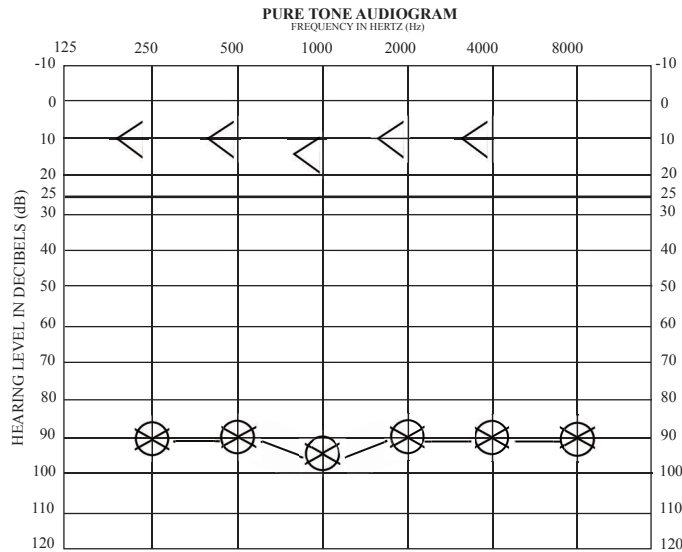
Audiogram: ____39____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Normal hearing sloping to a mild HL PTA: RE, 22 dB; LE, 22 dB No asymmetry
Determine the type of hearing loss	Conductive (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Conductive HL, bilaterally
Compare speech thresholds to pure tones	SDT is somewhat not in agreement with PTAs.
Assess word recognition score (Figure 2–3)	Excellent, bilaterally
Determine tympanometry type (Figure 3–1)	Type A tympanogram, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Present ARTs in both ears

Assessment: Normal hearing sloping to a mild conductive hearing loss with normal tympanograms, bilaterally. The 92-year-old man presented with no hearing complaints, but it seems he is experiencing a conductive hearing loss. This is likely caused by a collapsed ear canal from the pressure of the earphone against the pinna, which may result in the partial closure or complete collapse of the external auditory meatus. The audiologist will switch to insert earphones to alleviate this issue.

Recommendations: Once normal hearing is obtained, no further testing needed.

Audiogram 40: Intermediate

NAME: 40 DATE: _____ BY: _____Age: 22

SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	90		92%	110 dB HL
L	90		100%	110 dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD
 TEST METHOD: STD.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE		
MIDDLE EAR PRESSURE		
P.V.T		
COMPLIANCE		

Reflex Threshold

	Contralateral				Ipsilateral		
	5K	1K	2K	4K	5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Bilateral microtiaCOMMENTS: Annual audiologic evaluation. BAHS user.

Audiogram 40: Intermediate—Interpretation

Audiogram: ____40____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Profound HL, bilaterally PTA: RE, 92 dB, LE, 92 dB No asymmetry
Determine the type of hearing loss	Masking dilemma (see PEARL below) Likely conductive in both ears
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Bilateral conductive HL
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally.
Determine tympanometry type (Figure 3–1)	N/A
Check ear canal volume	N/A
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Testing reveals a maximal conductive hearing loss likely in both ears, consistent with bilateral microtia.

PEARL: In bilateral, moderately-severe to severe conductive hearing losses, the amount of masking needed results in crossover to the test ear (see Step 2, Masking).

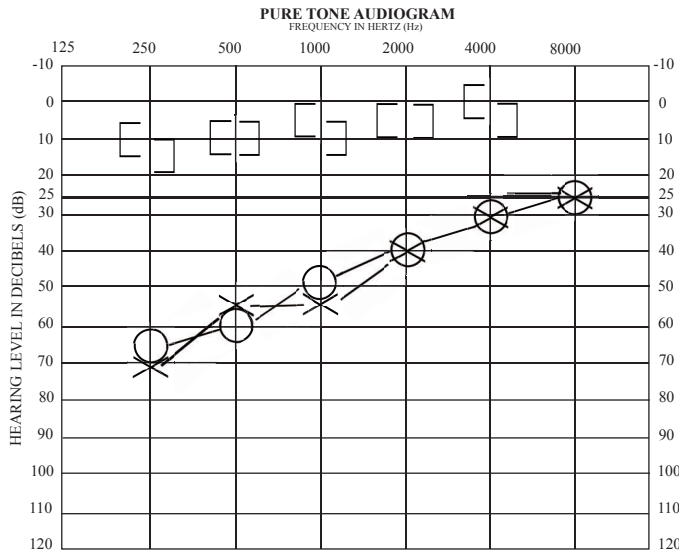
Recommendations: Return for annual audiologic evaluations. Continue full-time BAHS use.

ADVANCED AUDIOGRAMS

Audiogram 41: Advanced

NAME: 41 DATE: BY:

Age: 10



SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	45		100%	80 dB HL
L	45		100%	80 dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD Std.

TEST METHOD: _____

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		∇	∇	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	B	B
MIDDLE EAR PRESSURE	---	---
P.V.T	0.5	0.5
COMPLIANCE	---	---

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY:

RT.

LT.

REFERRED BY: _____

HISTORY: Patient with history of long-standing mild low-frequency conductive hearing loss (unknown etiology) is here today for his pre-op audiogram. Patient c/o recent decrease in hearing in both ears. Patient denies otorrhea, ear pain, and vertigo.

COMMENTS: _____

Audiogram 41: Advanced—Interpretation

Audiogram: ____41____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Moderately severe HL rising to mild HL, bilaterally PTA: RE, 50 dB; LE, 50 dB No significant asymmetry
Determine the type of hearing loss	Conductive (bilateral)
Check if audiogram fits a classic pattern (Figure 1–5 – 1–16)	Conductive HL, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally.
Determine tympanometry type (Figure 3–1)	Right Ear: Type B tympanogram Left Ear: Type B tympanogram
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	N/A

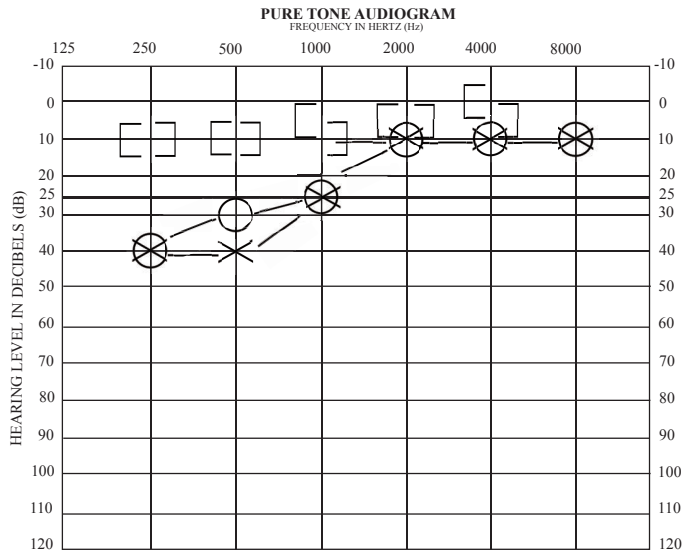
Assessment: Testing reveals a moderately severe rising to mild conductive hearing loss with abnormal tympanograms, bilaterally. Audiogram is consistent with chronic otitis media with bilateral effusions.

Recommendations: Follow up with ENT recommendation for evaluation and consideration of tympanostomy tube placement. Return to clinic for postoperative audiogram.

Audiogram 42: Advanced

NAME: 42 DATE: BY:

Age: 10



SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	20		100%	50 dB HL
L	15		100%	50 dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD Std.

TEST METHOD: _____

	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry		
	RIGHT	LEFT
TYMPANOGRAM TYPE	B	B
MIDDLE EAR PRESSURE	---	---
P.V.T	7.0	5.0
COMPLIANCE	---	---

Reflex Threshold							
	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient with long-standing mild low-frequency conductive hearing loss who received PETs in both ears, here today for his post-op audiogram. Patient denies recent otorrhea and ear pain. He reports he is hearing much better since surgery.

COMMENTS: _____

*See previous audio

Audiogram 42: Advanced—Interpretation

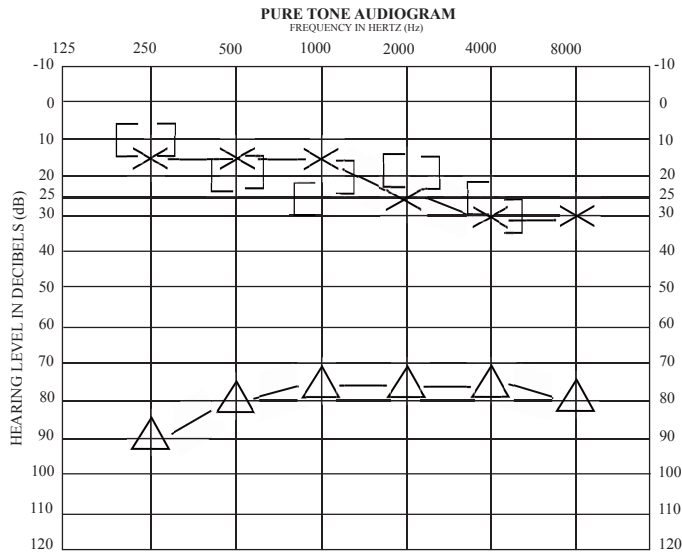
Audiogram: 42

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Mild rising to normal hearing, bilaterally PTA: RE, 22 dB; LE, 25 dB No significant asymmetry
Determine the type of hearing loss	Conductive (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Conductive hearing loss, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Excellent, bilaterally
Determine tympanometry type (Figure 3–1)	Right Ear: Type B tympanogram Left Ear: Type B tympanogram
Check ear canal volume	Large ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Testing reveals a mild conductive hearing loss rising to normal hearing with type B tympanograms with large ECVs, bilaterally. Results are consistent with patent PETs. Threshold improvement was noted at 250 to 8000 Hz, bilaterally.

Recommendations: Follow up with ENT every 6 to 9 months to evaluate status of PETs while in place.

Audiogram 43: Advanced

NAME: 43 DATE: _____ BY: _____Age: 42

SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	80*		80%	110* dB HL
L	20		96%	60 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD Std.

TEST METHOD: _____

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry		
	RIGHT	LEFT
TYMPANOGRAM TYPE	B	A
MIDDLE EAR PRESSURE	---	0
P.V.T	2.0	0.9
COMPLIANCE	---	0.6

Reflex Threshold							
	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE	NR	NR			NR	NR	
LE	NR	NR			85	90	

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with history of cholesteatoma in the right ear and is post-op mastoidectomy. Denies aural drainage.

COMMENTS: _____

Audiogram 43: Advanced—Interpretation

Audiogram: 43

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Severe HL Left Ear: Normal to mild HL PTA: RE, 77 dB; LE, 18 dB Asymmetric
Determine the type of hearing loss	Right Ear: Conductive Left Ear: Sensorineural
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral conductive HL
Compare speech thresholds to pure tones	SRT in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Good Left Ear: Excellent
Determine tympanometry type (Figure 3–1)	Right Ear: Type B tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Right Ear: Borderline normal ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	Right ipsilateral and contralateral responses are absent. Left ipsilateral responses are present and contralateral responses are absent.

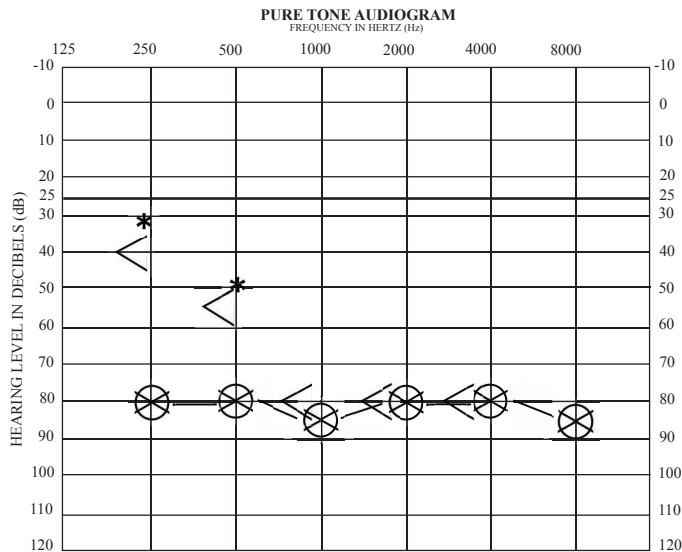
Assessment: Right ear reveals a severe conductive hearing loss with abnormal middle ear mobility, consistent with middle ear pathology (mastoidectomy). The left ear reveals normal hearing sloping to a mild sensorineural hearing loss with a normal tympanogram.

Recommendations: Consider amplification for right ear. ENT follow-up for serial postsurgical examination.

Audiogram 44: Advanced

NAME: 44 DATE: BY:

Age: 33



SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	85		36%	105 dB HL
L	80		32%	100 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
 TEST METHOD: ☐ SOUNDFIELD Std.

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	0.8	0.8
COMPLIANCE	0.4	0.5

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE					NR		
LE					NR		

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with history of meningitis and is here for a cochlear implant evaluation.

COMMENTS: _____

Audiogram 44: Advanced—Interpretation

Audiogram: 44

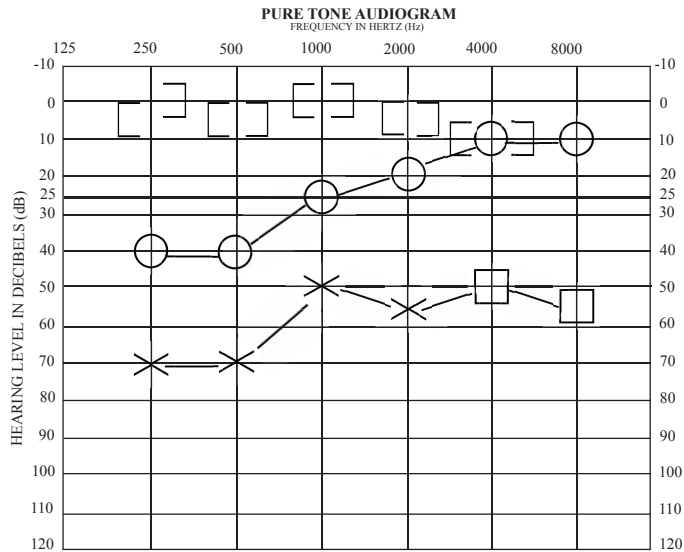
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Severe HL, bilaterally PTA: RE, 82 dB; LE, 82 dB Symmetric
Determine the type of hearing loss	Sensorineural (bilateral).
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Sensorineural HL, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Poor Left Ear: Poor
Determine tympanometry type (Figure 3–1)	Type A tympanogram, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Absent in both ears

Assessment: Severe sensorineural hearing loss with poor word recognition, bilaterally.

Recommendations: Referral to ENT for otologic workup and imaging. Referral to cochlear implant audiologist to discuss cochlear implant candidacy. Generally, the poorer hearing ear is implanted if candidacy is confirmed. Audiologically, this patient is a candidate in both ears.

Audiogram 45: Advanced

NAME: 45 DATE: _____ BY: _____
 Age: 14



SPEECH AUDIOMETRY				
			☒ RECORDED	☐ MLV
EAR	SRT	SDT	WRS	LEVEL
R	25		100%	60 dB HL
L	55		96%	85 dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD Std.

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD		S		
AIDED		A		
OTHER TESTS MASKED				*

IMMITTANCE TESTING

Tympanometry		RIGHT	LEFT
TYMPANOGRAM TYPE		B	B
MIDDLE EAR PRESSURE		---	---
P.V.T.		0.6	0.7
COMPLIANCE		---	---

Reflex Threshold							
Contralateral				Ipsilateral			
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents to clinic with a history of recurrent otitis media and PET placement. Recent complaints of decreased hearing, L > R. Mother reports foul smelling discharge in left ear.

COMMENTS: _____

Otoscopy:

Right Ear: plugged PET

Left Ear: otorrhea

Audiogram 45: Advanced—Interpretation

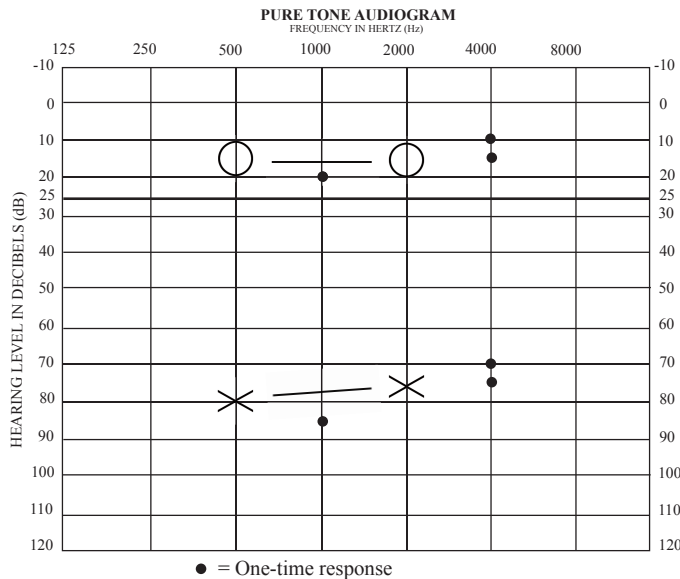
Audiogram: ____45____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Mild low frequency hearing loss rising to normal hearing Left Ear: Moderately severe rising to moderate HL PTA: RE, 28 dB; LE, 58 dB Asymmetry
Determine the type of hearing loss	Conductive (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Conductive HL, bilaterally
Compare speech thresholds to pure tones	SRT in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent in both ears.
Determine tympanometry type (Figure 3–1)	Type B tympanogram, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Testing reveals a mild low-frequency conductive hearing loss rising to normal hearing in the right ear and a moderately severe rising to moderate conductive hearing loss in the left ear. Right ear is suggestive of a plugged PET and the left ear is suggestive of a possible cholesteatoma.

Recommendations: Referral to ENT for microscopic evaluation and debridement. If cholesteatoma is noted, CT imaging and tympanomastoidectomy should be discussed. Will also address right ear tube replacement. Return to audiology clinic after medical management.

Audiogram 46: Advanced

NAME: 46 DATE: _____ BY: _____Age: 4

SPEECH AUDIOMETRY				
		☐ RECORDED		☒ MLV
EAR	SRT	SDT	WRS	LEVEL
R		20	94%	40 dB HL
L		60	60%	100 dB HL *
Closed set WRS test: WIP1				

DPOAEs present 1 -8 kHz in right and left ears.

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD
 TEST METHOD: CPA - GOOD TO FAIR

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING								
Tympanometry								
TYMPANOGRAM TYPE MIDDLE EAR PRESSURE P.V.T COMPLIANCE					RIGHT		LEFT	
					As		As	
					20		0	
					0.3		0.3	
					0.2		0.2	
Reflex Threshold								
	Contralateral				Ipsilateral			
	.5K	1K	2K	4K	.5K	1K	2K	
RE								
LE								
REFLEX DECAY:					RT. _____			
					LT. _____			

REFERRED BY: _____

HISTORY: Previous ABR indicates unilateral Auditory Neuropathy Spectrum Disorder (ANSD). Patient here today for behavioral testing for left ear. Parents report patient is doing well with hearing aid in the left ear.

COMMENTS: _____

Otoscopy: Stenotic ear canals, bilaterally.

Audiogram 46: Advanced—Interpretation

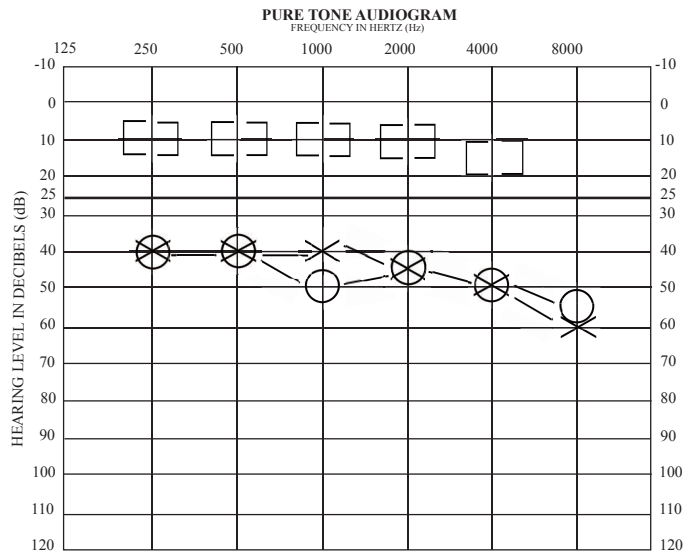
Audiogram: ____46____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Normal hearing Left Ear: Severe HL Significant asymmetry
Determine the type of hearing loss	Undetermined Previous reports of ANSD Masking not completed due to limited time and patient noncompliance
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral HL
Compare speech thresholds to pure tones	SDT is in agreement with PTs in the right ear. SDT in the left ear is likely a better indication of where pure-tone thresholds occur.
Assess word recognition score (Figure 2–3)	Right Ear: Excellent Left Ear: Fair WRS testing was used with the WIPI test—a picture-pointing task.
Determine tympanometry type (Figure 3–1)	Type A _s tympanograms, bilaterally A _s likely due to stenotic ear canals
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: The right ear suggests normal hearing sensitivity with a type A_s tympanogram and excellent WRS. The left ear suggests a severe hearing loss with a type A_s tympanogram and fair WRS.

Recommendations: Continue full-time hearing aid use, ensure preferential seating in the classroom, and consider FM system or other assistive technology for left ear.

Audiogram 47: Advanced

NAME: 47 DATE: _____ BY: _____Age: 55

SPEECH AUDIOMETRY				
EAR	SRT	SDT	WRS	LEVEL
R	40		100%	70 dB HL
L	40		100%	70 dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
 TEST METHOD: ☐ SOUNDFIELD Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	B	B
MIDDLE EAR PRESSURE	---	---
P.V.T	0.3	0.3
COMPLIANCE	---	---

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE	NR	---	---	---	NR	NR	---
LE	NR	---	---	---	NR	NR	---

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with excessive cerumen in both ears. Patient was sent to ENT for ear cleaning. ENT indicates patient presents with exostosis bilaterally. C/o occasional ear pain. Patient is an active swimmer.

COMMENTS: Otoscopy: cerumen impaction, AU

Audiometric responses are post ear cleaning. Otoscopy reveals exostosis, bilaterally, per ENT.

Audiogram 47: Advanced—Interpretation

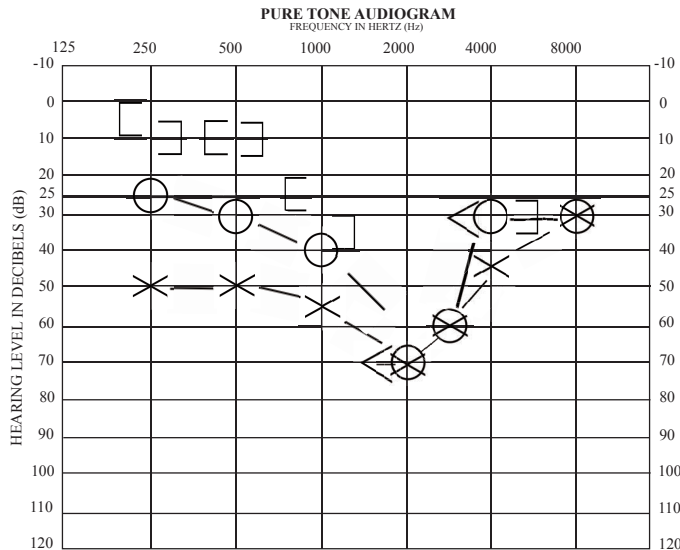
Audiogram: ____47____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Mild to moderate HL, bilaterally PTA: RE, 45 dB; LE, 41 dB Significant asymmetry
Determine the type of hearing loss	Conductive (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Conductive HL, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally.
Determine tympanometry type (Figure 3–1)	Type B tympanograms, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Absent ipsilateral and contralateral acoustic reflexes, bilaterally

Assessment: Post ear cleaning results indicate a mild to moderate conductive hearing loss with abnormal tympanograms, bilaterally, secondary to exostosis. Also known as “surfer’s ear.”

Recommendations: ENT referral for medical workup. ENT to discuss options of observation versus surgery (canalplasty) depending on desire to improve conductive hearing loss or prevent future otitis externa. Return to audiology clinic after medical management. Consider amplification if surgery is not an option at this time.

Audiogram 48: Advanced

NAME: 48 DATE: _____ BY: _____Age: 15

SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	45		92%	80 dB HL
L	55		92%	80 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
 TEST METHOD: Std.

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	B	C
MIDDLE EAR PRESSURE	---	-200
P.V.T	7.0	1.1
COMPLIANCE	---	0.6

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents to clinic with known history of 2000 Hz moderately-severe mixed hearing loss with PET placement in both ears. Patient denies otorrhea. Reports left PET has extruded.

COMMENTS: _____

Otoscopy:

Right Ear - PET visualized in TM

Left Ear - Clear canal

Audiogram 48: Advanced—Interpretation

Audiogram: 48

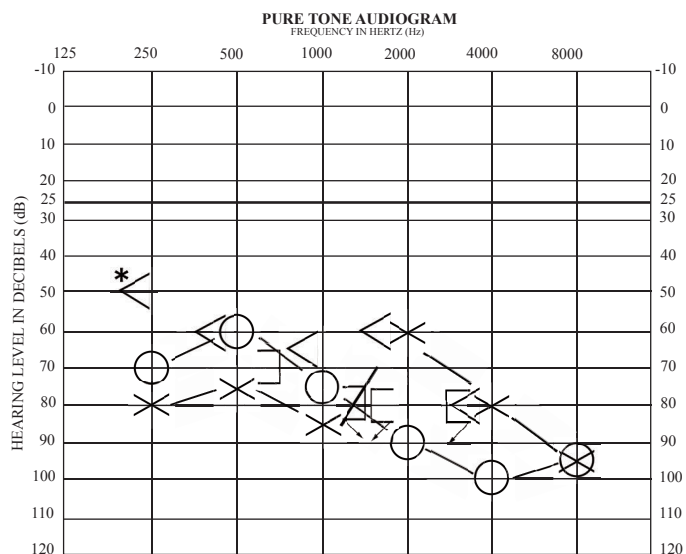
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Slight HL sloping to moderately-severe at 2000 Hz rising to mild HL from 4000 to 8000 Hz Left Ear: Moderate HL sloping to moderately-severe at 2000 Hz rising to mild HL at 8000 Hz PTA: RE, 47 dB; LE, 58 dB Asymmetric
Determine the type of hearing loss	Mixed (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Mixed hearing loss, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Excellent Left Ear: Excellent
Determine tympanometry type (Figure 3–1)	Right Ear: Type B tympanogram Left Ear: Type C tympanogram
Check ear canal volume	Right Ear: Large ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: The right ear indicates a slight HL sloping to moderately severe HL at 2000 Hz rising to mild mixed hearing loss, with a patent PET in the right ear. The left ear indicates a moderate HL sloping to moderately-severe at 2000 Hz rising to mild mixed hearing loss, with negative middle ear pressure. Conductive component in low frequencies is consistent with a patent PET in the right ear and eustachian tube dysfunction in the left ear.

Recommendations: ENT referral for evaluation of left ear and consideration of PET replacement. Return to audiology clinic after medical management. Consider amplification for both ears and preferential seating in the classroom.

Audiogram 49: Advanced

NAME: 49 DATE: _____ BY: _____
Age: 18



SPEECH AUDIOMETRY			☐ RECORDED	☐ MLV
EAR	SRT	SDT	WRS	LEVEL
R	75		72%	110 dB HL
L	80		80%	110 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED			△	
SOUND FIELD		S		
AIDED		A		
OTHER TESTS MASKED	*			

IMMITTANCE TESTING

Tympanometry		
	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	10	10
P.V.T	0.8	0.8
COMPLIANCE	0.8	0.7

Reflex Threshold						
	Contralateral				Ipsilateral	
	.5K	1K	2K	4K	.5K	1K 2K
RE		NR	_____	_____	NR	_____
LE		NR	_____	_____	NR	_____

REFLEX DECAY: RT. _____
LT. _____

REFERRED BY:

HISTORY: Pt. c/o tinnitus and sudden decrease in hearing.

COMMENTS: _____

Audiogram 49: Advanced—Interpretation

Audiogram: ____49____

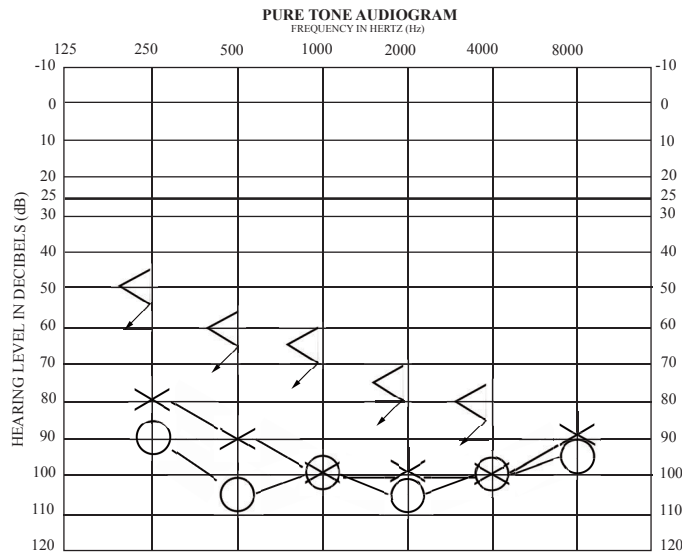
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Moderately severe to profound HL Left Ear: Severe rising to moderate at 2 kHz sloping to profound HL PTA: RE, 75 dB; LE, 73 dB Asymmetric
Determine the type of hearing loss	Sensorineural (bilateral)
Check if audiogram fits a classic pattern (Figure 1–5 – 1–16)	N/A
Compare speech thresholds to pure tones	SRT in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Fair Left Ear: Good
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Right Ear: Normal ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	Absent ipsilateral and contralateral acoustic reflexes, bilaterally

Assessment: Testing reveals a moderately severe to profound sensorineural hearing loss with a normal tympanogram and absent ARTs in the right ear. The left ear reveals a severe rising to moderate at 2 kHz sloping to profound sensorineural hearing loss with a normal tympanogram and absent ARTs.

Recommendations: ENT referral to discuss options for sudden sensorineural hearing loss (oral steroid taper or intratympanic steroid injections) and MRI to rule out retrocochlear lesions. Return to clinic after medical management.

Audiogram 50: Advanced

NAME: 50 DATE: _____ BY: _____
 Age: 18



SPEECH AUDIOMETRY ☐ RECORDED ☐ MLV				
EAR	SRT	SDT	WRS	LEVEL
R	95		24%	110 dB HL
L	90		28%	110 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD Std.

TEST METHOD: _____

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	-10	-15
P.V.T	0.8	0.8
COMPLIANCE	0.5	0.6

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE						NR	
LE						NR	

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient with history of tinnitus and sudden decrease in hearing. Patient received steroid treatment per
ENT 4 weeks ago and is here for his post-treatment audio. Hearing subjectively improved with steroids then
 COMMENTS: worsened again once therapy completed. ENT saw patient again and diagnosed Autoimmune Inner Ear Disease
(AIED)

Audiogram 50: Advanced—Interpretation

Audiogram: ____50____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Profound HL Left Ear: Severe to profound HL PTA: RE, 103 dB; LE, 96 dB Slight asymmetry
Determine the type of hearing loss	Sensorineural (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Bilateral SNHL
Compare speech thresholds to pure tones	SRT in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Poor Left Ear: Poor
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Right Ear: Normal ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	Absent ipsilateral and contralateral acoustic reflexes, bilaterally

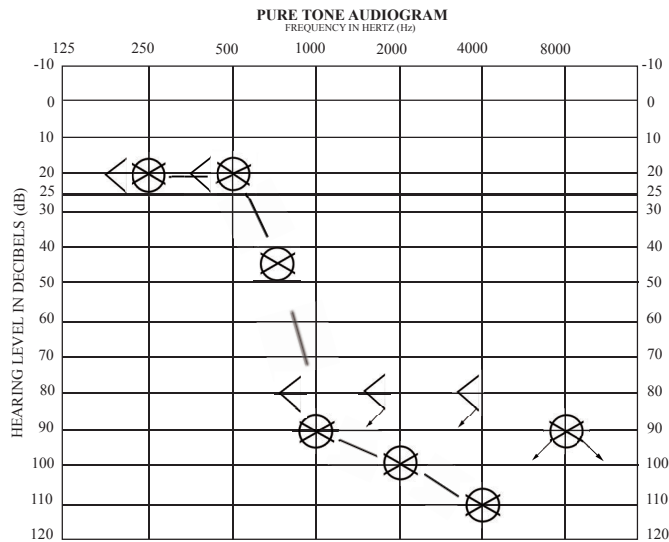
*See previous audio to compare pre and post treatment results (audio #49)

Assessment: Testing indicates a profound sensorineural hearing loss with a normal tympanogram and absent ARTs in the right ear. The left ear indicates a profound sensorineural hearing loss with a normal tympanogram and absent ARTs. Results are most suggestive of autoimmune sensorineural hearing loss.

Recommendations: Patient was initially steroid responsive but worsened again after therapy finished. Rheumatology referral should be considered for longer term steroid course or other immunosuppressive therapies. Serial audiograms may be needed and patient will need amplification and/or cochlear implant evaluation.

Audiogram 51: Advanced

NAME: 51 DATE: _____ BY: _____
 Age: 7



SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	75		24%	110 dB HL
L	80		32%	110 dB HL

DPOAE:

Right Ear: Absent 1 to 8 kHz

Left Ear: Absent 1 to 8 kHz

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD
 TEST METHOD: CPA

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0.6	0.6
P.V.T	1.0	1.0
COMPLIANCE	0.5	0.6

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: 7-year-old patient presents with medulloblastoma and is seen for ototoxic monitoring. Currently receiving chemotherapy with risk for ototoxicity.

COMMENTS: _____

Audiogram 51: Advanced—Interpretation

Audiogram: ____51____

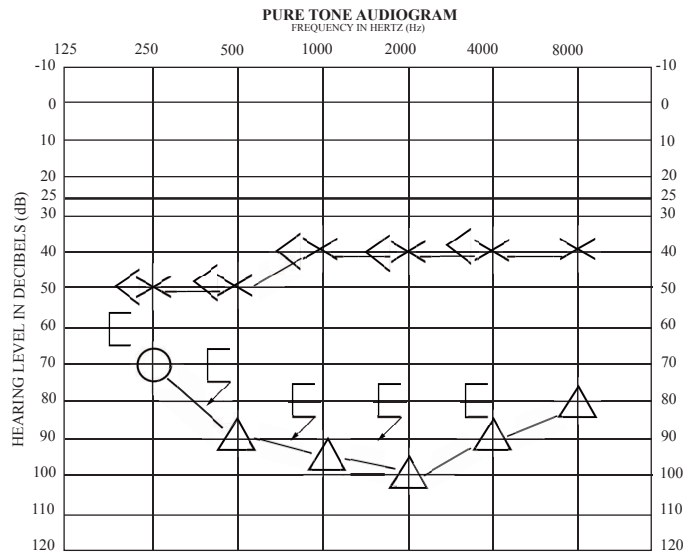
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Normal hearing precipitously sloping to profound HL, bilaterally PTA: RE, 70 dB; LE, 70 dB Symmetric
Determine the type of hearing loss	Sensorineural (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Sensorineural HL, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Poor Left Ear: Poor
Determine tympanometry type (Figure 3–1)	Type A tympanogram, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Normal hearing precipitously sloping to profound sensorineural hearing loss, bilaterally.

Recommendations: Follow up per oncology protocol. Consider amplification options (hearing aids or cochlear implants—hybrid).

Audiogram 52: Advanced

NAME: 52 DATE: _____ BY: _____
 Age: 60



SPEECH AUDIOMETRY				
			☒ RECORDED	☐ MLV
EAR	SRT	SDT	WRS	LEVEL
R	95	75	22%	110 dB HL*
L	45		84%	80 dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	0.5	0.6
COMPLIANCE	0.6	0.6

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE	NR	NR	NR		NR	NR	NR
LE	NR	NR	NR		NR	NR	NR

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: _____

COMMENTS: Patient with known moderate to mild sensorineural hearing loss, bilaterally, with brief episodes of dizziness only lasting 30 minutes. Patient c/o severe dizziness lasting about 4 hours, aural fullness, and intermittent "buzzing".
She reported a decrease in hearing in her right ear.

Negative Stenger

Audiogram 52: Advanced—Interpretation

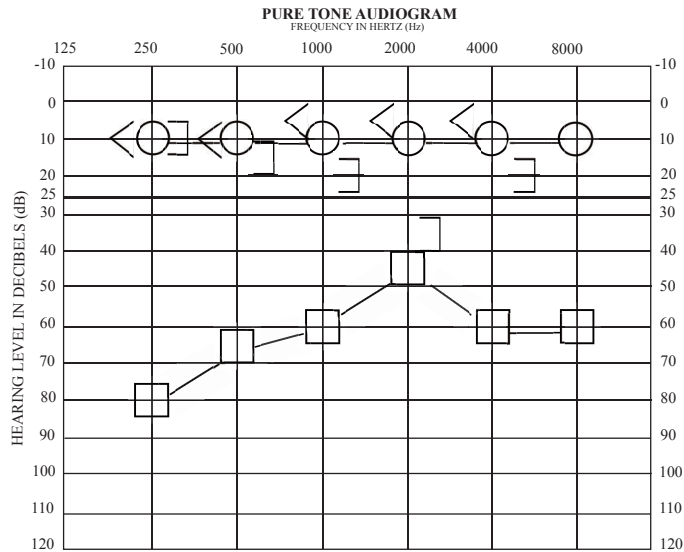
Audiogram: ____52____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Severe to profound HL Left Ear: Moderate to mild HL PTA: RE, 95 dB; LE, 43 dB Asymmetric
Determine the type of hearing loss	Sensorineural (asymmetrical)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Meniere’s disease SNHL
Compare speech thresholds to pure tones	SRT in agreement with PTs for the left ear and could not complete in the right ear
Assess word recognition score (Figure 2–3)	Right Ear: Poor Left Ear: Good
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Right Ear: Normal ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	Ipsilateral and contralateral responses are absent, bilaterally.

Assessment: Testing indicates a severe to profound sensorineural hearing loss in the right ear and a moderate to mild sensorineural hearing loss in the left ear. Tympanograms were normal in both ears. Case history, symptoms, and audiometric results are suggestive of Meniere’s disease.

Recommendations: ENT referral for discussion of treatment options for Meniere’s disease. Formal vestibular evaluation referral to evaluate vestibular function may be warranted.

Audiogram 53: Advanced

NAME: 53 DATE: _____ BY: _____Age: 55

SPEECH AUDIOMETRY

☒ RECORDED☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	10		100%	50 dB HL
L	55		100%	85 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD
 TEST METHOD: Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	1.2	1.0
COMPLIANCE	0.5	0.5

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE						85	90
LE						NR	NR

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents to clinic with gradual hearing loss in her left ear with tinnitus. She denies aural drainage, otalgia, and vertigo.

COMMENTS: _____

Audiogram 53: Advanced—Interpretation

Audiogram: ____53____

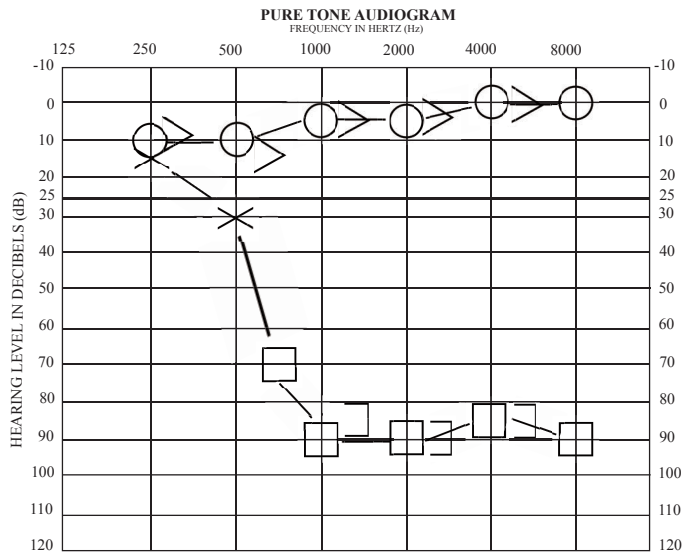
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Normal hearing Left Ear: Severe to moderate HL PTA: RE, 10 dB; LE, 56 dB Asymmetry
Determine the type of hearing loss	Conductive with sensorineural component at 2000 Hz. Also known as Carhart's notch.
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Otosclerosis
Compare speech thresholds to pure tones	SRT in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent in both ears.
Determine tympanometry type (Figure 3–1)	Type A tympanogram, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Present ipsilateral reflexes in the right ear and absent ipsilateral reflexes in the left ear

Assessment: Testing reveals normal hearing in the right ear and a severe conductive hearing loss rising to moderate sensorineural hearing loss sloping to moderately severe conductive hearing loss with a 2000-Hz notch (also known as Carhart's notch). Carhart's notch, hearing loss, and type A tympanogram suggest otosclerosis.

Recommendations: Referral to ENT to discuss suspected diagnosis of otosclerosis options of amplification (hearing aid or BAHS) versus surgery. If tuning forks flipped (AC < BC), then stapedectomy may be best option.

Audiogram 54: Advanced

NAME: 54 DATE: _____ BY: _____
 Age: 20



SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	15		100%	50 dB HL
L	60		20%	95 dB HL(MCL)

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD
 TEST METHOD: Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	1.0	1.0
COMPLIANCE	0.4	0.6

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE					95	A	A
LE					95	A	A

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient seen for urgent audio. Patient presents with sudden hearing loss in left ear. History includes congenital heart disease. Patient denies otorrhea. Reports tinnitus in left ear and occasional dizziness.

COMMENTS: _____

Audiogram 54: Advanced—Interpretation

Audiogram: ____54____

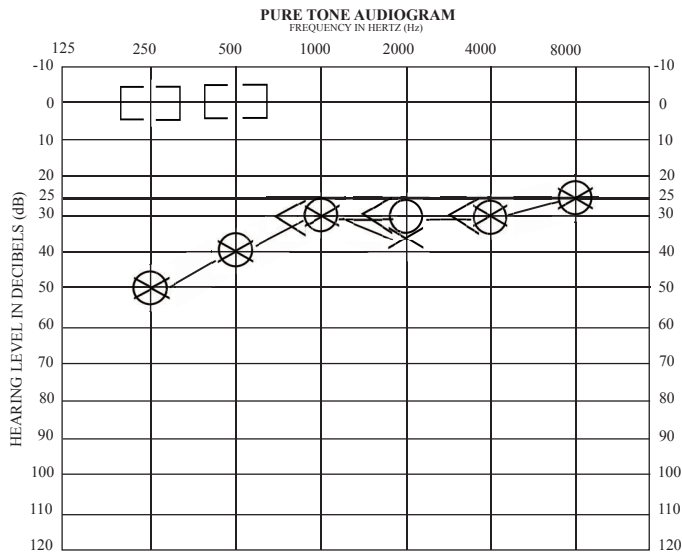
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Normal hearing Left Ear: Normal hearing at 250 Hz sloping to profound HL PTA: RE, 6 dB; LE, 70 dB Asymmetric
Determine the type of hearing loss	Sensorineural (unilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral SNHL
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Excellent Left Ear: Poor
Determine tympanometry type (Figure 3–1)	Type A tympanogram, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Present ipsilateral and contralateral reflexes at 500 Hz and absent at 1000 and 2000 Hz in both ears

Assessment: The right ear indicates normal hearing sensitivity with a normal tympanogram. The left ear indicates normal hearing at 250 Hz steeply sloping to profound sensorineural hearing loss with a normal tympanogram. Audiologic exam indicates for immediate referral to ENT.

Recommendations: ENT referral for consideration of high-dose steroid taper and MRI to rule out a retro-cochlear lesion. Repeat audiogram after treatment administered.

Audiogram 55: Advanced

NAME: 55 DATE: _____ BY: _____
 Age: 6



SPEECH AUDIOMETRY ☐ RECORDED ☒ MLV

EAR	SRT	SDT	WRS	LEVEL
R	30		92%	70 dB HL
L	30		96%	70 dB HL

DPOAE:

Right Ear: Absent 1-8 kHz

Left Ear: Absent 1-8 kHz

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
 TEST METHOD: CPA

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	-20	-30
P.V.T	0.5	0.5
COMPLIANCE	0.6	0.6

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient passed newborn hearing screening in both ears. Parents deny aural drainage, ear infections, and family history of congenital hearing loss.

COMMENTS: Patient reportedly has been saying "what" frequently. Hearing difficulty started shortly after falling off a moving wagon and hitting head on concrete. Patient was not admitted to the ER.

Audiogram 55: Advanced—Interpretation

Audiogram: ____55____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Moderate rising to mild hearing loss, bilaterally PTA: RE, 33 dB; LE, 35 dB Symmetric
Determine the type of hearing loss	Mixed (bilaterally)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Mixed HL, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Excellent Left Ear: Excellent
Determine tympanometry type (Figure 3–1)	Type A tympanograms, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	N/A

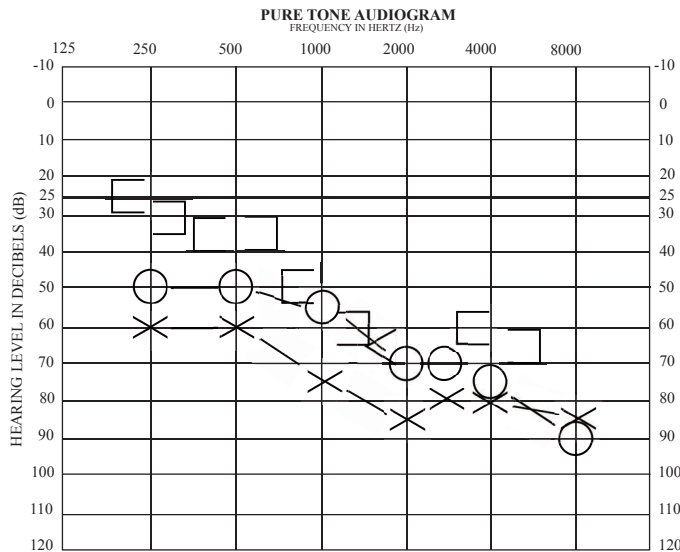
Assessment: Testing indicates a moderate rising to mild mixed hearing loss and normal tympanograms, bilaterally. Patient passed newborn hearing screening and is now experiencing hearing loss shortly after a minor head injury.

Recommendations: ENT referral for evaluation. CT scan to assess temporal bone structures.

PEARL: Key findings of EVA: low-frequency conductive component with type A tympanogram, sudden decrease in hearing after minor head injury, progressive or fluctuating HL. EVA can be a sign of a genetic disorder called Pendred syndrome.

Classic presentation of enlarged vestibular aqueduct (EVA). Recommend hearing aid evaluation to discuss amplification options. Careful fitting and close monitoring of amplification is also imperative due to the fluctuating and progressive nature of hearing loss.

Audiogram 56: Advanced

NAME: 56 DATE: _____ BY: _____Age: 70

SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	60		96%	95 dB HL
L	65		88%	100 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
 TEST METHOD: ☐ SOUNDFIELD Std.

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	B	B
MIDDLE EAR PRESSURE	---	---
P.V.T	1.4	1.5
COMPLIANCE	---	---

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents to clinic with known history of mild to moderately-severe sensorineural hearing loss in both ears and wears hearing aids. Patient c/o decrease in hearing in both ears.

COMMENTS: _____

Audiogram 56: Advanced—Interpretation

Audiogram: ____56____

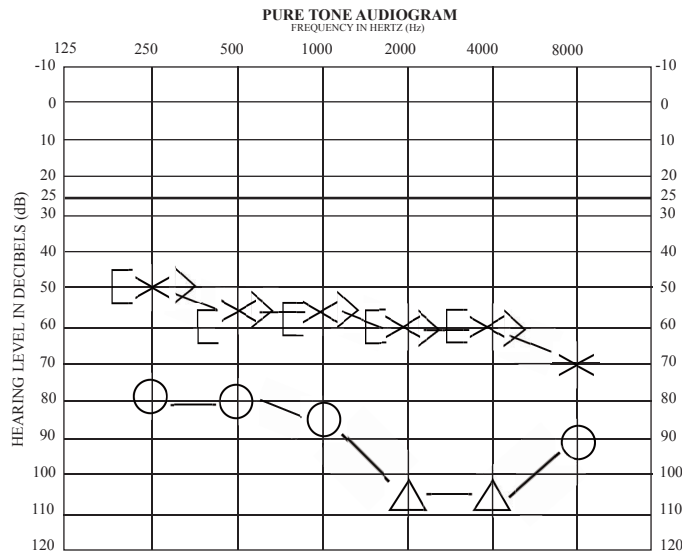
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Moderate to severe HL Left Ear: Moderately severe to severe HL PTA: RE, 58 dB; LE, 73 dB Asymmetry
Determine the type of hearing loss	Mixed (bilaterally)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Mixed hearing loss, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Excellent Left Ear: Good
Determine tympanometry type (Figure 3–1)	Type B tympanogram, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: The right ear indicates a moderate to severe mixed hearing loss with a type B tympanogram. The left ear indicates a moderately severe to severe mixed hearing loss with a type B tympanogram. Results are suggestive of probable middle ear pathology, bilaterally.

Recommendations: ENT referral; return to audiology clinic after medical management. Enable volume control on hearing aids to allow patient to adjust as needed.

This patient presents with a new conductive component to their hearing loss as evidenced by the air-bone gap that is now present. This is highly suggestive of a new bilateral middle ear pathology, such as serous otitis media.

Audiogram 57: Advanced

NAME: 57 DATE: _____ BY: _____Age: 60

SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	90		76%	110 dB HL
L	55		80%	95 dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD
 TEST METHOD: Standard

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	B	A
MIDDLE EAR PRESSURE	---	10
P.V.T	7.0	1.0
COMPLIANCE	---	0.2

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with unknown etiology to his bilateral moderate to moderately-severe sensorineural hearing loss. Patient recently went scuba diving and experienced severe otalgia (ear pain). Patient denies aural drainage, tinnitus, and vertigo. Patient wears binaural hearing aids.

COMMENTS: _____

Otoscopy:

Right Ear: TM perforation

Left Ear: clear canal

Audiogram 57: Advanced—Interpretation

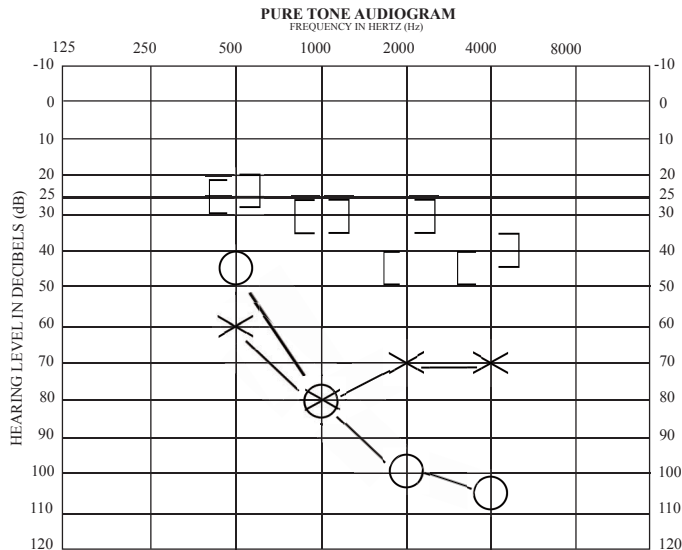
Audiogram: ____57____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Severe to profound HL Left Ear: Moderate to moderately severe HL PTA: RE, 90 dB; LE, 56 dB Asymmetric
Determine the type of hearing loss	Right Ear: Mixed Left Ear: Sensorineural
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Sensorineural HL with mixed HL in the opposite ear
Compare speech thresholds to pure tones	SRT in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Good Left Ear: Good
Determine tympanometry type (Figure 3–1)	Right Ear: Type B tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Right Ear: Large ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Testing reveals a severe to profound mixed hearing loss in the right ear with an abnormal (type B) tympanogram with a large ECV, consistent with a TM perforation. The left ear reveals a moderate to moderately severe sensorineural hearing loss.

Recommendations: ENT referral for microscopic examination and discussion of options (observation to await self-closure of perforation versus tympanoplasty). Patient to follow up with managing audiologist to have hearing aid temporarily adjusted for the right ear.

Audiogram 58: Advanced

NAME: 58 DATE: _____ BY: _____Age: 6

SPEECH AUDIOMETRY

☐ RECORDED☒ MLV

EAR	SRT	SDT	WRS	LEVEL
R	85	50	22%	100 dB HL
L	75	60	24%	95 dB HL
PBK-50 WRS				

Spondee Picture
Cards for SRT

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD **CPA**

TEST METHOD: _____

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	0.6	0.6
COMPLIANCE	0.4	0.4

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with history of Kabuki syndrome. CT scan reveals bilateral inner ear malformations.

COMMENTS: _____

Audiogram 58: Advanced—Interpretation

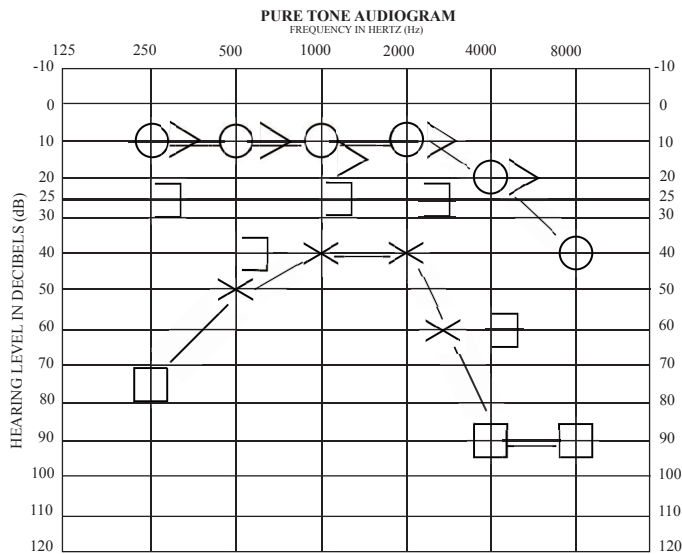
Audiogram: ____58____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Moderate to profound HL Left Ear: Moderately severe to severe HL PTA: RE, 75 dB; LE, 70 dB Asymmetric
Determine the type of hearing loss	Mixed (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Mixed HL, bilaterally
Compare speech thresholds to pure tones	SRT in agreement with PTs (used spondee card pictures) *Notice SDT is NOT in agreement with PTs. Why? Because oftentimes a person will detect a sound softer than they will be able to repeat a spondee with a sloping hearing loss.
Assess word recognition score (Figure 2–3)	WRS is poor, bilaterally.
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Testing indicates a moderate to profound mixed hearing loss in the right ear with a normal tympanogram. The left ear indicates a moderately severe to severe mixed hearing loss with a normal tympanogram. CT scan confirms abnormal inner ear structures, consistent with audiometric thresholds.

Recommendations: Consider amplification and follow up with ENT for evaluation and medical clearance.

Audiogram 59: Advanced

NAME: 59 DATE: _____ BY: _____Age: 64

SPEECH AUDIOMETRY

☒ RECORDED☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	15		100%	50 dB HL
L	50		88%	90 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD Std.

TEST METHOD: _____

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	B
MIDDLE EAR PRESSURE	0	---
P.V.T	1.0	6.0
COMPLIANCE	0.5	---

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with history of known sensorineural hearing loss in the left ear. Arrives to clinic with c/o decrease in hearing for the left ear.

COMMENTS: _____

Otoscopy: Non-occluding cerumen, AU

Audiogram 59: Advanced—Interpretation

Audiogram: ____59____

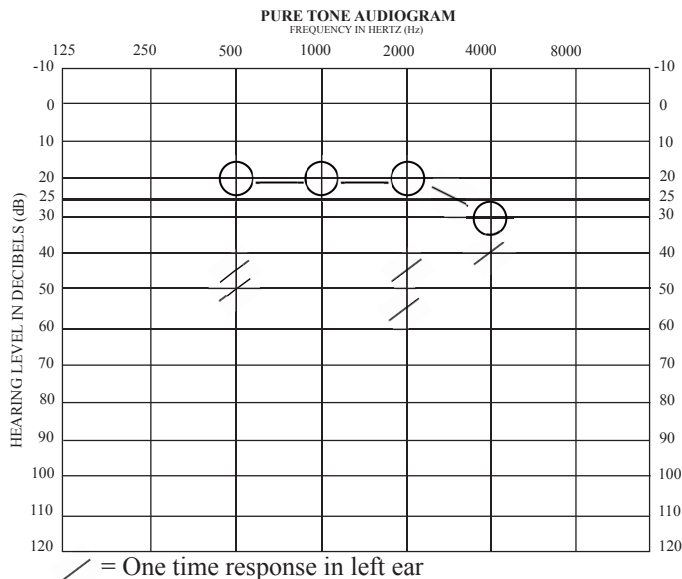
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Normal hearing sloping to mild hearing loss at 8000 Hz Left Ear: Severe rising to mild HL in mid-frequencies sloping to profound high-frequency HL PTA: RE, 10 dB; LE, 43 dB; asymmetric
Determine the type of hearing loss	Mixed (unilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral mixed HL
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Excellent Left Ear: Good
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type B tympanogram
Check ear canal volume	Right Ear: Normal ECV Left Ear: Large ECV
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Normal hearing from 250 to 4000 Hz sloping to a mild hearing loss at 8000 Hz in the right ear and a severe low-frequency HL rising to mild HL sloping to profound high-frequency mixed HL. Left ear thresholds are consistent with abnormal tympanogram with large ECV, suggesting TM perforation.

Recommendations: Referral to ENT to perform microscopic evaluation. Discuss options of observation to see if closure occurs versus tympanoplasty. Return to audiology clinic after medical management.

Audiogram 60: Advanced

NAME: 60 DATE: _____ BY: _____
 Age: 8



SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	25			
L	30			

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD
 TEST METHOD: CPA - Good to Fair

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED	*			

IMMITTANCE TESTING				
Tympanometry				
TYMPANOGRAM TYPE		RIGHT	LEFT	
		A	B	
MIDDLE EAR PRESSURE		0	---	
P.V.T		0.5	0.6	
COMPLIANCE		0.4	---	

Reflex Threshold						
	Contralateral				Ipsilateral	
	.5K	1K	2K	4K	.5K	1K 2K
RE						
LE						

REFLEX DECAY:	RT. _____
	LT. _____

REFERRED BY: _____

HISTORY: Previous ABR results from one year ago indicated normal hearing sensitivity, bilaterally. Patient here for routine hearing test. Patient presents with developmental delay.

COMMENTS: _____

Audiogram 60: Advanced—Interpretation

Audiogram: ____60____

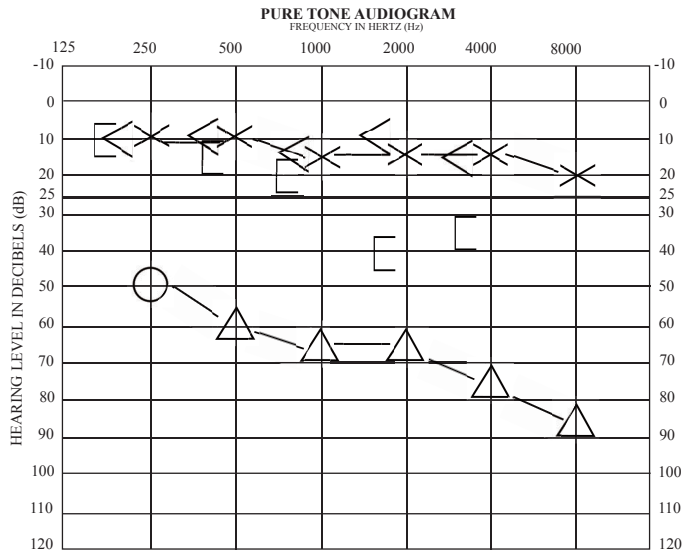
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Normal to mild HL Left Ear: One-time responses indicate possible mild HL. PTA: RE, 20 dB; LE, N/A Asymmetric
Determine the type of hearing loss	Undetermined
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	N/A
Compare speech thresholds to pure tones	SRT in agreement with PTs.
Assess word recognition score (Figure 2–3)	N/A
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type B tympanogram
Check ear canal volume	Right Ear: Normal ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Right ear indicates normal to mild hearing loss with normal middle ear mobility. The left ear indicates one-time responses in the mild hearing loss range. Patient fatigued during left ear testing as kids are usually unpredictable.

Recommendations: ENT follow-up as scheduled for examination to rule out middle ear disease on the left side given type B tympanogram.

Audiogram 61: Advanced

NAME: 61 DATE: _____ BY: _____
 Age: 70



SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	60		100%	100 dB HL
L	15		100%	55 dB HL

TRANSDUCER: ☒ HEADPHONES ☐ INSERTS
☐ SOUNDFIELD
 TEST METHOD: _____ Std. _____

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	B	A
MIDDLE EAR PRESSURE	---	0
P.V.T	1.5	1.5
COMPLIANCE	---	0.5

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE					NR	NR	NR
LE					85	85	85

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient c/o decreased hearing, ear pain, and pulsing tinnitus in her right ear.

COMMENTS: Otoscopy:
Right Ear: Red mass behind TM
Left Ear: Clear canal

Audiogram 61: Advanced—Interpretation

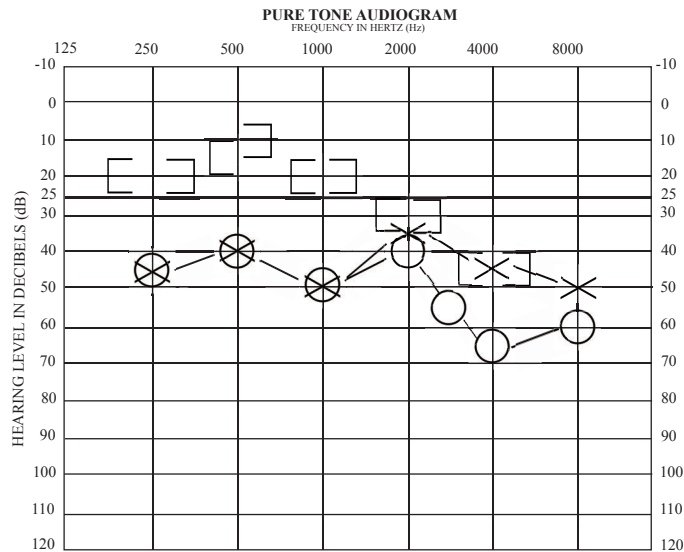
Audiogram: ____61____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Moderate sloping to severe HL Left Ear: Normal hearing PTA: RE, 63 dB; LE, 13 dB Asymmetry
Determine the type of hearing loss	Right Ear: Mixed Left Ear: Normal
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Unilateral mixed hearing loss
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally.
Determine tympanometry type (Figure 3–1)	Right Ear: Type B tympanogram Left Ear: Type A tympanogram
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Ipsilateral Right Ear: Absent Left Ear: Present

Assessment: Moderate sloping to severe mixed hearing loss in the right ear with an abnormal tympanogram suggesting probable middle ear pathology and normal hearing in the left ear. Otoscopy, symptoms, audiologic, and tympanometric results are suggestive of Glomus tumor.

Recommendations: Follow up with ENT to discuss surgical options after imaging obtained. Consider amplification if treatment is not an option at this time. Return to audiology clinic after medical management for repeat audiogram.

Audiogram 62: Advanced

NAME: 62 DATE: _____ BY: _____Age: 9

SPEECH AUDIOMETRY				
			☒ RECORDED	☐ MLV
EAR	SRT	SDT	WRS	LEVEL
R	45		100%	85 dB HL
L	40		100%	80 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
 TEST METHOD: ☐ SOUNDFIELD Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	B	B
MIDDLE EAR PRESSURE	---	---
P.V.T	0.8	0.8
COMPLIANCE	---	---

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE						CNT	

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: History of bilateral mild to moderate high-frequency sensorineural hearing loss and PE tubes in both ears.Patient c/o difficulty hearing in both ears.

COMMENTS: _____

Otoscopy:Right Ear: PET extruded in ear canal - Middle ear effusion noted in both ears.Left Ear: PET extruded in ear canal

Audiogram 62: Advanced—Interpretation

Audiogram: ____62____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Mild to moderately severe HL Left Ear: Mild to moderate HL PTA: RE, 40 dB; LE, 36 dB Asymmetric at 4 kHz
Determine the type of hearing loss	Mixed (bilateral).
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Mixed HL, bilaterally
Compare speech thresholds to pure tones	SRT in agreement with PTs.
Assess word recognition score (Figure 2–3)	WRS is excellent, bilaterally.
Determine tympanometry type (Figure 3–1)	Type B tympanogram, bilaterally
Check ear canal volume	Normal ECV, bilaterally
Review acoustic reflexes (Figure 3–8)	Could not test

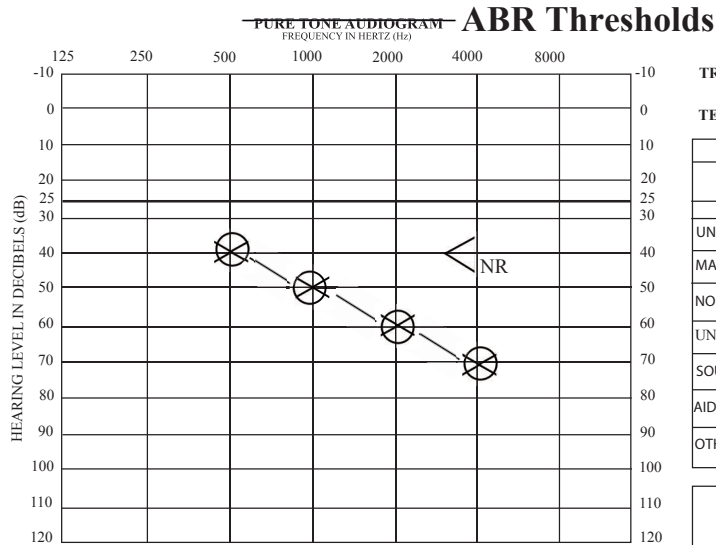
Assessment: Right ear indicates a mild to moderately severe mixed hearing loss with a type B tympanogram. The left ear indicates a mild to moderate mixed hearing loss with a type B tympanogram. Results are suggestive of probable middle ear pathology in both ears.

Recommendations: ENT referral for microscopic examination and consideration of interventions, which may include PET placement if effusions noted. Return to audiology clinic after medical management for repeat testing.

Audiogram 63: Advanced

NAME: 63 DATE: _____ BY: _____

Age: 1 month



SPEECH AUDIOMETRY ☐ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R				
L				

DPOAE:

Right Ear: Absent 1- 8 kHz

Left Ear: Absent 1 - 8 kHz

REFERRED BY: _____

HISTORY: One-month-old patient referred on his newborn hearing screening in both ears and is here for his subsequent ABR testing.
Patient was born full term, no birth or medical complications, no syndromic features, and no family history of congenital hearing loss.

COMMENTS: _____

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD

TEST METHOD: _____

	KEY			
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	□	□
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED	*			

IMMITTANCE TESTING

1000 Hz

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE		
P.V.T		
COMPLIANCE		

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

Audiogram 63: Advanced—Interpretation

Audiogram: 63

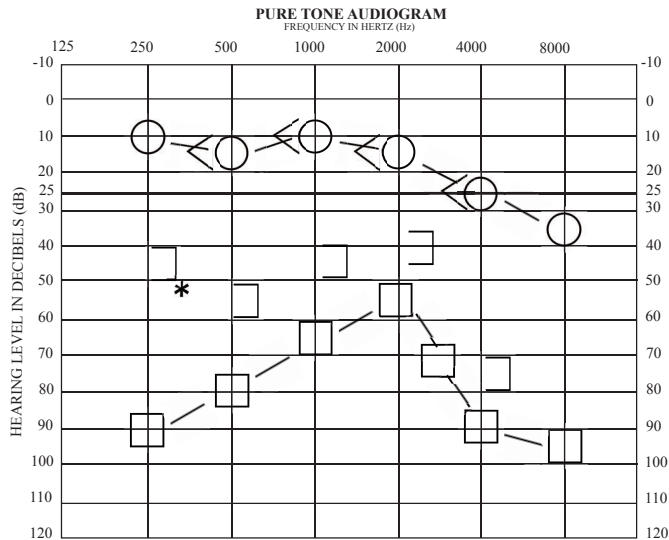
Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Mild sloping to moderately severe HL
Determine the type of hearing loss	Sensorineural—no response at 40 dB is sometimes the highest ABR equipment can test bone conduction. Cannot always rule out mixed hearing losses.
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	
Compare speech thresholds to pure tones	N/A
Assess word recognition score (Figure 2–3)	N/A
Determine tympanometry type (Figure 3–1)	Type A tympanogram, bilaterally
Check ear canal volume	N/A—using 1000-Hz tympanogram
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Results indicate a mild sloping to moderately severe sensorineural hearing loss, bilaterally. DPOAEs were absent in both ears, consistent with ABR responses. Tympanometry was normal, suggesting normal middle ear function. Mixed hearing loss cannot be ruled out.

Recommendations: ENT referral for medical workup for congenital causes of SNHL (CMV, ototoxicity, genetics etc.). Return to audiology clinic for a hearing aid evaluation. Consider genetic evaluation (connexin 26/30, etc.). Referral to state Early Intervention program.

Audiogram 64: Advanced

NAME: 64 DATE: _____ BY: _____
 Age: 26



SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	10		96%	50 dB HL
L	65		84%	95 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD
 TEST METHOD: Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↙	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMpanogram TYPE	A	B
MIDDLE EAR PRESSURE	5	---
P.V.T	1.3	5.0
COMPLIANCE	0.5	---

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	.5K	1K	2K
RE						80	NR
LE						NR	NR

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents with sudden decrease in hearing in the left ear. He reports it happened shortly after a gunshot went off unexpectedly next to his left ear. He experienced tinnitus and occasionally still does. He reports severe otalgia. Patient denies aural drainage and vertigo.

COMMENTS: _____

Otoscopy:

Right Ear - clear canal

Left Ear - perforation of TM visualized

Audiogram 64: Advanced—Interpretation

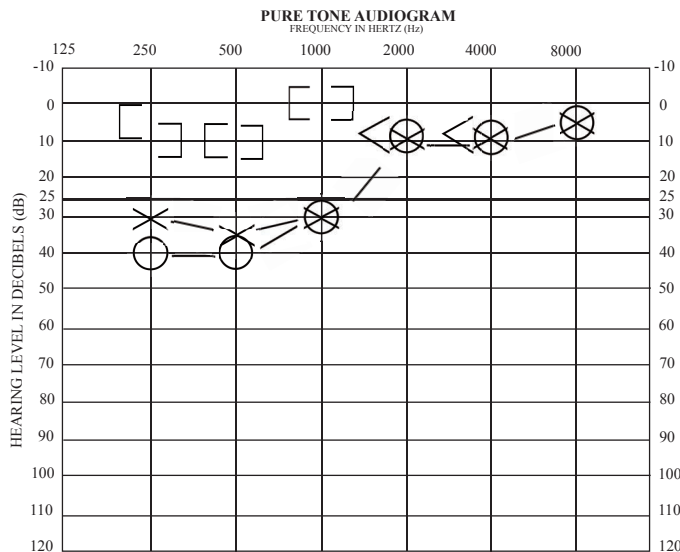
Audiogram: 64

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Right Ear: Normal hearing sloping to mild HL Left Ear: Severe rising to moderate at 2000 Hz sloping to profound HL at 8000 Hz PTA: RE, 13 dB; LE, 66 dB Asymmetric
Determine the type of hearing loss	Right Ear: Sensorineural Left Ear: Mixed
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	N/A
Compare speech thresholds to pure tones	SRT in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Excellent Left Ear: Good
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram Left Ear: Type B tympanogram
Check ear canal volume	Right Ear: Normal ECV Left Ear: Large ECV
Review acoustic reflexes (Figure 3–8)	Right ipsilateral and contralateral responses are present. Left ipsilateral and contralateral responses are absent.

Assessment: Testing indicates normal hearing sloping to mild SNHL with a normal tympanogram in the right ear. The left ear indicates a severe rising to moderate mixed HL at 2000 Hz sloping to profound mixed HL at 8000 Hz with an abnormal (type B) tympanogram with a large ECV, consistent with TM perforation.

Recommendations: ENT referral for examination of the perforation. It can be observed for self-resolution of the perforation and the conductive component. For any persistent hearing loss, consider amplification.

Audiogram 65: Advanced

NAME: 65 DATE: _____ BY: _____Age: 55

SPEECH AUDIOMETRY ☒ RECORDED ☐ MLV

EAR	SRT	SDT	WRS	LEVEL
R	25		92%	80 dB HL
L	25		96%	80 dB HL

TRANSDUCER: ☐ HEADPHONES ☒ INSERTS
☐ SOUNDFIELD

TEST METHOD: Std.

KEY				
	AIR		BONE	
	R	L	R	L
UNMASKED	○	×	<	>
MASKED	△	□	◻	◻
NO RESPONSE		↘	↗	
UNSPECIFIED				△
SOUND FIELD			S	
AIDED			A	
OTHER TESTS MASKED			*	

IMMITTANCE TESTING

Tympanometry

	RIGHT	LEFT
TYMPANOGRAM TYPE	A	A
MIDDLE EAR PRESSURE	0	0
P.V.T	1.0	1.1
COMPLIANCE	0.6	0.6

Reflex Threshold

	Contralateral				Ipsilateral		
	.5K	1K	2K	4K	5K	1K	2K
RE							
LE							

REFLEX DECAY: RT. _____
 LT. _____

REFERRED BY: _____

HISTORY: Patient presents to the clinic with c/o hearing loss in both ears, aural fullness, pressure induced vertigo, and pulsatile tinnitus.

COMMENTS: _____

Otoscopy:

Right Ear - Clear canal

Left Ear - Clear canal

Audiogram 65: Advanced—Interpretation

Audiogram: ____65____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	Mild low-frequency HL rising to normal hearing, bilaterally
Determine the type of hearing loss	Conductive (bilateral)
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	Conductive HL, bilaterally
Compare speech thresholds to pure tones	SRT is in agreement with PTs.
Assess word recognition score (Figure 2–3)	Right Ear: Excellent Left Ear: Excellent
Determine tympanometry type (Figure 3–1)	Right Ear: Type A tympanogram. Left Ear: Type A tympanogram
Check ear canal volume	Right Ear: Normal ECV Left Ear: Normal ECV
Review acoustic reflexes (Figure 3–8)	N/A

Assessment: Testing indicates a mild low-frequency conductive hearing loss rising to normal hearing with normal tympanograms, bilaterally. Patient’s symptoms and low-frequency HL with normal tympanograms may suggest superior semicircular canal dehiscence (SSCD).

Recommendations: ENT referral to discuss diagnosis and consideration for further workup with a thin-cut temporal bone CT scan. Several options for surgical intervention exist to help with symptom resolution.

APPENDIX A

AUDIOGRAM INDEX

BY DIAGNOSIS

Types of Hearing Loss

Asymmetric SNHL (21, 23, 24, 26, 31, 52, 54, 64)
Conductive HL (5, 8, 25, 28, 30, 32, 41, 42, 45, 47)
High-frequency SNHL (3, 7, 13, 15, 17)
Mixed HL (20, 22, 48, 55, 56, 57, 58, 59, 61, 62)
Normal hearing (1, 4, 10, 33)
Sensorineural HL (3, 7, 9, 11, 13, 15, 17, 18, 22, 44, 49, 50, 51, 63)
Unilateral CHL (6, 14, 16, 19, 27, 34, 43, 53)
Unilateral SNHL (21, 29, 34, 36, 54)

Etiologies

Anacusis (11, 18)
Atresia/microtia (12, 40)
Auditory neuropathy spectrum disorder (46)
Autoimmune inner ear disease—AIED (49, 50)
Barotrauma (57)
Cardiovascular/diabetes (26)
Genetic (connexin 26/30) (57)
Glomus tumor (61)
Head trauma (29)
Idiopathic (11)
Inner ear malformation (58)
Malingering (37)
Meniere's disease (52)
Meningitis (18, 44)
Middle ear disorders:
 Cholesteatoma (19, 45)
 Eustachian tube dysfunction (19)

Exostosis (47)
Mastoidectomy (27, 43)
Ossicular discontinuity (22)
Otitis externa (25)
Otitis media (5, 8, 14, 41, 45, 56)
Otosclerosis (20, 53)
Pressure equalization tubes (4, 8, 14, 30, 32, 33, 42, 48)
Tympanic membrane perforation (6, 16, 28, 57, 59)
Tympanosclerosis (6)
Noise-induced HL (13, 64)
Ototoxic (17, 51)
Presbycusis (3, 7, 9, 15)
Retrocochlear lesion (23, 24)
Sudden SNHL (54)
Superior semicircular canal dehiscence (65)

Congenital/Progressive/Syndromic HL

Difficult to test (35, 60)
EVA (55)

Headphone Challenges

Collapsing canals (39)
Masking dilemma (40)
Standing wave @ 8 kHz (38)

APPENDIX B

BEDSIDE HEARING TESTS

Outside of an audiogram, other tools exist to assess a patient's hearing status (whispered voice, finger rub, ticking watch, or tuning fork). Foremost of these tools is the tuning fork.^{1,2} Tuning fork testing has high value as it can be done independently without an audiogram. Generally, a 512-Hz tuning fork is used for standard testing, but tuning forks are available for 128 Hz to 2048 Hz if individual frequency testing is needed. Two main tests are utilized, the Rinne and Weber tests.³ Mastery of these tests can prove to be a valuable addition to a clinician's armamentarium as they serve to confirm audiometric testing and guide therapeutic discussions for diseases like otosclerosis.

Weber Testing

The Weber test is the mainstay of clinical hearing assessments as it provides simple, yet informative data on the patient's hearing status. In this test, a tuning fork is placed on a bony prominence in the midline of the skull (generally on the glabella or the central incisors), and the vibrating tuning fork transfers sound via bone conduction into the patient's bilateral cochlea. If the sound is heard in the midline, then one can assume normal or symmetric hearing loss. If the sound lateralizes to one side, then the patient is suspected to have an asymmetric hearing loss present.

To determine the type of hearing loss, the authors of this text like to preface each Weber test with a single, high-value question directed to the patient. When the patient is asked, "Which ear is your better hearing ear?" the clinician can properly interpret the information gathered during the Weber. If the sound lateralizes to the patient's poorer hearing ear, then an ipsilateral conductive hearing loss is suspected. When the sound lateralizes to the better hearing ear, a contralateral sensorineural hearing

loss is assumed. One would report the Weber testing as "midline" or "lateralizes left/right."

Rinne Testing

The Rinne test serves to differentiate between the bone and air conduction hearing thresholds of the tested ear. This maneuver is done by placing the end of the tuning fork on the mastoid tip and holding it in place until the auscultated sound is no longer appreciated. Then the tuning fork is held approximately 1 inch from the ipsilateral ear canal and the patient is asked if the sound is still heard. If the sound is heard, then air conduction is considered greater than bone conduction, and the test is reported to be "positive." This test can serve to confirm that the conductive system is functioning to some degree to amplify sound louder than the bone conduction system. Clinicians may opt to report the Rinne results in terms of air conduction (AC) versus bone conduction (BC) to avoid confusion with the "positive" or "negative" labels. A "positive" test would be written $AC > BC$ and a "negative" test would be written $BC > AC$.

A "negative" test will mainly denote the presence of an air-bone gap. While no specific quantitative information is derived from the Rinne test, it is estimated that when the conductive hearing loss or air-bone gap is less than ~18 dB, the Rinne will remain positive.³ Thus, whenever a negative result is noted ($BC > AC$), the clinician may assume an air-bone gap of at least 18 dB depending on the frequency. One exception to this assumption, however, is seen with unilateral severe to profound sensorineural hearing loss. Crossover stimulation (see Section 1: Pure-Tone Audiometry) from the contralateral normal hearing cochlea can paradoxically result in a louder bone conduction on the poorer side than air conduction even when no conductive hearing loss is present.

The Rinne test can also serve as a confirmation of the Weber testing. For example, if the Weber test lateralizes to the better hearing ear, then one would suspect sensorineural hearing loss is present in the poorer hearing ear and the Rinne should reflect that suspicion ($AC > BC$).

Limitations

While tuning fork testing can be used to verify the type of hearing loss present, these tests are not great at distinguishing the degree of hearing loss. For example, normal hearing and bilateral symmetric sensorineural hearing loss would result in identical fork results (ie, midline Weber and $AC > BC$ on Rinne bilaterally). Similarly, normal hearing ears would produce indistinguishable fork results as a bilateral mild conductive hearing loss. A patient's verbal history

of subjective hearing loss should help differentiate these situations. In the case of a mixed conductive and sensorineural deficit, the fork testing can vary dramatically depending on the severity of each type of loss present. If these situations are suspected, audiometric testing should absolutely be pursued.

References

1. Bagai A, Thavendiranathan P, Detsky AS. Does this patient have hearing impairment? *JAMA*. 2006;295(4):416–428.
2. Pirozzo S, Papinczak T, Glasziou P. Whispered voice test for screening for hearing impairment in adults and children: systematic review. *BMJ*. 2003;327(7421):967.
3. Jacob V, Alexander P, Nalinesha KM, Nayar RC. Can Rinne's test quantify hearing loss? *Ear Nose Throat J*. 1993;72(2):152.

APPENDIX C

OTHER AUDIOMETRIC TESTS

This workbook is dedicated to the battery of tests and techniques that comprise the standard audiogram. Yet, several other important audiometric tests and techniques exist. Given the introductory nature of our text, full attention to this testing was excluded from the primary chapters. Periodically, an audiologist may need to utilize these techniques to obtain further objective information to assist clinicians with more advanced decision making. A brief summary of these diagnostic techniques and tests is listed below. For more information, please refer to more advanced audiologic textbooks to guide your learning on these topics.

Sound Field Testing

Sound field testing is an alternative to the standard techniques for gathering audiometric data. For infant and toddler-aged patients, as well as individuals who are unable to tolerate inserts or headphones, pure-tone and speech threshold testing can be performed via sound field testing. Generally, the testing is obtained via sounds played through a single loudspeaker in the audio booth. Patient reactions to the stimuli are closely observed for any evidence of purposeful response. Responses seen are charted on a standard audiogram grid and denoted with an “S” and do not provide side-specific information, as both ears are present in the sound field created by the single speaker. Sound field audiometry can be used also for individuals of all ages who have special amplification devices (such as bone-anchored hearing aids or cochlear implants), with a goal to measure the functional gain while wearing them in the booth. This type of testing is mentioned in Section 1: Pure-Tone Audiometry.

Visual Reinforcement Audiometry (VRA)

During **visual reinforcement audiometry**, an infant or toddler sits with a test assistant in a sound booth while sounds are presented through speakers on each side or through headphones placed on the child’s ears. When the sound is presented, it is coupled to a lighted toy and the child is provided with positive reinforcement to encourage them to turn toward the sound when it is heard. As the name suggests, the child must be conditioned to perform this test reliably to estimate hearing threshold levels.

Conditioned Play Audiometry (CPA)

For children who have developed the ability to follow directions and focus on simple tasks, the audiologist can use **conditioned play audiometry** to obtain threshold data. Children who can undergo this type of testing are typically toddler to preschool age (between 2 and 5 years old). The test assistant will sit with the child in the soundproof booth and ask them to perform an activity, like placing pegs in a board, balls in a basket, or stacking blocks into a tower, whenever a sound is heard. Depending on how the stimuli are presented (via inserts, headphones, or bone oscillator), an audiologist can determine the type and degree of hearing loss present.

Otoacoustic Emissions

Otoacoustic emissions (OAEs) are measurable emissions produced by a healthy and functioning cochlea whenever sound stimuli are presented. These emissions are generated from the movement

of a subset of cochlear cells (outer hair cells) to sound stimuli and can be measured by placing a receiver into the ear canal. If OAEs are measured, one can then assume that the patient has a normal external auditory canal, a functioning middle ear without disease, as well as the presence of outer hair cells within the cochlea. Since OAE testing does not require behavioral responses from the patient, it is most commonly used as a newborn screening tool for hearing loss. Several other benefits qualify it to be the ideal universal newborn screening tool; OAEs are readily available, ear specific, inexpensive, painless, and able to be performed in any setting. However, any excess environmental noise or complicating patient factors (excessive cerumen, otitis media, middle ear effusions, noncompliant child, etc.) can lead to inconclusive results.

OAE testing involves placing a tiny probe into one or both ears, and while sound stimuli are emitted, a microphone will record the otoacoustic responses. There would be no emissions in a newborn with hearing loss, regardless of the type of hearing loss (conductive versus sensorineural). When this occurs, the newborn is considered to have “failed” the screening and is referred officially to an audiologist for further testing. Since the absence of OAEs cannot assess the specific type or quantify the degree of hearing loss that is present, the referral to an audiologist for further testing is paramount to ensure the patient receives the proper diagnosis and early interventions are devised for any hearing loss that is present. In doing so, the patient can avoid communication delays that can occur when hearing loss is overlooked.

There are many forms of OAEs, many of which are beyond the scope of this text. One of the most commonly utilized forms of emissions testing is **distortion product otoacoustic emissions** (DPOAEs). DPOAEs are measured by playing two simultaneous sounds at the varying frequencies (generally from 1000–8000 Hz) and calculating the cochlear responses. Many audiologists will use this during audiometric assessments to confirm obtained results, especially when a child is unable to cooperate fully

with the behavioral testing or questionable data are generated during standard testing. To find specific examples of DPOAE use, please refer to Section 4: Audiogram Workbook.

Auditory Brainstem Response

Auditory brainstem response (ABR) testing evaluates the cochlea and auditory pathway through objective measurements obtained via electrodes placed on the scalp. Computer-generated sounds are transmitted into headphones or canal inserts. The electrodes relay brain activity back to the computer as the sounds are played through the headphones. An audiologist performs this testing and will provide an interpretation of the responses as they are collected back into the computer. The softest intensity or loudness level (dB) at which the responses are measured approximates the patient's hearing thresholds at that frequency. By testing several sound intensity levels and frequencies, the audiologist can estimate the air or bone thresholds of the tested patient.

Since the measurements are obtained without relying on a patient's ability to respond, unlike much of the standard audiometric testing, this testing is most often performed for infants who fail their newborn hearing screen and congenital hearing loss is suspected. Generally, ABR testing is performed while the infant is napping in the audiology office. As it can take a few hours to perform, children older than ~6 months and younger than 6 or 7 years old usually will undergo anesthesia to have this testing done. Any child older than 7 years can typically undergo an ABR without sedation or anesthesia.

Sometimes, an adult will present with sensorineural hearing loss suspected to be from a retrocochlear pathology (brain, brainstem, or anywhere along the auditory pathway). In these cases, the audiogram may not provide sufficient information and further testing should be considered. In most cases, magnetic resonance imaging (MRI) is performed to evaluate the pathway, but ABRs are used if an MRI is not available, nonrevealing, or unable to be performed.

APPENDIX D COMMON AUDIOLOGIC ACRONYMS

AC	Air conduction	Hz	Hertz
AD	Right ear	IA	Interaural attenuation
AIED	Autoimmune inner ear disease	ITE	In-the-ear hearing instrument
ART	Acoustic reflex threshold	LF	Low frequency
AS	Left ear	MCL	Most comfortable loudness
AU	Both ears	MLV	Monitored live voice
BC	Bone conduction	mmhos	Millimhos
BTE	Behind-the-ear hearing instrument	mL	Milliliter
CNS	Central nervous system	NR	No response
CNT	Cannot test	OAE	Otoacoustic emission
C/O	Complains of	PET	Pressure equalization tube
CPA	Conditioned play audiometry	PTA	Pure-tone average
CPA	Cerebellopontine angle	PVT	Physical volume test (ear canal)
CROS	Contralateral routing of signals	SDT	Speech detection threshold
daPa	Decapascal	SNHL	Sensorineural hearing loss
dB	Decibel	SF	Sound field
DNT	Did not test	SPL	Sound pressure level
DPOAE	Distortion product otoacoustic emissions	SRT	Speech reception threshold
ECV	Ear canal volume	TM	Tympanic membrane
ENT	Ear, nose, and throat physician	UCL	Uncomfortable loudness
HAE	Hearing aid evaluation	VRA	Visual reinforcement audiometry
HF	High frequency	WRS	Word recognition score
HL	Hearing level	WRT	Word recognition testing

BLANK INTERPRETATION SHEETS

Audiogram: _____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	
Determine the type of hearing loss	
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	
Compare speech thresholds to pure tones	
Assess word recognition score (Figure 2–3)	
Determine tympanometry type (Figure 3–1)	
Check ear canal volume	
Review acoustic reflexes (Figure 3–8)	

Assessment:

Recommendations:

Audiogram: _____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	
Determine the type of hearing loss	
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	
Compare speech thresholds to pure tones	
Assess word recognition score (Figure 2–3)	
Determine tympanometry type (Figure 3–1)	
Check ear canal volume	
Review acoustic reflexes (Figure 3–8)	

Assessment:

Recommendations:

Audiogram: _____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	
Determine the type of hearing loss	
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	
Compare speech thresholds to pure tones	
Assess word recognition score (Figure 2–3)	
Determine tympanometry type (Figure 3–1)	
Check ear canal volume	
Review acoustic reflexes (Figure 3–8)	

Assessment:

Recommendations:

Audiogram: _____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	
Determine the type of hearing loss	
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	
Compare speech thresholds to pure tones	
Assess word recognition score (Figure 2–3)	
Determine tympanometry type (Figure 3–1)	
Check ear canal volume	
Review acoustic reflexes (Figure 3–8)	

Assessment:

Recommendations:

Audiogram: _____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	
Determine the type of hearing loss	
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	
Compare speech thresholds to pure tones	
Assess word recognition score (Figure 2–3)	
Determine tympanometry type (Figure 3–1)	
Check ear canal volume	
Review acoustic reflexes (Figure 3–8)	

Assessment:

Recommendations:

Audiogram: _____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	
Determine the type of hearing loss	
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	
Compare speech thresholds to pure tones	
Assess word recognition score (Figure 2–3)	
Determine tympanometry type (Figure 3–1)	
Check ear canal volume	
Review acoustic reflexes (Figure 3–8)	

Assessment:

Recommendations:

Audiogram: _____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	
Determine the type of hearing loss	
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	
Compare speech thresholds to pure tones	
Assess word recognition score (Figure 2–3)	
Determine tympanometry type (Figure 3–1)	
Check ear canal volume	
Review acoustic reflexes (Figure 3–8)	

Assessment:

Recommendations:

Audiogram: _____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	
Determine the type of hearing loss	
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	
Compare speech thresholds to pure tones	
Assess word recognition score (Figure 2–3)	
Determine tympanometry type (Figure 3–1)	
Check ear canal volume	
Review acoustic reflexes (Figure 3–8)	

Assessment:

Recommendations:

Audiogram: _____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	
Determine the type of hearing loss	
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	
Compare speech thresholds to pure tones	
Assess word recognition score (Figure 2–3)	
Determine tympanometry type (Figure 3–1)	
Check ear canal volume	
Review acoustic reflexes (Figure 3–8)	

Assessment:

Recommendations:

Audiogram: _____

Method Steps	Interpretation
Assess for hearing loss (Figure 1–4)	
Determine the type of hearing loss	
Check if audiogram fits a classic pattern (Figures 1–5 to 1–16)	
Compare speech thresholds to pure tones	
Assess word recognition score (Figure 2–3)	
Determine tympanometry type (Figure 3–1)	
Check ear canal volume	
Review acoustic reflexes (Figure 3–8)	

Assessment:

Recommendations:

- Acoustic decay:** Inability to maintain the stapedial reflex in the setting of a sustained or repeated stimulus, suggestive of a retrocochlear lesion, most likely an eighth cranial nerve tumor.
- Acoustic neuroma:** Benign tumor of the eighth cranial nerve, which usually stems from the vestibular portion of the vestibulocochlear nerve. For this reason, it is more accurately referred to as a vestibular schwannoma.
- Acoustic reflex:** Automatic bilateral contraction of the stapedius muscle in response to unilateral loud sound stimuli.
- Acoustic reflex testing:** The portion of the immittance testing that measures the minimal sound stimuli needed to prompt reflexive stapedial contraction.
- Acoustic reflex threshold:** Least intense sound stimuli that elicit stapedius muscle contraction, normally between 85 and 100 dB.
- Admittance:** Measure of how much sound passes through the tympanic membrane, typically measured as a part of immittance testing.
- Air-bone gap:** Calculation obtained by subtracting the air conduction thresholds from the bone conduction thresholds at each individual frequency, suggestive of a conductive hearing loss.
- Air conduction threshold:** Determination of the patient's lowest level (dB) of hearing using pure-tone audiometry delivered through the ear canal at which $\geq 50\%$ of perception occurs.
- Anacusis:** Total loss of hearing. No bone or air conduction thresholds measured during audiometric testing.
- Audiometer:** A standardized machine used in hearing evaluations through which an audiologist can perform all basic aspects of audiometric assessments, including pure-tone and speech testing.
- Audiometric zero:** Horizontal line at "0" on the audiogram grid. This value is a calculated relative threshold obtained by averaging the lowest sound pressure level heard at each frequency in thousands of non-pathologic ears.
- Auditory brainstem response (ABR):** An objective audiometry test used to evaluate the cochlea and auditory pathway. Monitoring is accomplished by recording brain activity via electrodes placed on the scalp while varying sound stimuli are provided to the patient.
- Autophony:** Hearing one's own voice louder than usual. This can be a symptom seen with various ear pathologies.
- Bell's palsy:** Idiopathic facial (seventh) nerve paralysis.
- Bone conduction threshold:** Calculated as the lowest sound pressure level (dB) at which 50% of the pure tones introduced via the bone oscillator overlying the mastoid are perceived.
- Broca's aphasia:** Occurs as a result of an injury to Broca's area of the brain, leading to an inability to produce language without extreme effort and considerable halting.
- Carhart notch:** An improvement seen in the air conduction thresholds at 2000 Hz, classically found in otosclerosis patients.
- Cholesteatoma:** Middle ear disease involving keratin debris buildup that can be slowly erosive into important bony structures and lead to chronic infection.
- Cochlea:** Hearing organ located in the inner ear.
- Compliance:** Reciprocal of stiffness. A reflection of the mobility of the tympanic membrane and is determined during tympanometry.
- Conditioned play audiometry:** Type of audiometric testing (typically for toddler to preschool-aged children). Hearing threshold measurements are obtained by the child performing certain tasks whenever sound stimuli are heard.
- Conductive hearing loss:** Measured by the presence of an air-bone gap. Signals the presence of pathology in the external or middle ear that is obstructing normal sound transmission into the cochlea.
- Contralateral ear:** Ear opposite to the ear being tested.
- Crossover hearing:** When the non-test ear detects sound from the tested ear.
- Decibel (dB):** Logarithmic unit of measure of sound pressure or "loudness." See pascals.
- Decapascal (daPa):** Unit of measure used in tympanometry to signify the amount of air pressure emitted from the probe. Also can be measured in millimeters of water (mm H₂O).
- Distortion product otoacoustic emissions (DPOAEs):** Common form of OAE testing. DPOAEs are measured by playing two simultaneous sounds at the varying frequencies (generally from 1000–8000 Hz) and calculating the resulting cochlear responses.
- Eustachian tube dysfunction:** Disorder of the cartilaginous and bony tube that connects the middle ear to the nasopharynx, often manifesting as ear fullness, fluid sensation, and hearing loss.
- Frequency:** Refers to pitch or tone of a sound. Found on the x-axis on an audiogram and is measured in hertz.
- Hertz:** Unit of frequency on the x-axis on an audiogram.

- Impedance:** Refers to the opposition of the acoustic flow of energy. Used in the context of tympanometry and acoustic reflex testing. Opposite of admittance.
- Immittance:** General term used to indicate the principle of evaluating acoustic energy flow through the tympanic membrane and middle ear space. It is derived from the words *impedance* and *admittance*.
- Immittance testing:** In the audiologic context, refers to tympanometry and acoustic reflex testing.
- Interaural attenuation:** The amount of energy that is lost when sound travels through the skull from one ear to the contralateral side. Important to determine when performing masking.
- Ipsilateral ear:** The tested ear.
- Masking:** Utilized when performing air or bone conduction testing to help determine the tested ear's true thresholds and avoid crossover hearing.
- Masking dilemma:** The inability to successfully determine true bone thresholds for each ear due to the presence of moderate to severe conductive hearing loss in both ears.
- Meniere's disease:** A disease affecting the inner ear from unknown etiology leading to episodic constellation of four symptoms: tinnitus, ear fullness, hearing loss, and vertigo lasting minutes to hours.
- Middle ear pressure:** Measured during tympanometry testing. Normal range is from -100 to 0 daPa.
- Millimeters of water (mm H₂O):** Unit of measure used in tympanometry to signify the amount of air pressure emitted from the probe. Also can be measured in decapascal (daPa).
- Millimhos (mmhos):** Unit of measure used in tympanometry to signify the compliance or admittance of the tympanic membrane. Also can be measured in milliliters (mL).
- Mixed hearing loss:** Evident on pure-tone testing. Denotes hearing loss with both sensorineural and conductive components.
- Monitored live voice:** A technique used during speech testing in which the examiner controls his or her voice intensity to ensure each spondee syllable has equal balance.
- Monomeric membrane:** Thin, weaker area of the tympanic membrane where only one or two layers are present that forms after a perforation heals.
- Monosyllabic words:** Used in word recognition testing to measure the ability of patients to discriminate speech with their current hearing impairment.
- Most comfortable listening level:** Suprathreshold level at which a person most comfortably hears speech.
- Myringotomy:** Incision made into the tympanic membrane to allow air pressure equalization or direct access to the middle ear space.
- Otoacoustic emissions:** Measurable emissions produced by healthy and functioning cochlear outer hair cells whenever sound stimuli are presented. Common test used for newborn hearing screens.
- Otosclerosis:** Autosomal dominant disease with incomplete penetrance that often results in fixation of the stapes footplate at the oval window causing a conductive hearing loss.
- Ototoxicity:** Refers to the detrimental result on the hearing or balance functions of the inner ear by a medication or chemical agent.
- Oval window:** Underneath the stapes footplate through which sound is transmitted via vibration into the cochlea.
- Pascals:** Unit of measure of sound pressure or "loudness." See decibels.
- Peak compliance (Y_{tm}):** Point of maximal mobility on tympanometry. In a normal tympanic membrane, the maximal compliance usually is centered over 0 daPa.
- Peak static acoustic admittance (Y_{tm}):** See peak compliance.
- Practice effect:** Falsely elevated speech scores due to undergoing several audiograms over a short period of time.
- Presbycusis:** Age-related changes in hearing, typically associated with a gradually progressive downsloping or high-frequency sensorineural hearing loss.
- Pseudohypoacusis:** Nonorganic cause to hearing loss noted on audiogram. Most commonly seen with malingering.
- Pure-tone audiometry:** One of three portions of any standard audiogram. Recorded pure tones are employed to determine levels of air and bone conduction at 250, 500, 1000, 2000, 4000, and 8000 Hz at variable sound pressure level intensities.
- Pure-tone average:** Calculated average using air thresholds garnered at the frequencies at 500, 1000, and 2000 Hz. This average should be very close to the speech reception and detection thresholds.
- Recruitment:** Phenomenon in cochlear hearing loss whereby a small increase in sound stimuli leads to a disproportionate increase in sound perception. Also known as loudness recruitment.
- Reliability:** Assessment made by the audiologist regarding the consistency and validity of the obtained pure-tone and speech testing data.
- Retrocochlear hearing loss:** Hearing loss that is caused by a lesion beyond the cochlea (ie, vestibular schwannoma).
- Rollover:** Phenomenon in retrocochlear impairment whereby further increases in intensity cause decreased speech reception scores.
- Sensorineural hearing loss:** Evident on pure-tone testing. Denotes pathology either of the cochlea or auditory nerve.
- Sound-field testing:** Speech detection threshold obtained when sound is delivered through loudspeakers on either side of the patient.
- Sound pressure:** Ambient air pressure change measured when a sound is emitted. Measured in pascals.
- Sound pressure level:** The y-axis on the audiogram grid. It is calculated on a logarithmic scale using the sound pressure (pascals) relative to the audiometric zero. A decibel (dB) is the unit of measurement for this calculation.
- Speech audiometry:** One of three portions of any standard audiogram. Recorded speech is used to obtain speech reception or detection thresholds, as well as word recognition testing.

- Speech detection threshold:** Level at which a listener can identify the presence of speech 50% of the time.
- Speech discrimination testing:** Type of test whereby a list of monosyllabic words is presented to the patient near the most comfortable listening level (MCL) and the patient is asked to repeat them. Same as word recognition testing.
- Speech reception threshold:** Lowest level at which a listener can identify and repeat bisyllabic words correctly 50% of the time.
- Spondee words:** Bisyllabic words where each syllable has the same weight. Used in speech reception threshold testing.
- Stapedius muscle:** Attaches to the stapes via the neck from the pyramidal eminence, innervated by the facial nerve. Movement of this muscle is detected as part of the acoustic reflex test.
- Static immittance:** Representation of ear canal volume, usually expressed in cubic centimeters.
- Stenger test:** Test used when malingering or pseudohypacusis is suspected.
- Sudden sensorineural hearing loss:** Rapid unexplained hearing loss in which bone thresholds are elevated.
- Superior olivary nucleus:** Important brainstem relay station involved in the auditory pathway and acoustic reflex.
- Tympanic membrane:** Boundary between the external ear canal and middle ear. Important to the amplification and conduction of sound pressure waves. Also called the "eardrum."
- Tympanometry:** Measurement of tympanic membrane mobility at various air pressures in the external ear.
- Tympanosclerosis:** Scarring and calcification of the tympanic membrane, often causing a conductive hearing loss. May be also referred to as myringosclerosis.
- Tympanostomy tube:** Small tube placed into a myringotomy incision to allow long-term communication between the external auditory canal and the middle ear space. Also known as pressure equalization tube (PET).
- Uncomfortable loudness level:** Lowest sound pressure level at which the stimuli are painful.
- Vestibular schwannoma:** Benign tumor of the vestibulocochlear (eighth) cranial nerve. Also known as acoustic neuroma.
- Vestibulocochlear (eighth) nerve:** Cranial nerve arising from the brainstem responsible for hearing and balance. Splits into the cochlear and vestibular nerves prior to entering the internal auditory canal.
- Visual reinforcement audiometry:** Type of audiometric testing (typically for infant to toddler-aged children). Threshold measurements are obtained by reinforcing a behavior—like turning head toward a lighted toy—whenever a sound stimulus is heard.
- Word recognition scores:** A calculated percentage representing the correct number of monosyllabic words recognized from a list of prerecorded words. How the word recognition testing is reported on an audiogram.
- Word recognition testing:** Same as speech discrimination testing.

Note: Page numbers in **bold** reference non-text material.

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